

The Astrophysics of Isotopes

Conroy & Claude

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Preamble

This memo was written by Claude (Opus 5.5), working in Claude Code with Charlie Conroy over several sessions in September and October 2026. Conroy chose the subject, approved an outline built around the three roles of an isotope ratio, and directed the scope, emphasis, figures, and style through many rounds of comments, while Claude drafted the text and made the figures. Most of the literature research was delegated to subagents, each of which surveyed one part of the subject through ADS and arXiv and wrote a dossier in which every number is tied to a primary source that was opened and read (bibcode and table, equation, or page). No number was taken from the model's memory. The figures were made for this memo from published data and public codes (e.g., PRyMordial for Big Bang nucleosynthesis, SYNTHÉ for the synthetic spectra, MIST for the stellar tracks, and IntCal20 for radiocarbon), and the provenance of each data file is recorded with it.

The draft then went through several rounds of review by independent agents: section-by-section audits of accuracy, checks of consistency and style, reviews of coverage and of the physical explanations, and a verification pass after each large batch of edits, in which every new sentence was compared with its sources. These reviews found errors, including misattributions introduced while earlier corrections were being applied, and each correction was itself checked against the sources. However, the memo has not been refereed, and readers should consult the cited papers before relying on a specific number.

1. Introduction

Most of what we know about the chemical composition of the Universe is expressed as in terms of elemental abundances. However, an elemental abundance is a sum over isotopes, which can discard important information. The isotopes of one element are often made by different nuclear processes, in different kinds of stars, and at different times. An isotope ratio recovers that information. Some isotopes are also radioactive, and physical and chemical processes can separate the isotopes of an element. Isotope ratios can thus measure time and trace those processes as well.

In this memo we review how isotope ratios are measured in astronomy and what can be learned from them. Our goal is to identify the physical ideas that connect a very broad literature.

We find it useful to organize the subject around the three roles that an isotope ratio can play:

- A **fingerprint** of a nuclear process. When two isotopes of an element are made in different places, their ratio identifies the process or the site. The ratio $^{12}\text{C}/^{13}\text{C}$ compares helium burning with the CN cycle, the odd/even ratio of barium compares the r and s processes, and the $^{17}\text{O}/^{16}\text{O}$ of a presolar grain gives the mass of the star in which it formed. The ratio of a secondary to a primary isotope is a gauge of chemical enrichment, because it grows with the metallicity of the stars that made the secondary isotope.
- A **clock** or heat source. Radioactive isotopes decay at rates set by nuclear physics. Their abundances measure time (radiocarbon dating is the familiar example), and their decay energy powers supernova light curves and melted the first planetesimals. Isotopes that are released only after a known stellar lifetime (e.g., $^{25,26}\text{Mg}$ from intermediate-mass asymptotic giant branch, or AGB, stars) turn an enrichment gauge into a slower clock.
- A **tracer of fractionation**. Isotopes of the same element have slightly different chemical and physical behavior, because of their different masses, and processes that act on those differences (low-temperature chemistry, photodissociation, diffusion, and atmospheric escape) leave their marks in the isotope ratios.

Two physical ideas connect the three roles. The first is that the most informative isotope ratios are those in which the quantities we do not know cancel. The isotopes of an element share their chemistry, so their ratio is almost insensitive to the chemical history of the element. Other ratios are chosen so that the distance to a source, the amount of a radioactive parent, or the time between nucleosynthesis and the formation of the Solar System cancels. What cannot cancel is the reference with which we compare a measured ratio (the production ratio of a radioactive clock, the solar composition, or a laboratory standard), and many of the disputes we describe concern that reference. The second idea is that a nuclide made and destroyed at steady rates settles at an abundance equal to its production rate times its lifetime, so that a ratio of two such abundances measures a ratio of rates. This steady state governs the isotopes made in the CN cycle and the s process, the radioactive isotopes in the Galaxy, and the decay chains of the heavy elements.

Deuterium shows the range of these effects in a single element, and in Fig. 1 we collect its abundance across environments. The primordial D/H ratio of $(2.527 \pm 0.030) \times 10^{-5}$, measured in distant quasar absorption systems (Cooke et al. 2018), is a fingerprint of the first minutes of the Universe

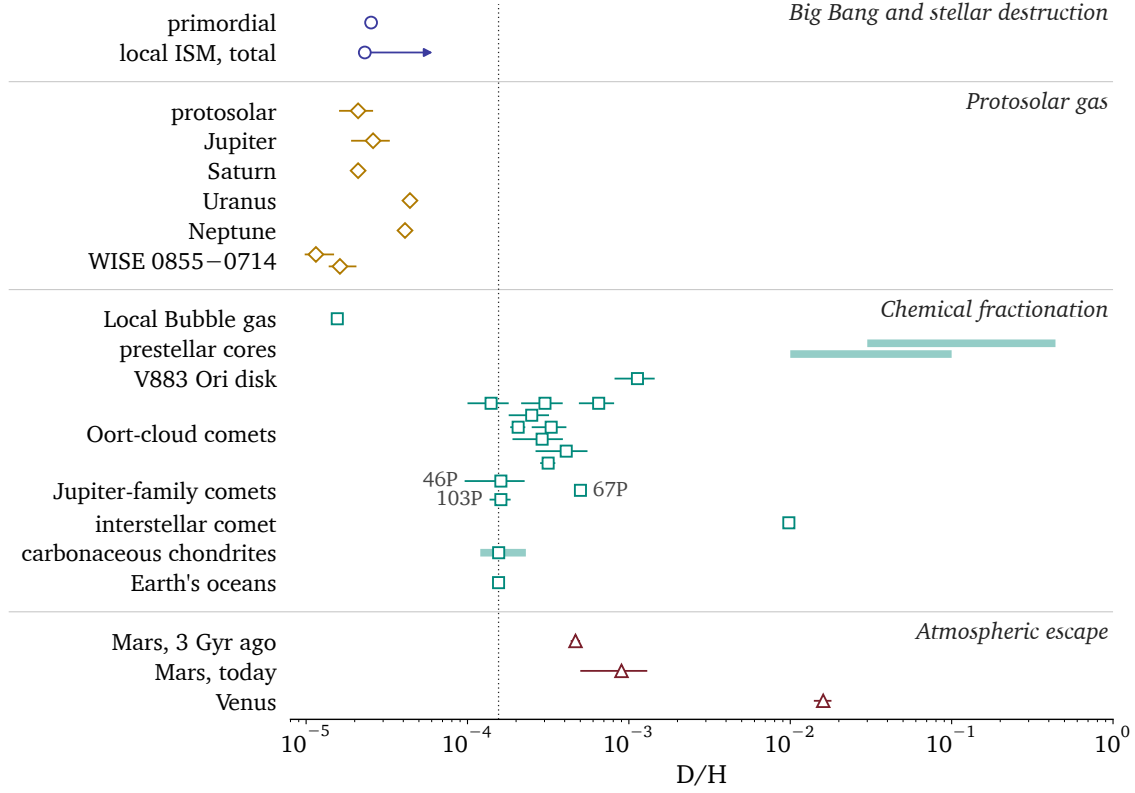


Figure 1: D/H per atom across environments, grouped by the process that sets it; Local Bubble gas is the gas-phase value, the two WISE 0855–0714 points are from two *JWST* data sets, and the interstellar comet is 3I/ATLAS. Markers are published values with 1σ errors (3σ for 1P/Halley), light bars are ranges (molecules with one H atom in prestellar cores; bulk carbonaceous chondrites around their mean), the arrow is a lower limit, and the dotted line marks terrestrial ocean water.

and one of the most precise measurements of the cosmic baryon density (Section 4.1). Stars destroy deuterium, so D/H also records how much gas has been cycled through stars. The total (gas plus dust) D/H of the local interstellar medium, $\geq 2.3 \times 10^{-5}$ (Linsky et al. 2006), is within $\sim 10\%$ of the primordial value.

However, in cold molecular gas, chemistry raises the D/H of individual molecules to 0.01–0.44, and the D/H of water in the Solar System, from the Earth’s oceans to comets, carries a diluted imprint of that chemistry. Atmospheric escape then raised the D/H of Venus to 100 times that of the Earth (Section 6.4). Notice that the full range covers more than four decades in a single element, and that the largest excursions have nothing to do with nucleosynthesis.

A second theme is that isotopes are difficult to observe. The isotopes of an element have almost identical atomic spectra, and for most elements the isotope shifts of atomic lines are much smaller than the widths of the lines in stellar spectra. As a result, the isotope ratios that we can measure are determined by spectroscopic physics as much as by astrophysical interest.

Molecules have much larger isotope shifts than atoms, and they supply most of the isotope ratios of the light elements. A few heavy elements have large hyperfine structure or field shifts. We can identify radioactive isotopes by their γ -ray lines. Finally, we can measure samples that reach the

Table 1. The isotope ratios discussed in this memo, with the principal ways in which each is measured and the roles it plays.

isotopes	principal measurements	roles	sections
D/H	Lyman lines, molecules, samples	fingerprint, fractionation	4.1, 6.2, 6.4–6.7
$^3\text{He}/^4\text{He}$	$^3\text{He}^+$ 8.7 GHz, He I, solar wind	fingerprint, fractionation	4.1, 4.2, 6.5, 6.8
$^6\text{Li}/^7\text{Li}$	Li I 6708 Å	fingerprint	3.1, 4.1
$^{10}\text{B}/^{11}\text{B}$	B II 1362 Å (interstellar)	fingerprint	2.2, 3.1
$^{12}\text{C}/^{13}\text{C}$	CO, CN, CH, C ₂ ; mm; grains	fingerprint, fractionation	4.2, 4.4, 4.5, 6.2, 6.6
$^{14}\text{N}/^{15}\text{N}$	CN, HCN, HNC, NH ₃ ; samples	fingerprint, fractionation	4.4, 6.2, 6.3–6.7
$^{16}\text{O}/^{17}\text{O}/^{18}\text{O}$	CO; samples	fingerprint, fractionation	4.2, 4.4, 4.5, 6.3
$^{20}\text{Ne}/^{22}\text{Ne}$	cosmic-ray detectors	fingerprint	5.4
$^{24,25,26}\text{Mg}$	MgH bands	fingerprint, clock	4.3
$^{32}\text{S}/^{34}\text{S}$	CS isotopologues	fingerprint	4.4
Ba, Eu isotopes	Ba II 4554 Å, Eu II lines	fingerprint	4.3
Ca, Hg, Pt isotopes	Ca II, Hg II, Pt II lines	fractionation	6.8
Tc, Th, U	stellar lines	clock	5.2
^{56}Co , ^{44}Ti , ^{26}Al , ^{60}Fe	γ -ray lines, light curves	clock, heat	5.3, 5.4
^{10}Be , ^{60}Fe	cosmic-ray detectors	clock	5.4, 5.5
^{26}Al , ^{129}I , ^{247}Cm , U, Pb	meteorites	clock, heat	5.1, 5.5
^{182}Hf , ^{182}W	meteorites, terrestrial rocks	clock	5.5
^{60}Fe , ^{244}Pu	deep-sea deposits	clock	5.5
^{232}Th , ^{238}U	geoneutrinos	heat	5.5.5
^{14}C	tree rings, organic samples	clock	5.1, 5.5

laboratory (meteorites, presolar grains, and the solar wind) with a precision that no telescope can approach. We develop this physics in Section 3, and Table 1 lists the isotope ratios discussed in this memo together with the ways in which each one is measured.

The memo is organized as follows. In Section 2 we describe the nuclear origins of the isotopes, and in Section 3 we describe how isotopes are measured. Sections 4, 5, and 6 take up the three roles in turn, and in Section 7 we summarize the main lessons and the open problems.

2. Nuclear origins of the isotopes

We begin with a map of which nuclei exist and how each one was made. The chemistry of an atom is set by its electrons, so the isotopes of an element behave almost identically in a stellar atmosphere or a molecular cloud (we discuss the exceptions in Section 6). An isotope ratio measured in either place preserves the nuclear history of the material.

2.1. The chart of the nuclides

We specify a nucleus by its proton number Z and neutron number N , with mass number $A = Z + N$. NUBASE2020 (Kondev et al. 2021) lists 254 nuclides as stable. A further 32 radioactive nuclides have half-lives long enough that they survive from before the formation of the Solar System (e.g., ^{40}K , ^{87}Rb , ^{232}Th , and ^{238}U). These 286 **primordial nuclides** belong to 83 elements. Every element from hydrogen to lead except Tc and Pm has a stable isotope, and Bi, Th, and U survive as long-lived radioactive nuclides.

In Fig. 2a we plot the neutron excess $N - Z$ of each primordial nuclide vs. its proton number, so that the isotopes of each element form a column. The nuclides occupy a narrow **valley of stability** that stays at $N - Z \approx 0$ up to ^{40}Ca and then climbs toward neutron-rich nuclei. The bend occurs because the Coulomb repulsion among the protons grows faster with Z than the nuclear binding does. At fixed A the most stable heavy nuclei therefore have extra neutrons. The ratio N/Z reaches 1.15 for ^{56}Fe , 1.54 for ^{208}Pb , and 1.59 for ^{238}U .

We can also see the effects of pairing and of closed shells in the chart. Nucleons of the same kind bind more tightly in pairs, so of the 254 stable nuclides 147 are even–even and only five are odd–odd (^2H , ^6Li , ^{10}B , ^{14}N , and the isomer ^{180m}Ta). As a consequence, no odd- Z element has more than two stable isotopes. From Na to Tl the odd- Z elements are also less abundant than the geometric mean of their two even- Z neighbors, by 0.15–2.5 dex with a median of 0.8 dex in the protosolar mixture (Lodders 2021).

The **magic numbers** 2, 8, 20, 28, 50, 82, and 126 mark closed nuclear shells (Mayer 1949). Nuclei with a magic number of neutrons have unusually small neutron-capture cross sections, because the next neutron is weakly bound and the compound nucleus forms at a low excitation energy, where there are few levels to capture into. Closed neutron shells produce the abundance peaks of the heavy elements, as we will see below. Finally, there is no stable nucleus with $A = 5$ or $A = 8$. These two gaps stop Big Bang nucleosynthesis at ^7Li and force stars to build carbon through the triple- α reaction, in which a third α particle must be captured within the brief lifetime of ^8Be .

2.2. Nucleosynthetic processes

Burbidge et al. (1957) (hereafter B²FH) argued that “at least eight different types of synthesizing processes are demanded” to explain the solar-system abundances, and they assigned every known isotope to one of them. Cameron (1957) reached the same picture independently. The modern list adds the neutrino process, the γ process, and neutron-rich nuclear statistical equilibrium, and replaces the α process of B²FH with carbon, neon, and oxygen burning (Woosley et al. 2002). We list the processes, with their sites and physical conditions, in Table 2. In Fig. 2a each nuclide is colored by its dominant process. Notice that the cells in one column of Fig. 2a, i.e., the isotopes

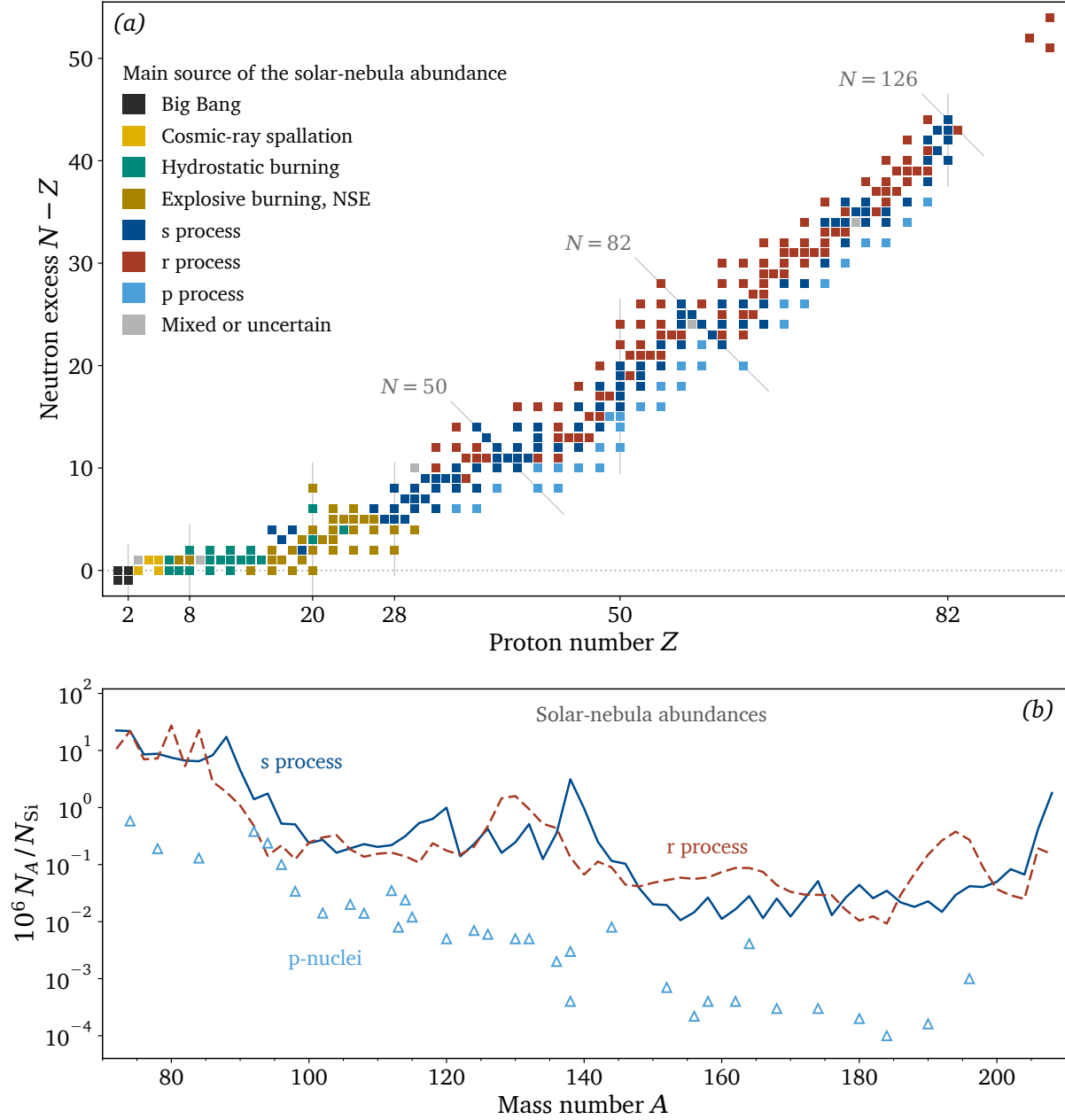


Figure 2: (a) The 286 primordial nuclides as neutron excess $N - Z$ vs. proton number, so that each element forms a column, colored by the process that dominates its abundance in the solar nebula (for $A \leq 68$ the first origin listed by Woosley et al. 2002, with ^{11}B assigned to spallation, and for heavier nuclides the s/r decomposition of Prantzos et al. 2020 and the p-nuclei of Rauscher et al. 2013; gray marks mixed or uncertain origins); gray segments mark the magic proton numbers (vertical) and the closed neutron shells $N = 50, 82$, and 126 (diagonal), and the dotted line is $N = Z$. (b) s-process (solid) and r-process (dashed) abundances in the solar nebula per 10^6 Si atoms at each even mass number, summed over isobars (protosolar abundances from Lodders 2021, divided into s and r parts by Prantzos et al. 2020), with the 35 p-nuclei as triangles (Rauscher et al. 2013).

Table 2. The nucleosynthetic processes. Burning temperatures are central values in a $25 M_{\odot}$ model (Woosley et al. 2002), except for novae and the γ process; neutron densities are from Käppeler et al. (2011), Hampel et al. (2016), Cowan et al. (2021), and Woosley et al. (2002).

process	principal sites	conditions	characteristic products
Big Bang	early Universe	...	^2H , ^3He , ^4He , some ^7Li
spallation	cosmic rays in the ISM	...	^6Li , ^9Be , ^{10}B , ^{11}B
ν process	core-collapse supernovae	$kT_{\nu} \approx 6\text{--}8 \text{ MeV}$	^{11}B , ^{138}La , ^{180m}Ta
H burning	cores, shells, novae	$3.8 \times 10^7 \text{ K}$; $2\text{--}3 \times 10^8 \text{ K}$	^{13}C , ^{14}N , ^{15}N , ^{17}O , ^{26}Al
He burning	cores and shells	$2.0 \times 10^8 \text{ K}$	^{12}C , ^{16}O , ^{18}O , ^{22}Ne
C, Ne, O burning	massive stars	$0.84\text{--}2.1 \times 10^9 \text{ K}$	^{20}Ne , ^{24}Mg , ^{28}Si , ^{32}S
Si burning, NSE	massive stars, supernovae	$3.7 \times 10^9 \text{ K}$	Fe peak (as ^{56}Ni)
main s process	AGB stars	$n_n = 10^6\text{--}10^8 \text{ cm}^{-3}$	Sr to Pb
weak s process	massive stars	$n_n \approx 10^8 \text{ cm}^{-3}$	Cu, Ga, Ge, As
i process	H ingested into He shells	$n_n \lesssim 10^{15} \text{ cm}^{-3}$	CEMP-r/s patterns
r process	neutron-star mergers	$n_n \gtrsim 10^{26} \text{ cm}^{-3}$	Eu, Pt, actinides
γ process	core-collapse supernovae	$2\text{--}3 \times 10^9 \text{ K}$	most of the 35 p-nuclei

of a single element, frequently have different colors (43 of the 62 elements with more than one primordial isotope have isotopes in more than one color). This is the basic reason that isotope ratios carry information. We now step through the chart from light to heavy nuclei.

2.2.1. Light nuclei

The lightest nuclides are relics of the Big Bang and of cosmic-ray spallation. Big Bang nucleosynthesis made ^1H , ^2H , ^3He , ^4He , and some ^7Li (Section 4.1). At the *Planck* baryon density (Fields et al. 2020), the Big Bang supplied 89% of the Sun’s ^4He by mass and $\sim 60\%$ of its ^3He (Lodders 2021). The primordial $^7\text{Li}/\text{H}$ is 23% of the protosolar value, and once the destruction of lithium in stars is included, the Big Bang supplied only $\sim 12\%$ of the Sun’s ^7Li (Prantzos 2012). Approximately 80% of the primordial deuterium survived to the birth of the Sun.

The fragile nuclei ^6Li , ^9Be , and ^{10}B are destroyed in stellar interiors, so we must look elsewhere for their origin. They are made when cosmic-ray protons and α particles break up C, N, and O nuclei in the interstellar medium (Reeves et al. 1970). Spallation produces $^{11}\text{B}/^{10}\text{B} = 2.8$, below the meteoritic value of 4.0, and so accounts for approximately 70% of ^{11}B (Prantzos 2012). The remainder comes from the **neutrino process**, in which the neutrinos from a collapsing core spall nuclei in the overlying shells (Woosley et al. 1990). The neutrino process also makes most of the rare odd–odd nuclei ^{138}La and ^{180m}Ta (Heger et al. 2005). The origin of most of the solar ^7Li is not settled. The Big Bang, cosmic rays, and neutrinos together supply at most half of it, and Prantzos (2012) argues that the rest must come from low-mass stars or novae (we return to novae in Section 5.3).

2.2.2. Hydrogen burning

Hydrogen burning converts four protons into a helium nucleus through the pp chains or, in stars more massive than $\approx 1.2 M_{\odot}$ (Karakas & Lattanzio 2014), through the CNO cycle, in which C, N, and O act as catalysts. The catalysts are rearranged along the way. In the CN cycle, $^{12}\text{C}(p,\gamma)^{13}\text{N}(\beta^+\nu)^{13}\text{C}(p,\gamma)^{14}\text{N}$, and the slowest step, $^{14}\text{N}(p,\gamma)$, accumulates the catalysts as ^{14}N . In Fig. 3a we draw the cycle on the chart of the nuclides. A proton capture moves a nucleus up by one row, a β^+ decay moves it diagonally down and to the right, and the $^{15}\text{N}(p,\alpha)^{12}\text{C}$ reaction closes the loop by moving two columns left and one row down. Within the cycle the two carbon isotopes come into equilibrium with each other, and if we set the rate of change of the ^{13}C abundance to zero we find:

$$\frac{dn_{13}}{dt} = n_p (n_{12}\langle\sigma v\rangle_{12} - n_{13}\langle\sigma v\rangle_{13}) = 0 \implies \left(\frac{^{12}\text{C}}{^{13}\text{C}}\right)_{\text{eq}} = \frac{\langle\sigma v\rangle_{13}}{\langle\sigma v\rangle_{12}}, \quad (1)$$

where $\langle\sigma v\rangle_{12}$ and $\langle\sigma v\rangle_{13}$ are the proton-capture rates of the two isotopes. Notice that the equilibrium ratio is independent of density, because both rates are proportional to the proton density, and that it depends only weakly on temperature. Both carbon isotopes tunnel through the same Coulomb barrier, so the Gamow factor, which rises by $\sim 3 \times 10^{10}$ between 20 and 140 MK, differs between them by only 5–11% and largely cancels in the ratio. The ratio is then set by the low-energy cross sections of the two reactions. The LUNA underground measurements of these cross sections give $(^{12}\text{C}/^{13}\text{C})_{\text{eq}} = 3.6 \pm 0.4$ between 20 and 140 MK (Skowronski et al. 2023). Material in which the CN cycle has reached equilibrium has a carbon isotope ratio near 3.6, compared with 89 in terrestrial and meteoritic carbon. We will use this contrast in Section 4.2 as a probe of mixing in red giants. Eq. 1 is also our first example of a steady state, in which each nucleus in a chain of reactions accumulates until it is destroyed as fast as it is made, so that its abundance equals its production rate times its lifetime against destruction. The level populations of an atom in statistical equilibrium are likewise set by ratios of rates.

The CNO cycle also makes ^{17}O and destroys ^{18}O , and the NeNa and MgAl chains, which operate at higher temperatures, make ^{23}Na , ^{26}Al , and ^{27}Al (Karakas & Lattanzio 2014). Classical novae burn hydrogen explosively at $2\text{--}3 \times 10^8$ K and overproduce ^{13}C , ^{15}N , and ^{17}O . Chemical evolution models without novae fall short of the solar ^{15}N abundance by a factor of 6–10 (Kobayashi et al. 2011). Novae are an example of a stellar source identified purely from isotope ratios.

2.2.3. Helium and later burning stages

Helium burning at 2×10^8 K builds ^{12}C through the triple- α reaction and ^{16}O through $^{12}\text{C}(\alpha,\gamma)^{16}\text{O}$. The ^{14}N left behind by the CNO cycle is converted to ^{18}O by $^{14}\text{N}(\alpha,\gamma)^{18}\text{F}(\beta^+\nu)^{18}\text{O}$, and then to ^{22}Ne . In Fig. 3a this path appears as the dashed arrows, alongside the ON cycle of hydrogen burning, which makes ^{17}O . We can now see that each of the C and O isotope ratios compares the products of two different burning processes: ^{12}C and ^{16}O come from helium burning, ^{13}C and ^{17}O from hydrogen burning, and ^{18}O from the helium burning of nitrogen. Helium burning also releases the free neutrons that drive the s process, through $^{22}\text{Ne}(\alpha,n)^{25}\text{Mg}$ in the cores of massive stars and $^{13}\text{C}(\alpha,n)^{16}\text{O}$ in AGB stars (Käppeler et al. 2011).

Stars more massive than $\sim 8 M_{\odot}$ go on to burn carbon, neon, oxygen, and silicon, each stage generally hotter, denser, and shorter than the last. Oxygen burning leaves mostly ^{28}Si and ^{32}S .

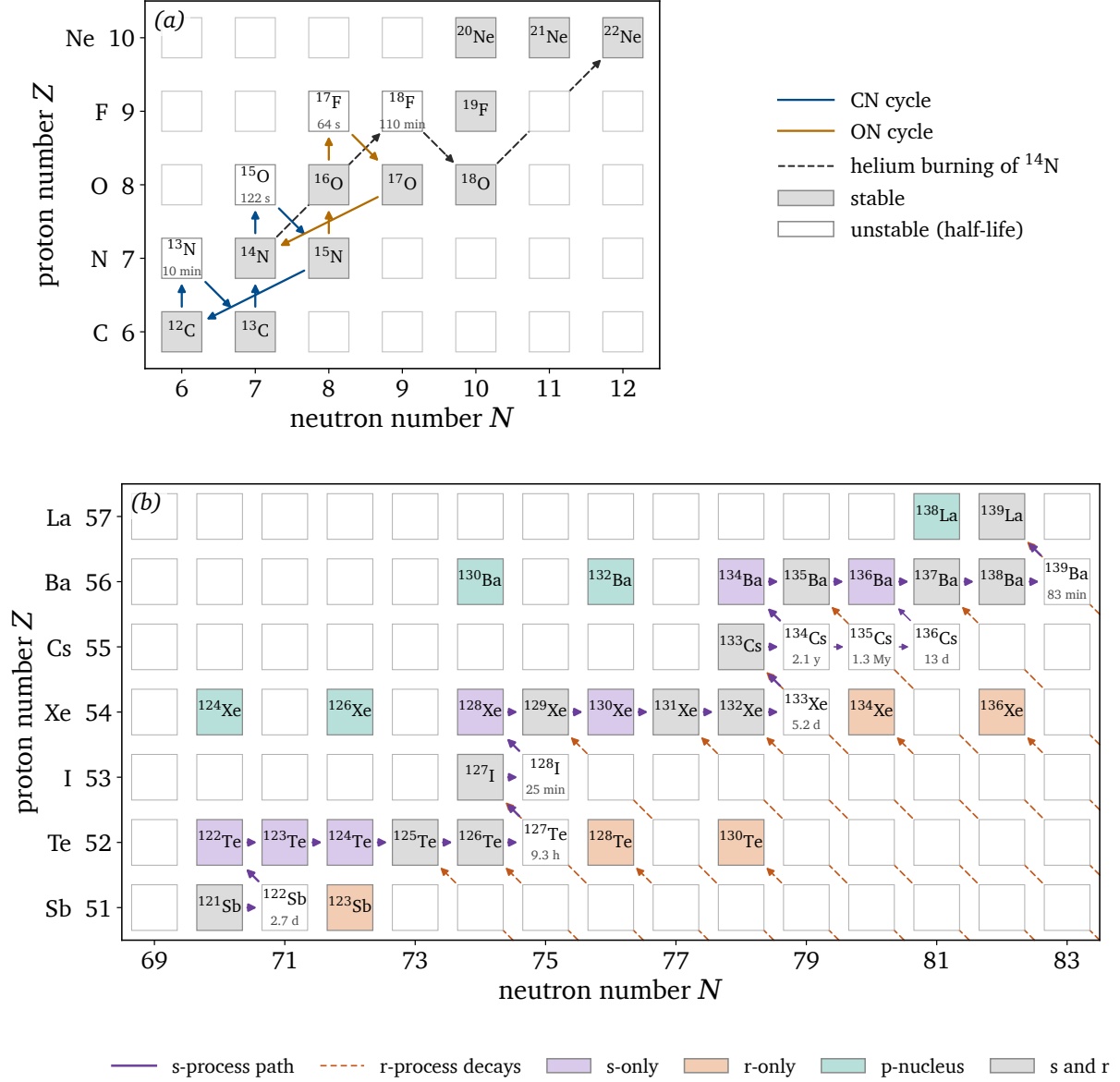


Figure 3: Two zooms on the chart of the nuclides, with stable nuclides shaded and the half-lives of the unstable nuclides along each path (NUBASE2020). (a) The CN and ON cycles and the helium burning of ^{14}N . (b) The s-process path from Sb to La, including the branch at ^{134}Cs , and the r-process β -decay chains that arrive along isobars from the neutron-rich side; shading distinguishes the s-only, r-only, and p-nuclei.

Silicon burning then drives the matter toward **nuclear statistical equilibrium** (NSE), in which the abundances are set by nuclear binding energies and the neutron excess.

NSE is the nuclear analog of the Saha equation. Captures of protons, neutrons, and α particles are balanced by the photodisintegrations that reverse them, so the abundance of each nucleus is set by its binding energy, as an ionization balance is set by the ionization potential (Cowan et al. 2021). The weak interactions, the only reactions that convert protons and neutrons into each other, are much slower. NSE is therefore established at a ratio of neutrons to protons that changes only gradually

(Cowan et al. 2021), as the molecular equilibrium of a cool stellar atmosphere is established at fixed total abundances of carbon and oxygen. The binding energy per nucleon is largest in the iron group, reaching 8.795 MeV for ^{62}Ni and 8.79 MeV for ^{58}Fe and ^{56}Fe (Wang et al. 2021), but these nuclei have more neutrons than protons. An important point is that matter with equal numbers of neutrons and protons cannot consist mostly of these nuclei, and NSE then favors ^{56}Ni (8.643 MeV per nucleon), the most tightly bound nucleus with $N = Z$. Most of the iron in the Universe was made as ^{56}Ni and decayed through ^{56}Co to ^{56}Fe , and we will see in Section 5.3 that this decay chain powers the light curves of supernovae.

2.2.4. The *s* process

Charged-particle burning ends in the iron group, where the binding energy per nucleon peaks, and beyond the iron peak the growing Coulomb barrier blocks charged-particle reactions. The heavy nuclei are built by neutron capture, which faces no barrier. In the **s process** (slow neutron capture) the neutron density is low enough that an unstable nucleus almost always β -decays before it captures another neutron, so the flow follows the valley of stability. If we write N_A for the abundance of the nucleus of mass A on the path, σ_A for its neutron-capture cross section, and $\tau = \int n_n v_T dt$ for the neutron exposure, then the flow obeys:

$$\frac{dN_A}{d\tau} = \sigma_{A-1}N_{A-1} - \sigma_A N_A, \quad (2)$$

and in steady flow $\sigma_A N_A \approx \sigma_{A-1} N_{A-1}$, i.e., the *s* process leaves abundances inversely proportional to the capture cross sections. This is the steady state of Eq. 1 again, with neutron captures in place of proton captures. The magic nuclei with $N = 50, 82$, and 126 have very small cross sections and act as bottlenecks. We can see the result in Fig. 2b, where the *s*-process abundances pile up at ^{88}Sr , ^{138}Ba , and ^{208}Pb .

The main *s* component is made in low-mass AGB stars, where $^{13}\text{C}(\alpha, n)$ burns at 0.9×10^8 K and provides $n_n = 10^6$ – 10^8 cm $^{-3}$. This component requires a seed of only 0.04% of the solar iron, with each seed nucleus capturing ~ 15 neutrons (Käppeler et al. 2011). The weak component, made in massive stars, supplies most of the solar Cu, Ga, Ge, and As (Käppeler et al. 2011). In Fig. 3b we show the *s*-process path from Sb through Te, I, Xe, Cs, and Ba. Each neutron capture moves the nucleus one column to the right, and whenever the product is unstable it β^- -decays diagonally up and to the left, back to the valley of stability. Where the path meets an unstable nucleus with a long enough half-life, the flow branches. A branch opens above the critical neutron density at which neutron captures compete with the β decay (Käppeler et al. 2011), as collisions compete with radiative decay above the critical density of a spectral line. Consider ^{134}Cs ($t_{1/2} = 2.06$ yr). At low neutron densities this nucleus decays to ^{134}Ba , while at higher densities some of the flow captures further neutrons and reaches ^{136}Ba through ^{135}Cs and ^{136}Cs . The ratio of the *s*-only isotopes $^{134}\text{Ba}/^{136}\text{Ba}$ therefore measures the neutron density (Section 4.5).

2.2.5. The *r*, *p*, and *i* processes

The **r process** (rapid neutron capture) operates at neutron densities of 10^{26} cm $^{-3}$ and higher (Cowan et al. 2021). Captures then outpace β decays, and the flow runs far out on the neutron-rich side of the chart, along a path set by the balance between neutron captures and photodisintegrations. Once

the neutrons are exhausted, the nuclei decay back toward stability. The magic numbers also produce peaks in the r-process abundances. At each closed neutron shell the r-process path bends toward stability, where the β decays are slow, and matter accumulates at these **waiting points** (Cowan et al. 2021). The waiting points lie at lower Z than the stable magic nuclei, so after decay the pile-ups land at $A \approx 80, 130$, and 195 (8–13 mass units below the s-process peaks at 88, 138, and 208). We can see this doubling of each peak in Fig. 2b. Suess & Urey (1956) noticed the doubling in the abundance curve and attributed it to slow and rapid neutron capture, and B²FH developed the theory of the two processes.

By number, the protosolar nuclei with $A \geq 69$ are $\approx 56\%$ s process, 44% r process, and 0.5% p process in the decomposition of Prantzos et al. (2020). The r-process sites include neutron-star mergers and possibly rare classes of core-collapse supernovae (Arcones & Thielemann 2023). Mergers were identified as a site through the strontium in the kilonova that followed GW170817 (Watson et al. 2019) and an emission line attributed to tellurium in the kilonova of GRB 230307A (Levan et al. 2024). The proton-rich edge of the chart holds 35 **p-nuclei**, from ^{74}Se to ^{196}Hg , that neither neutron-capture process can reach. They are 1–2 dex less abundant than their neighbors and are thought to be made in the γ **process**, the photodisintegration of pre-existing heavy nuclei at $2\text{--}3 \times 10^9$ K in the O/Ne layers of exploding massive stars (Woosley et al. 2002; Rauscher et al. 2013). No element is dominated by its p isotopes, so we know the p-process abundances only from solar-system material. Finally, in the **i process** (intermediate neutron capture), protons mixed into a helium-burning convective zone produce neutron densities up to 10^{15} cm^{-3} (Cowan & Rose 1977; Hampel et al. 2016).

2.3. Primary and secondary isotopes

A **primary** isotope is made from hydrogen and helium within the star that ejects it, so its yield does not depend on the metallicity the star was born with. A **secondary** isotope requires seed nuclei inherited from earlier generations (Arcones & Thielemann 2023). For example, ^{13}C and ^{14}N are made by CN cycling of a star’s initial carbon, ^{17}O by proton capture on ^{16}O , ^{18}O and ^{22}Ne from the initial ^{14}N , and the s process requires iron seeds. The minor isotopes of C, O, and Mg are therefore largely secondary, while ^{12}C , ^{16}O , and ^{24}Mg are primary.

We can see why the distinction matters with the simplest model of chemical evolution, the closed box, in which gas forms stars and the stars return their ejecta to the same gas at once. Let ds be the mass of gas locked into long-lived stars per unit mass of remaining gas. The mass fraction of a primary isotope then grows as $dZ_P = y_P ds$, where y_P is its yield. A secondary isotope grows as $dZ_S = \alpha Z_P ds$, since its production is proportional to the abundance of seed nuclei, which we take to scale with Z_P . Dividing the second growth law by the first, we find:

$$\frac{dZ_S}{dZ_P} = \frac{\alpha Z_P}{y_P} \implies Z_S = \frac{\alpha}{2y_P} Z_P^2, \quad \frac{Z_S}{Z_P} \propto Z_P, \quad (3)$$

whatever the star-formation history. The ratio of a secondary to a primary isotope of the same element therefore grows in proportion to the metallicity. In the model of Kobayashi et al. (2011) for the solar neighborhood, the ratio $^{16}\text{O}/^{17}\text{O}$ falls from 190 000 to 1790 between $[\text{Fe}/\text{H}] = -2.6$ and 0, $^{16}\text{O}/^{18}\text{O}$ from 14 000 to 460, and $^{24}\text{Mg}/^{25}\text{Mg}$ from 250 to 8.5. Real stars depart from this simple scaling, for example through hot-bottom burning in intermediate-mass AGB stars, which converts

freshly dredged-up ^{12}C into ^{13}C and ^{14}N and makes both partly primary. In this model $^{12}\text{C}/^{13}\text{C}$ falls from 4200 to 89, but not monotonically, as the AGB stars begin to contribute. Delays of this kind, in which an isotope is released only after the lifetime of an intermediate-mass star, allow an enrichment gauge to serve as a clock (Section 4.3).

2.4. The solar-system reference pattern

Almost every isotope ratio in astronomy is quoted relative to the Solar System, so we need to know where the solar values come from.

2.4.1. Protosolar isotope ratios

The standard compilation is Lodders (2021), which gives protosolar isotopic abundances, i.e., the composition of the solar nebula 4.567 Gyr ago (with radioactive nuclides corrected for decay to that epoch). For most elements the isotopic compositions are terrestrial (Meija et al. 2016). The exceptions are H, C, N, O, and the noble gases, whose terrestrial samples are not representative of the solar nebula and whose protosolar values come from the solar wind and from Jupiter. The terrestrial and protosolar values of these elements can differ substantially. The Sun, as inferred from solar-wind samples, has $^{14}\text{N}/^{15}\text{N} = 440$ (Lodders 2021) while terrestrial air has 272 (Füri & Marty 2015), and solar oxygen is $\approx 6\%$ poorer in ^{17}O and ^{18}O than terrestrial oxygen (McKeegan et al. 2011). For carbon, Lodders (2021) adopts $^{12}\text{C}/^{13}\text{C} = 95.6$ from the solar wind, compared with 89 in terrestrial and meteoritic carbon, and we use the latter as the solar-system reference in this memo. The nitrogen and oxygen differences probably record chemistry in the solar nebula or its parent cloud, and we return to them in Section 6.3.

2.4.2. The s, r, and p contributions

The structure of the chart also allows us to separate the s-, r-, and p-process contributions isotope by isotope, and Fig. 3b shows how. The r process delivers matter from the neutron-rich side by β^- decay, and each decay chain stops at the first stable isobar it reaches. A nucleus with a stable isobar at lower Z is **shielded** from the r process, and a shielded nucleus that also lies on the s-process path is **s-only**, made only by the s process (e.g., ^{134}Ba and ^{136}Ba , shielded by ^{134}Xe and ^{136}Xe). Shielded nuclei that lie off the s-process path, such as ^{130}Ba and ^{132}Ba in Fig. 3b, are p-nuclei. Conversely, the most neutron-rich isotopes of an element are bypassed by the s process, which β -decays to the next element before reaching them. These isotopes are **r-only** (e.g., ^{130}Te and ^{136}Xe).

Shielding requires two stable isobars, which pairing makes possible. Because nucleons bind more tightly in pairs, the odd–odd nuclei along an even- A isobar near the valley of stability lie above their even–even neighbors in mass, so an even–even nucleus can be stable even when an isobar two units of charge away is more tightly bound. In Fig. 3b, ^{134}Cs lies 1.23 MeV above ^{134}Xe and 2.06 MeV above ^{134}Ba (Wang et al. 2021), so ^{134}Xe cannot β -decay, and the double β decay to ^{134}Ba is extremely slow (half-life $> 10^{16}$ yr; Kondev et al. 2021). As a result, 27 of the 30 s-only nuclei and 31 of the 35 p-nuclei are even–even (Prantzos et al. 2020; Rauscher et al. 2013). The 30 s-only isotopes calibrate the s-process contribution, and subtracting that contribution from the solar abundances leaves the r-process residuals. These residuals are the standard reference pattern for r-process-enhanced stars (Snedden et al. 2008; Prantzos et al. 2020). (Branchings leak some s-process flow into the r-only nuclei, the least secure of the three labels.)

3. Measuring isotopes

King & Birge (1929) found ^{13}C in the laboratory as a faint copy of a C_2 Swan band, $7.5 \text{ \AA}$ to the red of the head at $4737 \text{ \AA}$. In the same year Sanford (1929) identified the band in carbon stars, where the isotope band is comparable in strength to the main band, although in a furnace the isotope band is very weak. In hindsight this was the first evidence for a non-terrestrial isotope ratio in a star, found in a molecule within a year of the discovery of ^{13}C . However, Sanford looked to the physical conditions in the stars for the explanation. ^{18}O was found in weak doublets of the atmospheric oxygen bands in the solar spectrum (Giauque & Johnston 1929), and deuterium through the mass shift of the Balmer lines of hydrogen that had been enriched by evaporating liquid hydrogen (Urey et al. 1932).

The isotopes of an element differ in three properties that affect a spectrum: the mass of the nucleus, its size, and its spin. The mass enters through a reduced mass (of the electrons in an atom, or of the vibrating nuclei in a molecule), the size through the overlap of the electron wavefunctions with the nucleus, and the spin through the hyperfine interaction. Each effect displaces or splits a line by a characteristic fraction of its wavelength.

Whether we can measure an isotope ratio depends on how that fraction compares with the width of the line. For a stellar absorption line the width is set by thermal and turbulent motions. At 5000 K with a microturbulence of 1.5 km s^{-1} the Doppler width (FWHM) is $\Delta\lambda/\lambda = 2.1 \times 10^{-5}$ for lithium, 1.1×10^{-5} for $A = 50$, and 9.3×10^{-6} for $A = 150$, comparable to the resolution element of an $R = 10^5$ spectrograph. In Fig. 4 we compare each of the isotope effects with this width.

3.1. Atomic isotope shifts

3.1.1. Mass shifts

We begin with the mass shift of an atomic line. The Rydberg constant is proportional to the reduced mass of the electron, $\mu = m_e M / (m_e + M)$, so the transition frequencies of two isotopes with nuclear masses $M_1 < M_2$ differ by:

$$\frac{\delta\nu}{\nu} \approx \frac{m_e}{M_1} - \frac{m_e}{M_2} = \frac{m_e}{m_u} \frac{A_2 - A_1}{A_1 A_2}, \quad (4)$$

where m_u is the atomic mass unit. This is the **normal mass shift**, and it makes the heavier isotope bluer. For deuterium and hydrogen the shift is 2.72×10^{-4} . The $\text{Ly}\alpha$ line of D therefore lies $0.33 \text{ \AA}$ (82 km s^{-1}) blueward of that of H, and the two are easily separated in quasar absorption systems (Section 4.1). However, the normal mass shift falls as A^{-2} . Already for lithium the shift is comparable to the Doppler width, and for heavier atoms it is orders of magnitude smaller, as the solid black curve in Fig. 4a shows.

Two further terms contribute to the isotope shift of a line, and we take them in turn. The nuclear recoil energy is $P^2/2M$, where $P = -\sum_i \mathbf{p}_i$ is the momentum of the nucleus and the $\mathbf{p}_i$ are the momenta of the electrons. If we expand the square, we recover the normal mass shift plus a cross term $(1/M) \sum_{i < j} \mathbf{p}_i \cdot \mathbf{p}_j$ that depends on the correlations between the electrons. This **specific mass shift** can exceed the normal mass shift by an order of magnitude and can have either sign. The specific mass shift is also difficult to calculate, since first- and second-order perturbation theory cannot guarantee even its sign (Berengut et al. 2003).

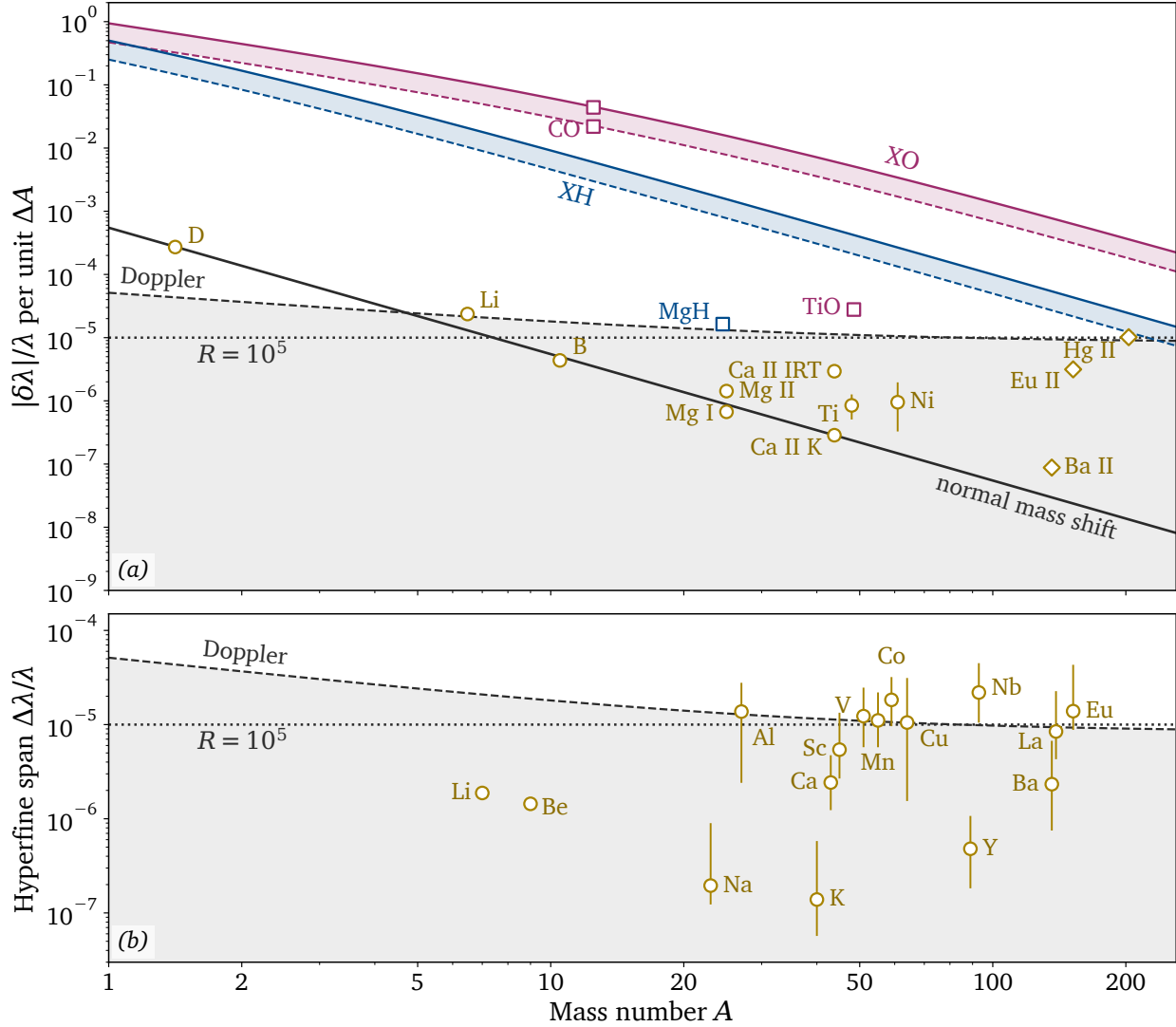


Figure 4: (a) Fractional isotope shift per unit mass number vs. mass number: measured values for atoms (circles: mass shift dominant; diamonds: field shift dominant) and molecules (squares), and bands for diatomic molecules XO and XH whose atom X has mass A , bounded by the rotational (solid) and vibrational (dashed) shifts. (b) Median hyperfine span of each element's lines at 300–1000 nm vs. the mass number of its odd- A isotopes, with bars spanning the 16th–84th percentiles of the lines in Kurucz's atomic line list. In both panels the dashed line is the Doppler width for 5000 K with 1.5 km s^{-1} microturbulence, the gray region lies below it, and the dotted line is the resolution element of an $R = 10^5$ spectrograph.

The Ca II lines show how large and how variable the specific mass shift can be. In the infrared triplet (a $3d-4p$ transition) the specific mass shift is ~ 10 times the normal mass shift and of opposite sign, so ^{48}Ca lies $0.200 \text{ \AA}$ redward of ^{40}Ca at $8498 \text{ \AA}$. In the K line (a $4s-4p$ transition), on the other hand, ^{48}Ca lies $0.009 \text{ \AA}$ blueward, as the normal mass shift predicts (Ryabchikova et al. 2008). Note that the isotope shift is a property of a transition, so one element can show shifts of opposite sign in different lines. For the Li I $6708 \text{ \AA}$ resonance line the ^6Li component lies $0.16 \text{ \AA}$ (7.15 km s^{-1}) redward of ^7Li , 1.8 times the normal mass shift and ≈ 1.1 Doppler widths (Cayrel et al. 2007).

3.1.2. Field shifts

The third term is the **field shift** (or volume shift). The nuclear charge is spread over a finite volume, so an electron that penetrates the nucleus sees a potential that depends on the nuclear radius, and isotopes with different radii have slightly different level energies. This shift is proportional to the change in the mean square nuclear charge radius, $\delta\langle r^2 \rangle$, and to the electron density at the nucleus, so the field shift is carried mainly by s electrons and grows steeply with Z . In Ba II, for example, the field-shift constant of the $6s$ level is 17–26 times larger in magnitude than those of the $6p$ levels (Berengut et al. 2003). Collecting the three terms, we can write the shift of a line between isotopes A and A' as:

$$\delta\nu^{A'A} = (k_{\text{NMS}} + k_{\text{SMS}}) \left(\frac{1}{A'} - \frac{1}{A} \right) + F \delta\langle r^2 \rangle^{A'A}, \quad (5)$$

where $k_{\text{NMS}} = -\nu m_e/m_u$ and F is the field-shift constant of the transition (Berengut et al. 2003). For heavy elements the field shift dominates. The isotopes of Hg II 3984 Å are spaced by 0.07–0.08 Å for every two mass units, 600–760 times the normal mass shift (White et al. 1976), and in Eu II 4129 Å ^{153}Eu lies 26 mÅ redward of ^{151}Eu , 130 times the normal mass shift (Lawler et al. 2001).

3.1.3. Measured atomic isotope ratios

In Fig. 4a we see that atomic isotope shifts fall below the Doppler width for almost every element. As a result, we have measured only a few atomic isotope ratios in stars. These are $^6\text{Li}/^7\text{Li}$, $^3\text{He}/^4\text{He}$ in a few chemically peculiar stars, the Hg and Pt isotopes (whose field shifts are large), and the Ca isotopes in the infrared triplet (whose specific mass shift is large, and whose isotopes span eight mass units). Hyperfine structure and field shifts add Ba, Eu, and Sm (Sections 3.2 and 4.3).

Because the limit is set by the ratio of the shift to the width of the line, the same atomic lines become usable where the gas is cold. Interstellar Li I components have Doppler parameters of only 0.7–2.1 km s^{−1}, so the 7 km s^{−1} isotope shift separates ^6Li from ^7Li . Toward σ Per, Knauth et al. (2000) found $^7\text{Li}/^6\text{Li} \approx 2$, compared with 12.3 in the Solar System, the signature of lithium freshly made by cosmic-ray spallation. Lambert et al. (1998) measured $^{11}\text{B}/^{10}\text{B} = 3.4 \pm 0.7$ from ultraviolet spectra of the interstellar B II 1362 Å line toward three stars, consistent with the solar-system value. They argue that a ratio above the ≈ 2.5 made by high-energy cosmic rays needs an additional source of ^{11}B , such as the neutrino process (Section 2.2) or low-energy cosmic rays.

However, even a shift far below the line width matters in searches for a variation of the fine-structure constant α in quasar absorption systems, which compare the positions of lines from different ions to better than 1 km s^{−1}. The Mg II 2796 Å line has ^{26}Mg 8.0 mÅ (0.86 km s^{−1}) blueward of ^{24}Mg (Murphy et al. 2003). A non-solar Mg isotope ratio in an absorber would shift the Mg lines and mimic a change in α (Murphy et al. 2003).

3.2. Hyperfine structure

A nucleus with spin $I \neq 0$ couples to the total angular momentum J of the electrons, and each level splits into components labeled by $F = |I - J|, \dots, I + J$. To first order the splitting follows the interval rule $E_F - E_{F-1} = \mathcal{A}F$, where $\mathcal{A}$ is the magnetic dipole constant of the level. Nuclei with an odd number of protons or neutrons have non-zero spin. Every line of an odd- Z element is therefore split (Sc, V, Mn, Co, Cu, Nb, La, Eu, etc.), while for an even- Z element only the lines of

its odd- A isotopes are split (e.g., ^{135}Ba and ^{137}Ba). In Fig. 4b we see that the typical hyperfine span of the optical lines of V, Mn, Co, Cu, Nb, and Eu is comparable to the Doppler width.

Hyperfine structure can distinguish the isotopes of an element, because each odd- A isotope has its own hyperfine pattern. In Ba II 4554 Å the even isotopes lie within 1.6 mÅ of each other, while the hyperfine components of the odd isotopes span 56 mÅ (Gallagher et al. 2010). The width and shape of the line measure the fraction of odd isotopes, and hence the ratio of r-process to s-process barium (Section 4.3). Because spreading a line over several components desaturates it, the odd isotopes also change the strength of the line. In our SYNTHE models for the metal-poor giant HD 122563, switching from pure s-process to pure r-process barium raises the equivalent width of the line from 108 to 114 mÅ, equivalent to 0.06 dex in the Ba abundance. In Eu II 4129 Å the hyperfine pattern of ^{151}Eu spans 180 mÅ and that of ^{153}Eu 78 mÅ, so the shape of the blended profile carries the isotope ratio (Snedden et al. 2002; Mucciarelli et al. 2008).

3.3. Molecular isotope shifts

The isotope shifts of molecular vibration and rotation have the same origin as the normal mass shift of an atom, a change in a reduced mass, but they are far larger. The difference lies in how far the isotopic nucleus moves. Consider two bodies of masses M and m held together by a force that does not depend on either mass. Their relative motion is that of a single body with the reduced mass $\mu = mM/(m + M)$, and if we change M by δM we find:

$$\frac{\delta\mu}{\mu} = \frac{m}{m + M} \frac{\delta M}{M}. \quad (6)$$

The factor $m/(m + M)$ is the fraction of the separation that the body of mass M covers as the pair moves about its center of mass. In an atom the partner is an electron, and this factor is $m_e/M \approx 5.5 \times 10^{-4}/A$. The nucleus barely moves (in hydrogen the proton circles the center of mass on an orbit 1836 times smaller than that of the electron), so its mass enters the electronic energy levels only as a small recoil correction, and Eq. 6 with $m = m_e$ reproduces Eq. 4.

In a molecule the electrons supply the restoring force, and the vibration and rotation are motions of the nuclei themselves. In the Born–Oppenheimer approximation the electrons adjust at once to the positions of the nuclei, so the potential in which the nuclei move is the same for every isotopologue, and the bond acts as a spring whose stiffness does not depend on the isotope. When the partner is another heavy atom the factor $m/(m + M)$ is of order unity, and the fractional shift is of order $\delta M/M$ itself, larger than the atomic shift by a factor of order M/m_e . A mass on a spring vibrates at a frequency $\omega \propto \mu^{-1/2}$, and a rigid rotor has a rotational constant $B \propto \mu^{-1}$. For a diatomic molecule made of an atom of mass A and a partner of mass m_p , a change ΔA in the mass of the first atom therefore shifts the vibrational and rotational frequencies by:

$$\left| \frac{\delta\omega}{\omega} \right| = \frac{1}{2} \frac{m_p \Delta A}{A(A + m_p)}, \quad \left| \frac{\delta B}{B} \right| = \frac{m_p \Delta A}{A(A + m_p)}. \quad (7)$$

For ^{12}CO and ^{13}CO ($m_p = 16$) the vibrational shift is 2.2‰ and the rotational shift 4.4‰ (exact values; the first-order Eq. 7 gives 2.4‰ and 4.8‰), three to four orders of magnitude larger than an atomic mass shift. In Fig. 4a the molecular bands lie above the Doppler width at almost every mass. This is the main reason that we measure the isotope ratios of C, N, O, Mg, and Ti from molecules.

Hydride shifts are smaller, because the light hydrogen atom does most of the moving. With $m = m_{\text{H}} \ll M$ the factor in Eq. 6 becomes m_{H}/M , the same form as in an atom with m_{H} in place of m_e . The XH band in Fig. 4a therefore falls as A^{-2} , like the normal mass shift, and at large A its rotational edge approaches $m_{\text{H}}/m_e \approx 1837$ times the atomic normal mass shift (the vibrational edge half that), a factor of ≈ 16 below the XO band. The normal-mass-shift curve and the XH and XO bands in Fig. 4a all follow from Eq. 6, with partner masses m_e , m_{H} , and m_{O} .

Consider next the lines of an electronic band, which are shifted only by the difference between the isotope effects in the upper and lower states. These shifts are smaller than those of pure vibration or rotation, and they can have either sign. In the A–X (0,0) band of MgH near 5140 Å, ^{25}MgH and ^{26}MgH lie ≈ 0.08 and $0.15\text{--}0.17$ Å redward of ^{24}MgH (McWilliam & Lambert 1988), approximately one Doppler width per mass unit, and appear as a red asymmetry of each ^{24}MgH line (Thygesen et al. 2016). In the $\gamma(0,0)$ band of TiO near 7080 Å each added neutron shifts a line by ≈ -0.2 Å, i.e., the heavier isotopes are bluer (Wyckoff & Wehinger 1972; Clegg et al. 1979). In the Q branch of the CH G band the shifts are too small to use, although ^{13}CH can be measured in the P and R branches (Snedden et al. 1986). C_2 , with a much larger change in reduced mass, is preferred where it is available (Cohen et al. 2006).

The first-overtone ($\Delta v = 2$) bands of CO in cool stars are the extreme case, with the (2,0) bandheads of ^{12}CO and ^{13}CO at 2.2935 and 2.3448 μm in vacuum (2.2929 and 2.3442 μm in air), 513 Å apart (Kleinmann & Hall 1986). At millimeter wavelengths the $J = 1\text{--}0$ lines of ^{12}CO , ^{13}CO , C^{18}O , and C^{17}O lie at 115.271, 110.201, 109.782, and 112.359 GHz.

For molecules the main difficulty is optical depth, because the lines of the most abundant isotopologue are usually saturated (e.g., the ^{12}CN lines near the ^{13}CN feature at 8004 Å; Gilroy 1989). A ratio of line strengths is then not a ratio of abundances. In the interstellar medium “in large measure, the value of any isotopic CO study hinges upon successful resolution of the optical depth problem” (Langer & Penzias 1990), and in stars we obtain the ratio by spectral synthesis of weak lines.

The saturation of the main isotopologue also makes the rare isotopologues useful as tools. In molecular clouds the ^{13}CO and C^{18}O lines are often optically thin when the ^{12}CO lines are not, so their intensities scale with column density. Dickman (1978) found $N(\text{H}_2) = (5.0 \pm 2.5) \times 10^5 N(^{13}\text{CO})$ in dark clouds with $1.5 \lesssim A_V \lesssim 10$, and Frerking et al. (1982) found $N(\text{C}^{18}\text{O}) = 1.7 \times 10^{14} (A_V - 1.3) \text{ cm}^{-2}$ in Taurus for $4 < A_V < 21$. In this use the isotope ratio is an input. Dickman’s lower limit on the fraction of the gas-phase carbon in CO, for example, is 12% for $^{12}\text{C}/^{13}\text{C} = 40$ and 27% for the terrestrial 89. Likewise, the ratio of an optically thick to an optically thin line gives the optical depth only if the abundance ratio is known and fractionation is negligible (Mangum & Shirley 2015). The fractionation and selective photodissociation of Section 6 then enter as systematic errors. Frerking et al. (1982) found $N(^{13}\text{CO})/N(\text{C}^{18}\text{O})$ of 10–60 in ρ Oph, where the terrestrial isotope ratios give 5.5.

In Fig. 5 we show synthetic spectra for four of the cases discussed above, which we computed with SYNTHE for the model atmospheres of real stars, with the contribution of each isotope or isotopologue drawn separately. In Fig. 5a, at the solar-system $^6\text{Li}/^7\text{Li}$ ratio, the ^6Li component depresses the red wing of the ^7Li line by at most 1.1% and moves the centroid of the feature by 0.5 km s^{-1} .

In Fig. 5b the hyperfine pattern of ^{151}Eu forms a broad, shallow trough, while ^{153}Eu forms a narrower

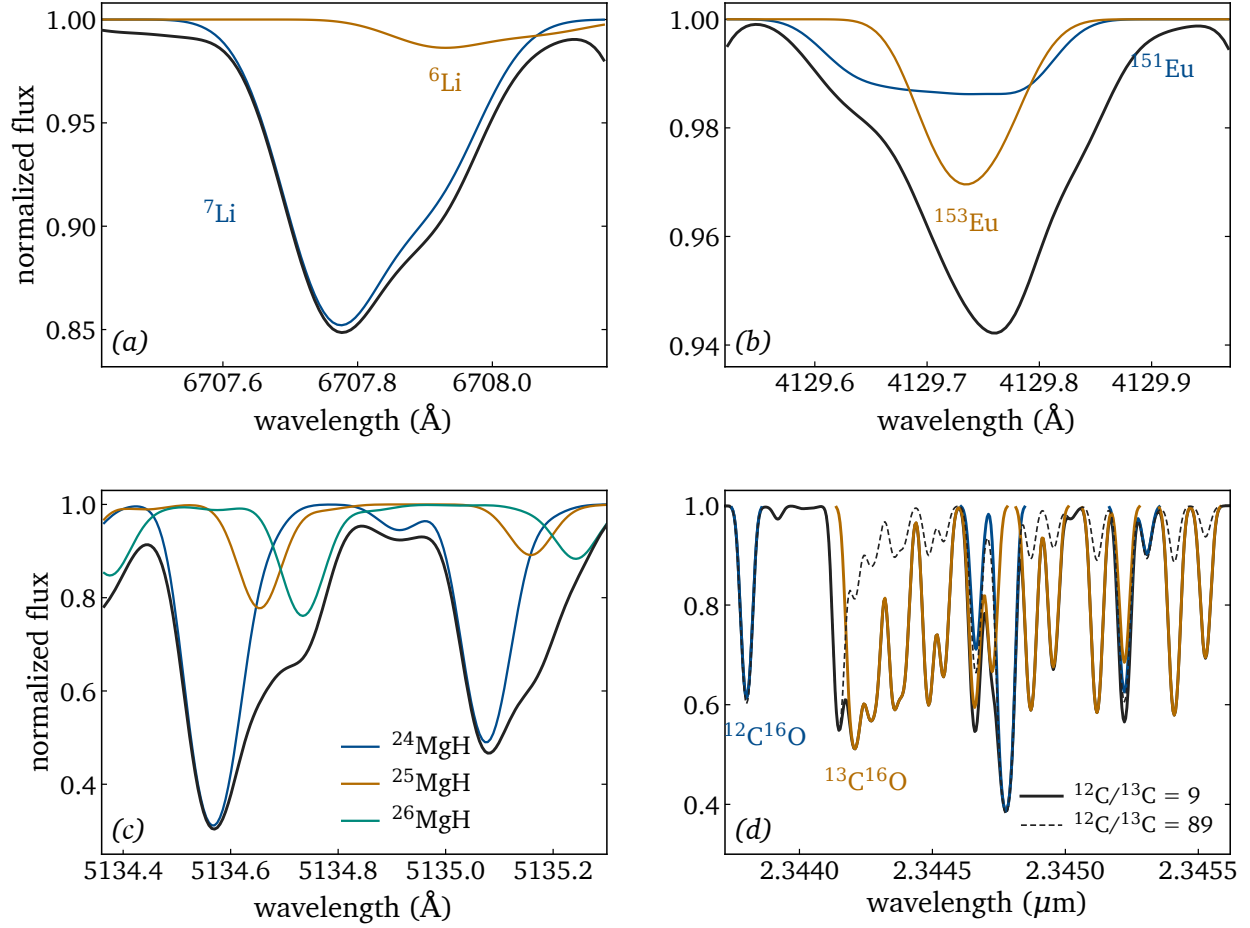


Figure 5: Synthetic spectra computed with SYNTHE at $R = 100\,000$ (air wavelengths): (a) Li I 6708 Å in the solar model with $A(\text{Li}) = 2.20$ and the solar-system ${}^6\text{Li}/{}^7\text{Li}$, (b) Eu II 4129.7 Å in the metal-poor giant HD 122563, (c) MgH lines near 5135 Å in Arcturus, and (d) the ${}^{13}\text{CO}$ (2,0) bandhead region in Arcturus. Black is the full synthesis, and each colored curve is a synthesis in which only the labeled isotope or isotopologue absorbs. In (b) the black profile includes a blended SiH line, and in (d) the line just blueward of the ${}^{13}\text{CO}$ head is not CO.

line. A blended SiH line supplies $\approx 40\%$ of the full profile and moves its minimum to the red. In Fig. 5c the ${}^{25}\text{MgH}$ and ${}^{26}\text{MgH}$ lines produce the red shoulders on which measurements of the Mg isotopes rest. Finally, Fig. 5d shows the ${}^{13}\text{CO}$ bandhead in Arcturus. At the measured ${}^{12}\text{C}/{}^{13}\text{C} = 9$ (Abia et al. 2012) the normalized flux at the bandhead is 0.51, compared with 0.81 at the solar ratio. The bandhead is a very sensitive measure of the carbon isotopes in cool giants.

3.4. Radio lines, γ -ray lines, and laboratory samples

The radio, γ -ray, and laboratory measurements avoid the problem of small shifts, because in each case the isotope determines which line or signal exists. Detectors in space that measure the charge and mass of each arriving cosmic-ray nucleus avoid it in the same way (Section 5.4).

3.4.1. Radio lines

The ground-state hyperfine (spin-flip) lines of hydrogen-like atoms depend on the magnetic moment of the nucleus, so each isotope has its own line: H I at 1420.406 MHz (21 cm), D I at 327.384 MHz (92 cm; Karshenboim 2000), and $^3\text{He}^+$ at 8665.65 MHz (3.46 cm; Balser & Bania 2018). These lines are never blended with the lines of the other isotopes, and the main difficulty is sensitivity. The Einstein A coefficient of the D I line is 61 times smaller than that of the 21 cm line and D/H is $\approx 2 \times 10^{-5}$. The line has been detected only with long integrations (Rogers et al. 2005).

The HD molecule shows that isotopic substitution can even create a line. H_2 has no dipole moment, but the asymmetry of HD gives it one (8.56×10^{-4} debye). The resulting rotational line at $112 \mu\text{m}$ has been used to weigh a protoplanetary disk (Bergin et al. 2013).

3.4.2. γ -ray lines

We can identify radioactive isotopes by the γ -ray lines emitted in their decays, whose energies are set by the nuclear levels of the daughter nucleus. The principal lines are at 1808.65 keV for ^{26}Al , 1173 and 1332 keV for ^{60}Fe (from its daughter ^{60}Co), 68 and 78 keV for ^{44}Ti (and 1157 keV from its daughter ^{44}Sc), 847 and 1238 keV for ^{56}Co , and 478 keV for ^7Be . The ^{60}Fe lines and the 1157 keV ^{44}Ti line follow the decay of a short-lived daughter (^{60}Co and ^{44}Sc). A daughter whose half-life is much shorter than that of its parent decays as fast as the parent supplies it, a steady state known as **secular equilibrium**, so the line of the daughter measures the decay rate of the parent. The identification is unambiguous, and the limitations are sensitivity and angular resolution. The germanium spectrometer SPI on *INTEGRAL* resolves 3 keV at 1.8 MeV ($E/\Delta E \approx 600$), enough to measure the Doppler shifts of the ^{26}Al line from Galactic rotation (Diehl et al. 2006). We return to these lines in Section 5.

3.4.3. Laboratory samples

Finally, we can analyze material that reaches the laboratory (meteorites, interplanetary dust, the solar wind captured by *Genesis*, and samples returned from comets and asteroids) with mass spectrometers, which separate the isotopes by mass directly. Ion microprobes (NanoSIMS) reach a spatial resolution of 100–150 nm (Nittler et al. 2018) and measure isotope ratios in single presolar grains of $1 \mu\text{m}$ to 1–2% (1σ ; Liu et al. 2021), and bulk measurements of rocks reach parts per million (Young et al. 2016). This precision exceeds anything available to a telescope by orders of magnitude, so much of what we know about isotopes comes from a few grams of meteorites.

Laboratory isotope ratios are quoted relative to a standard. A mass spectrometer itself fractionates isotopes, and measuring each sample against a standard cancels this bias. We express the result in the δ **notation**, as the fractional deviation from the standard, $\delta^{18}\text{O} = R_{\text{sample}}/R_{\text{std}} - 1$, where $R = ^{18}\text{O}/^{16}\text{O}$. The deviations are small, and we quote them in units of 10^{-3} . The standard for H and O is Vienna Standard Mean Ocean Water (VSMOW, with $\text{D}/\text{H} = 155.76 \times 10^{-6}$; Hagemann et al. 1970). For an element with three isotopes, such as oxygen, we plot $\delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ in a **three-isotope plot**. Any mass-dependent process moves a sample along a line of slope ≈ 0.52 , the **terrestrial fractionation line** (TFL; we derive its slope in Section 6.1). The departure from this line, $\Delta^{17}\text{O} = \delta^{17}\text{O} - 0.52 \delta^{18}\text{O}$, is therefore unchanged by such processes (Clayton 1993).

4. Isotopes as fingerprints of nucleosynthesis

In this section we treat isotope ratios as records of the nuclear processes that made them. We proceed roughly in order of increasing complexity. We begin with the Big Bang, a single event governed by one free parameter, and then turn to the interiors of individual stars, the neutron-capture elements, and the chemical evolution of galaxies. We end with single grains of stardust, in which we can measure the isotopes of one star in the laboratory.

4.1. Primordial isotopes

We begin with Big Bang nucleosynthesis (BBN), the cleanest application of isotope ratios in astronomy. The ratios D/H, $^3\text{He}/^4\text{He}$, and $^6\text{Li}/^7\text{Li}$ were set in the first few minutes by a process whose physics is known and whose only free parameter, in the Standard Model, is the baryon-to-photon ratio $\eta = n_b/n_\gamma$.

4.1.1. Predicted abundances

We can understand the outcome in terms of two timescales (Pitrou et al. 2018). At $kT_F \approx 0.8$ MeV the weak interactions that convert neutrons and protons into each other freeze out, leaving $n/p = e^{-\Delta/kT_F} \approx 0.2$, where $\Delta = 1.293$ MeV is the neutron–proton mass difference. The neutrons then decay (with a lifetime of 878 s) until they can be locked into nuclei. The first step, $n + p \rightarrow \text{D} + \gamma$, is reversed by photodissociation as long as there are enough photons above the deuteron binding energy $B_D = 2.22$ MeV. Since there are $\sim 10^9$ photons per baryon, this remains true until the temperature has fallen far below B_D . A Saha-type balance gives the temperature at which deuterium survives:

$$kT_{\text{nuc}} \approx \frac{B_D}{\ln \eta^{-1} + \frac{3}{2} \ln(m_N/kT_{\text{nuc}})} \approx 0.06 \text{ MeV}, \quad (8)$$

where m_N is the nucleon mass (the numerical prefactor of the Saha equation, omitted here, raises the estimate to 0.066 MeV, and the full calculation gives 0.078 MeV, at $t \approx 200$ s; Pitrou et al. 2018). Once the “deuterium bottleneck” opens, essentially all of the remaining neutrons end up in ^4He . By then decay (and the tail of the weak interactions) has lowered n/p to $\approx 1/7$, so the helium mass fraction is $Y_p \approx 2(n/p)/(1 + n/p) \approx 0.25$. A small fraction of the deuterium and ^3He is left unburned, along with traces of ^7Li and ^7Be .

This picture explains why the different isotopes measure different quantities. The helium abundance depends on the freeze-out temperature and the neutron lifetime, and only logarithmically on η . Near the observed baryon density helium is almost insensitive to η , with $Y_p \propto \eta^{0.039} \tau_n^{0.73}$ (Fields et al. 2020). Freeze-out occurs when the weak rate, which scales as T^5 , falls below the expansion rate, which scales as T^2 , so in a faster-expanding Universe the weak interactions freeze out earlier and leave more neutrons. Helium is a probe of the expansion rate of the Universe (e.g., the number of neutrino species).

Deuterium, in contrast, is an intermediate product that is destroyed by D+D and D+p reactions at a rate proportional to the baryon density. As a result $\text{D}/\text{H} \propto \eta^{-1.6}$, and a 1% measurement of D/H determines η to 0.6%. This steep dependence makes D/H the “baryometer” of cosmology. Finally, ^7Li rises with η ($\propto \eta^{2.1}$) because at high baryon density mass 7 is made mostly as ^7Be (95% at the observed η), which later captures an electron to become ^7Li .

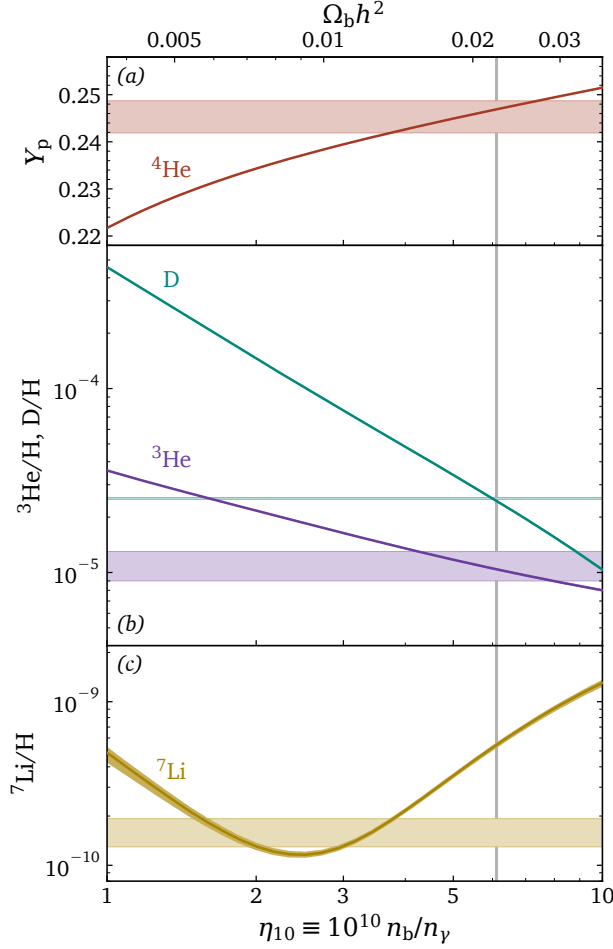


Figure 6: Big Bang nucleosynthesis yields vs. the baryon-to-photon ratio $\eta_{10} = 10^{10}\eta$ (with $\Omega_b h^2$ on the top axis), computed with PRyMordial and the PRIMAT rates; tinted bands show the 68% range from nuclear-rate and neutron-lifetime uncertainties. Light boxes are the observed values (Y_p from Aver et al. 2021; the ^3He box is an upper limit; the ^7Li box is the Spite plateau, Sbordone et al. 2010), and the gray band is the *Planck* baryon density.

In Fig. 6 we show the predicted abundances as a function of η , which we computed for this memo with the PRyMordial code (Burns et al. 2024) and the PRIMAT reaction rates (Pitrou et al. 2021). The *Planck* measurement of the baryon density from the cosmic microwave background (CMB), $\Omega_b h^2 = 0.02237 \pm 0.00015$ (Planck Collaboration et al. 2020), corresponds to $\eta = (6.124 \pm 0.041) \times 10^{-10}$ and fixes the predictions: $Y_p = 0.2469$, $D/H = (2.446 \pm 0.037) \times 10^{-5}$, $^3\text{He}/H = 1.04 \times 10^{-5}$, and $^7\text{Li}/H = 5.4 \times 10^{-10}$.

4.1.2. Deuterium

Deuterium is destroyed in stars and is not made anywhere else in significant amounts, so we can take the most primitive gas to have the primordial value. Cooke et al. (2018) measured D/H in seven quasar absorption systems at $z = 2.5\text{--}3.1$ with oxygen abundances between 1/45 and 1/630 of solar. They used the D I Lyman lines, which lie 82 km s^{-1} blueward of those of H (Section 3.1), and found $(D/H)_p = (2.527 \pm 0.030) \times 10^{-5}$. The mean of eleven such systems, $(2.55 \pm 0.03) \times 10^{-5}$, gives $\eta_{10} = 6.15 \pm 0.19$ with the BBN rates of Fields et al. (2020). The CMB by itself, with Y_p left free, gives 6.104 ± 0.058 (Fields et al. 2020). This agreement is one of the great successes of modern cosmology, since the comparison involves nuclear physics in the first few minutes on one side and the acoustic oscillations of the plasma 380 000 years later on the other.

However, a small tension remains at the percent level. With the PRIMAT rates the predicted D/H is 3.3% below the observed value, an offset smaller than the width of the deuterium curve in Fig. 6. An updated mean of all systems, $(2.533 \pm 0.024) \times 10^{-5}$, is in $\approx 2\sigma$ tension with the prediction (Kislitsyn et al. 2024). The LUNA underground measurement of $d(p,\gamma)^3\text{He}$ removed what had been the most uncertain nuclear input (Mossa et al. 2020). The remaining differences come from the $d(d,n)^3\text{He}$ and $d(d,p)t$ cross sections, and other evaluations of these give agreement to within 1% (Yeh et al. 2022; Burns et al. 2024). The limiting uncertainty in cosmology’s most precise isotope ratio therefore lies in laboratory measurements of nuclear cross sections.

4.1.3. Helium

The primordial helium mass fraction Y_p comes from the recombination lines of metal-poor H II regions. Helium is essentially all ^4He , so Y_p is an elemental abundance, and we show the value of Aver et al. (2021) in Fig. 6 for reference only.

We measure the rarer isotope ^3He through the 8.7 GHz hyperfine line of $^3\text{He}^+$ in Galactic H II regions, and Bania et al. (2002) derived an upper limit to the primordial abundance of $(1.1 \pm 0.2) \times 10^{-5}$. These measurements also sharpened the “ ^3He problem”, first raised by Rood et al. (1976). Standard stellar models predict that low-mass stars are net producers of ^3He , which would have raised its abundance by a large factor over the history of the Galaxy, but H II regions show little enrichment (Balser & Bania 2018). We will see in Section 4.2 that the solution involves the stellar mixing that lowers $^{12}\text{C}/^{13}\text{C}$ in red giants. A new optical method measures $^3\text{He}/^4\text{He}$ directly from the isotope shift of metastable He I absorption lines, and gives $(1.77 \pm 0.13) \times 10^{-4}$ toward the Orion Nebula (Cooke et al. 2022).

4.1.4. Lithium

The metal-poor dwarfs of the Galactic halo show a constant lithium abundance, the Spite plateau (Spite & Spite 1982), a factor of 3.4 below the BBN prediction (Sbordone et al. 2010). This “lithium problem” remains unexplained (Fields 2011). Because it concerns the elemental abundance, we do not pursue it here.

The lithium isotopes offer a cautionary tale for the measurement of isotope ratios in stellar spectra. Standard BBN predicts $^6\text{Li}/\text{H} \approx 10^{-14}$, far too little to detect. Nevertheless, Asplund et al. (2006) reported ^6Li in nine of 24 halo stars and suggested a ^6Li plateau ~ 500 times above the BBN value. The ^6Li component lies only 0.16 Å (7.15 km s^{-1}) redward of each ^7Li component (Section 3.1), comparable to the thermal width of the line, so ^6Li appears as a slight excess of absorption in the red wing, as we saw in Fig. 5a.

Cayrel et al. (2007) then showed that the convective motions in the stellar atmosphere, which produce asymmetric line profiles, generate a red-wing excess of the same size. With 3D hydrodynamic model atmospheres and non-LTE line formation, Lind et al. (2013) found no significant detection in any of their four stars. In HD 84937, $^6\text{Li}/^7\text{Li}$ fell from 0.060 in 3D LTE to $0.017 \pm 0.007 \pm 0.006$ in 3D non-LTE. This history closely parallels that of the barium isotopes in HD 140283 (Section 4.3). Both episodes show that an isotope ratio derived from a subtle line asymmetry is only as good as the model of the line profile.

4.1.5. Astration

Recall that D/H decreases as gas is cycled through stars, which destroy deuterium. The protosolar D/H of 2.0×10^{-5} (Asplund et al. 2009; Lodders 2021) implies that $\approx 20\%$ of the primordial deuterium had been destroyed by the time the Sun formed. In the local interstellar medium the gas-phase D/H varies from place to place, and Linsky et al. (2006) argue that the variations are caused by the depletion of deuterium onto dust grains, a form of chemical fractionation. The total (gas plus dust) value of $\geq 2.3 \times 10^{-5}$ is similar to the protosolar value, which suggests that most of the destruction of deuterium in the local disk occurred before the Sun formed (Linsky et al. 2006).

4.2. Stellar interiors and mixing

The surface composition of a star records how deeply its envelope has mixed with material processed by nuclear burning. The carbon isotope ratio is the most sensitive gauge of this mixing, because we can compare it with three well-separated reference values: the initial ratio (89 for solar-system material), the CN-cycle equilibrium ratio of 3.6 (Eq. 1), and the ratio after the first dredge-up.

4.2.1. Carbon isotopes and extra mixing

When a star leaves the main sequence its convective envelope deepens into layers where partial CN burning has converted some ^{12}C into ^{13}C and ^{14}N . Standard models predict $^{12}\text{C}/^{13}\text{C} = 18\text{--}26$ after this first dredge-up for stars of $1\text{--}7 M_{\odot}$ (Charbonnel 1994), and ≈ 40 for a metal-poor star of $0.8 M_{\odot}$ (Charbonnel 1995).

Observations agree with these predictions on the lower giant branch and fall below them higher up. In Fig. 7 we show metal-poor field stars from several studies (Gratton et al. 2000; Keller et al. 2001; Spite et al. 2006). The bump on the red giant branch (RGB) marks the point where the hydrogen-burning shell reaches the composition discontinuity left by the first dredge-up. Below $\log L/L_{\odot} \approx 1.8$ the moderately metal-poor stars in Fig. 7a have only lower limits (mostly $> 30\text{--}40$), and the two stars at $\log L/L_{\odot} \approx 1.8$ have ratios of 25 and 30, consistent with the first dredge-up. All 22 of these stars brighter than $\log L/L_{\odot} = 2$ have ratios of 3–10. The extremely metal-poor giants in Fig. 7b show the same drop at $\log L/L_{\odot} \approx 2.6$, from 14–30 to 4–7 in all but two stars. Their ratios below the bump are close to the 33 measured in the unevolved subgiant HD 140283 (Spite et al. 2021), so their initial ratio was already far below solar.

Open clusters show a similar drop (e.g., from 21–25 on the lower giant branch of M67 to 11–15 in its red clump; Gilroy & Brown 1991), and 96% of low-mass evolved stars have ratios below the first-dredge-up prediction (Charbonnel & Do Nascimento 1998). We conclude that a second mixing episode, absent from standard models, brings CN-processed material to the surface once a star passes the bump. Oxygen is not depleted in these stars, so the mixing reaches only regions of partial CN burning (Gratton et al. 2000).

The leading explanation involves ^3He , which low-mass stars make on the main sequence and which the first dredge-up spreads through the envelope. Above the hydrogen-burning shell the reaction $^3\text{He}(^3\text{He},2p)^4\text{He}$ turns two nuclei into three, which lowers the mean molecular weight of the gas by $\Delta\mu/\mu \sim 10^{-4}$ (Eggleton et al. 2006). Gas with lower μ lying below gas with higher μ is unstable to thermohaline (or “fingering”) convection. This is the same instability that produces “salt fingers” where warm, salty water lies above cold, fresh water in the ocean. The instability sets in only at the

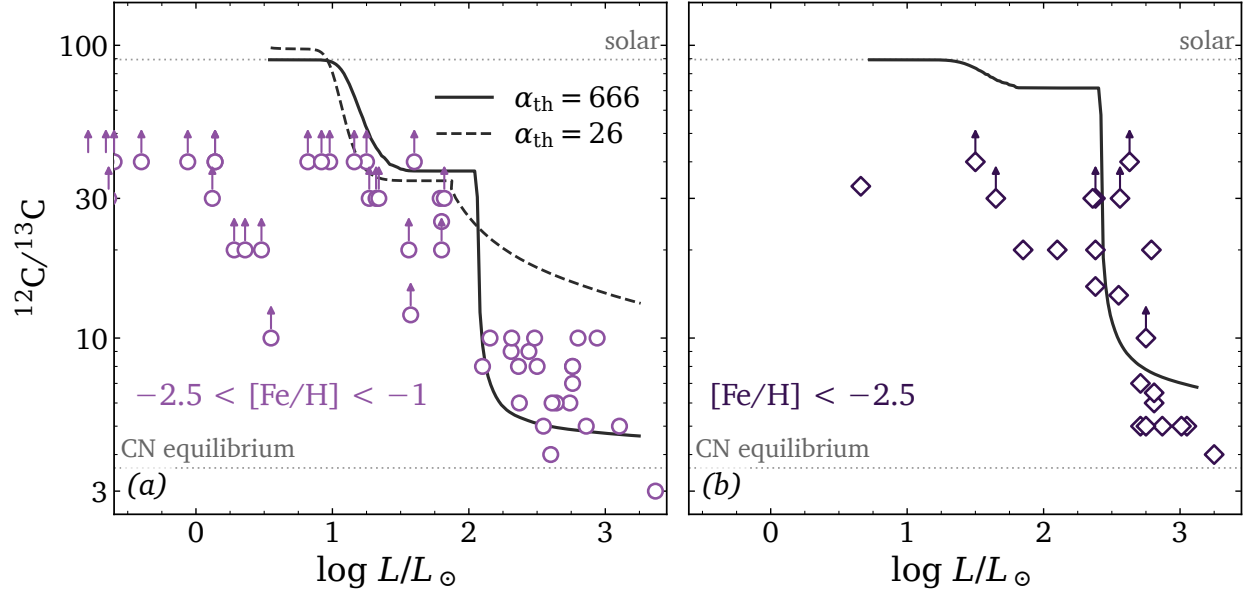


Figure 7: Carbon isotope ratio vs. luminosity for metal-poor field stars with (a) $-2.5 < [\text{Fe}/\text{H}] < -1$ (Gratton et al. 2000; Keller et al. 2001; Spite et al. 2006) and (b) $[\text{Fe}/\text{H}] < -2.5$ (Spite et al. 2006, 2021); arrows mark lower limits, and red horizontal-branch and AGB stars are omitted. Luminosities follow from each study’s T_{eff} and $\log g$ for $0.85 M_{\odot}$. The black curves are $0.8 M_{\odot}$ MIST tracks from the end of the main sequence to the tip of the RGB, for $[\text{Fe}/\text{H}] = -1.5$ in (a) and -3.0 in (b), with thermohaline mixing of efficiency $\alpha_{\text{th}} = 666$ (solid; MIST v1.2, Choi et al. 2016) or 26 (dashed; MIST 2.2 development models). Dotted gray lines mark the solar and CN-equilibrium (Skowronski et al. 2023) ratios.

bump, because before that point the gradient in μ left behind by the first dredge-up stabilizes the region.

Charbonnel & Zahn (2007) showed that the resulting mixing reproduces the observed $^{12}\text{C}/^{13}\text{C}$. They wrote the diffusion coefficient in terms of the aspect ratio α (length/width) of the fingers, with an efficiency $\alpha_{\text{th}} = (8/3)\pi^2\alpha^2$, and adopted $\alpha_{\text{th}} = 1000$, i.e., $\alpha \approx 6$. The MIST models in Fig. 7 use $\alpha_{\text{th}} = 666$, i.e., $\alpha \approx 5$ (Choi et al. 2016). At $[\text{Fe}/\text{H}] = -1.5$ their ratio stays at the first-dredge-up value of 37 up to the bump, falls to 6 by $\log L/L_{\odot} = 2.2$, and reaches 4.6 at the tip of the RGB, in agreement with the observed stars. With fingers as long as they are wide ($\alpha_{\text{th}} = 26$), the ratio declines only slowly after the bump and reaches 13 at the tip.

This mechanism also resolves the ^3He problem for most stars, since this mixing destroys most of the ^3He that the star made, although low-mass stars remain small net producers (Charbonnel & Lagarde 2010; Balser & Bania 2018). However, hydrodynamic simulations produce fingers with length-to-diameter ratios $\lesssim 0.5$ (Denissenkov 2010; Traxler et al. 2011), while the observations require $\alpha \approx 5\text{--}7$ (Charbonnel & Zahn 2007; Denissenkov 2010). The efficiency of the mixing remains unexplained.

4.2.2. Oxygen isotopes

The oxygen isotopes of giants record the stellar mass, because partial hydrogen burning below the envelope makes ^{17}O and destroys ^{18}O . The more massive the star, the hotter this region and the

more ^{17}O the first dredge-up brings to the surface. The predicted $^{16}\text{O}/^{17}\text{O}$ (by number) falls from ≈ 2600 initially to 170–260 at $2\text{--}2.5 M_{\odot}$, and then rises again at higher mass because ^{17}O is itself destroyed at higher temperatures (El Eid 1994; Lebzelter et al. 2015). Harris & Lambert (1984) measured $^{16}\text{O}/^{17}\text{O}$ between 160 and 1000 in seven bright giants from the $5 \mu\text{m}$ CO lines. We will see in Section 4.5 that the first dredge-up also sets the $^{17}\text{O}/^{16}\text{O}$ ratios of presolar oxide grains.

Hot-bottom burning in a $6 M_{\odot}$ AGB star drives $^{16}\text{O}/^{18}\text{O}$ above 10^6 and $^{12}\text{C}/^{13}\text{C}$ to ≈ 3 (Karakas & Lattanzio 2014). In the opposite direction, the R Coronae Borealis stars and hydrogen-deficient carbon stars have $^{16}\text{O}/^{18}\text{O}$ between 0.2 and 12, i.e., ^{18}O enhanced by factors of 40–2500 (Clayton et al. 2007). These stars are thought to form in the merger of a carbon–oxygen and a helium white dwarf, in which partial helium burning converts ^{14}N into ^{18}O .

4.3. Neutron-capture and magnesium isotopes in stars

Beyond the CNO elements, we have isotope ratios from stellar spectra for only a handful of elements, and each measurement rests on a different part of the spectroscopic physics of Section 3. We read magnesium isotopes from molecular lines of MgH, barium isotopes from the hyperfine broadening of a single resonance line, and europium isotopes from the combined hyperfine structure and field shift of Eu II.

4.3.1. Magnesium

The protosolar magnesium is 79.0% ^{24}Mg , 10.0% ^{25}Mg , and 11.0% ^{26}Mg (Asplund et al. 2021). ^{24}Mg is a primary product of carbon burning, while ^{25}Mg and ^{26}Mg are made in massive stars by $^{22}\text{Ne}(\alpha, n)^{25}\text{Mg}$ and $^{22}\text{Ne}(\alpha, \gamma)^{26}\text{Mg}$. The ^{22}Ne comes from the star’s initial CNO nuclei, so the heavy isotopes are secondary and their yields grow with metallicity. Intermediate-mass AGB stars add more of them after a delay of 0.1–0.3 Gyr (Meléndez & Cohen 2007).

We measure the heavy isotopes from the A–X band of MgH near 5140 Å. As we saw in Fig. 5c, the ^{25}MgH and ^{26}MgH lines are never resolved from the ^{24}MgH lines but add a red asymmetry to each one. We obtain the ratios by fitting the line profiles at $R \approx 6 \times 10^4\text{--}10^5$ (McWilliam & Lambert 1988; Yong et al. 2003b).

Meléndez & Cohen (2007) found that halo dwarfs with $-2.6 \leq [\text{Fe}/\text{H}] \leq -1.3$ have $^{25,26}\text{Mg}/\text{Mg}$ below 5% (for G 69-18 at $[\text{Fe}/\text{H}] = -2.6$, ^{25}Mg and ^{26}Mg are $0.9 \pm 2.0\%$ and $1.5 \pm 2.2\%$ of the total). The AGB contribution appears only above $[\text{Fe}/\text{H}] \approx -1.5$. Since the relevant stars ($3\text{--}6 M_{\odot}$) live 0.1–0.3 Gyr, the halo must have reached that metallicity within $\sim 0.1\text{--}0.3$ Gyr, if the AGB yields are right and the halo evolved as a single system (Meléndez & Cohen 2007). In this way the magnesium isotopes act as a clock for the early halo.

The magnesium isotopes vary much more widely within globular clusters. In the giants of NGC 6752, $^{24}\text{Mg}:^{25}\text{Mg}:^{26}\text{Mg}$ ranges from 83:10:7 to 53:9:39, and the ^{26}Mg abundance correlates with that of Al (Yong et al. 2003a). The accompanying Na and Al variations are also present at the main-sequence turnoff (Gratton et al. 2001). The stars must therefore have inherited them from an earlier generation of polluting stars. In M71 the first stellar population has $^{26}\text{Mg}/\text{Mg} \approx 4\%$ and the second $\approx 8\%$ (Meléndez & Cohen 2009).

Destroying ^{24}Mg by proton capture while preserving $^{25,26}\text{Mg}$ requires hydrogen burning at ~ 70 MK, hotter than the interior of any first-ascent red giant. Hot-bottom burning in intermediate-mass

AGB stars reaches such temperatures and was the favored polluter (Yong et al. 2003a). However, no proposed polluter reproduces all of the abundance patterns of globular clusters, and the origin of their multiple populations remains unsolved (Bastian & Lardo 2018).

4.3.2. Barium

Of the five abundant isotopes of barium, ^{134}Ba and ^{136}Ba are s-only (Section 2.4), and the odd isotopes ^{135}Ba and ^{137}Ba have hyperfine structure that the even isotopes lack. The odd fraction $f_{\text{odd}} = [N(^{135}\text{Ba}) + N(^{137}\text{Ba})]/N(\text{Ba})$ is a direct measure of the r-process share: $f_{\text{odd}} = 0.11$ for pure s-process barium and 0.46 for pure r-process barium, compared with the solar value of 0.178 (Arlandini et al. 1999). The s-process value is low because of the steady flow of Eq. 2. The magic nucleus ^{138}Ba ($N = 82$) has a very small capture cross section, so the flow piles up there (76% of s-process barium is ^{138}Ba in the model of Arlandini et al. 1999), while the odd isotopes, which capture neutrons more readily, are depleted. In practice we measure f_{odd} from the width of the Ba II 4554 Å resonance line, which the hyperfine components of the odd isotopes broaden. The measurement is therefore degenerate with every other source of line broadening. Lambert & Allende Prieto (2002) showed that for the metal-poor subgiant HD 140283 the inferred f_{odd} falls by 0.17 for every 0.33 km s^{-1} added to the assumed Gaussian broadening (macroturbulence, rotation, and the instrument).

The history of this one star illustrates the problem. Magain (1995) found $f_{\text{odd}} = 0.08 \pm 0.06$, consistent with pure s-process barium, and a 1D LTE analysis gave 0.02 ± 0.06 (Gallagher et al. 2010). A 3D hydrodynamic model atmosphere, which predicts the velocity field from first principles, then gave $f_{\text{odd}} = 0.38 \pm 0.02 \pm 0.06$ in LTE, i.e., 77% r process (Gallagher et al. 2015). For twenty years a systematic error in the line broadening had been mistaken for an s-process signature. The strongly r-process-enhanced star CS 31082-001 has $f_{\text{odd}} = 0.43 \pm 0.09$, consistent with pure r process (Meng et al. 2016).

4.3.3. Europium

The ground configuration of Eu II has an unpaired 6s electron, which produces large hyperfine splitting and a large field shift between ^{151}Eu and ^{153}Eu (Section 3). Europium isotopes are relatively easy to measure, and Sneden et al. (2002) found $f_{151} = 0.5 \pm 0.1$ in three r-process-rich halo stars, in agreement with the solar value of 0.478. Unfortunately the two processes make similar isotopic mixtures ($f_{151} = 0.54$ for the s process and 0.47 for the r process; Arlandini et al. 1999), so europium is a weak discriminant between them. The s-rich carbon-enhanced stars do show slightly higher values, 0.55–0.60 (Aoki et al. 2003), and the samarium isotopes separate the two processes more cleanly (Roederer et al. 2008).

4.4. Isotope ratios across the Galaxy and beyond

In molecular clouds we measure isotope ratios from the rotational lines of isotopologues at millimeter wavelengths. The main isotopologue is usually optically thick, so we take the ratio either between two rare isotopologues (e.g., $^{13}\text{C}^{34}\text{S}$ and $^{12}\text{C}^{34}\text{S}$) or with a “double-isotope” method, in which we multiply a ratio of rare species by an adopted $^{12}\text{C}/^{13}\text{C}$.

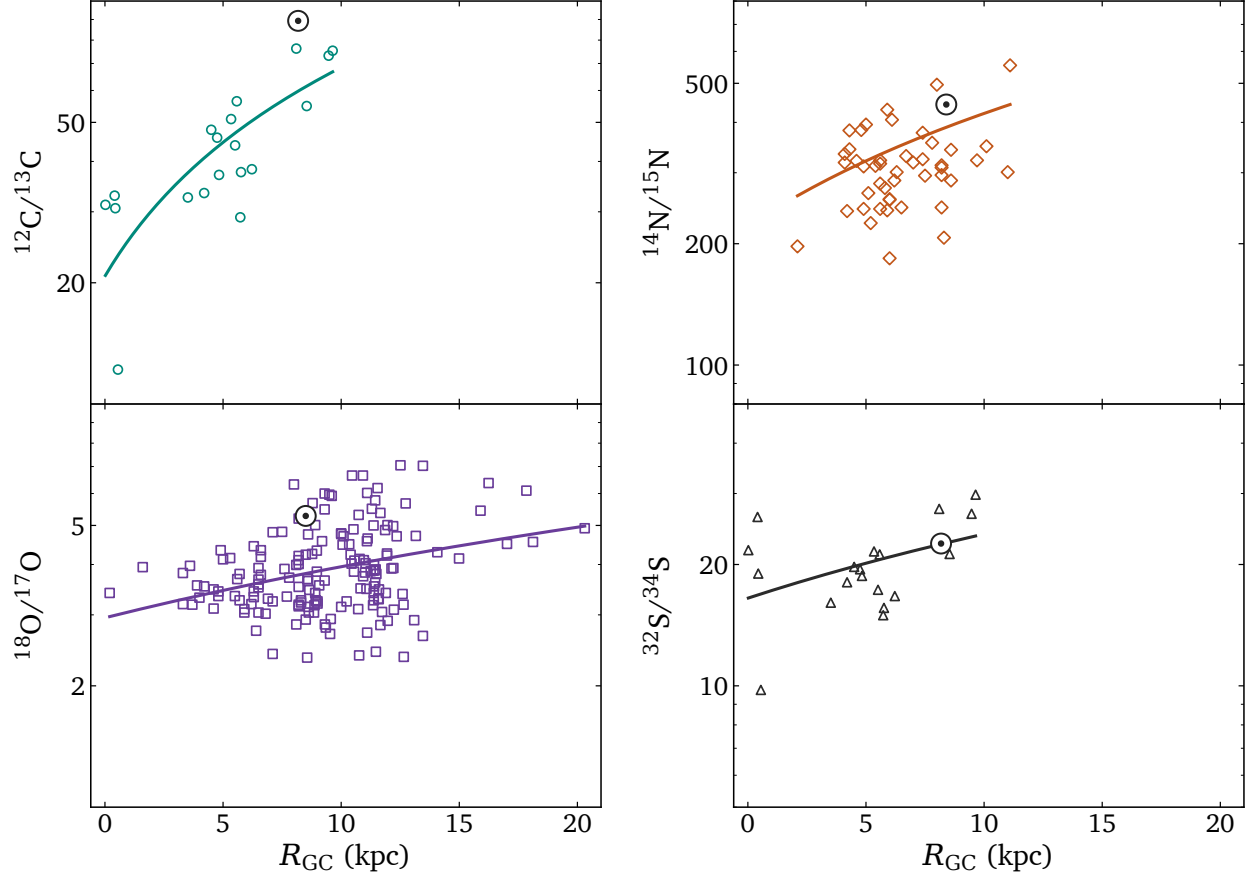


Figure 8: Isotope ratios vs. galactocentric radius in molecular clouds: $^{12}\text{C}/^{13}\text{C}$ and $^{32}\text{S}/^{34}\text{S}$ from $^{13}\text{C}^{34}\text{S}$ -based measurements (Yan et al. 2023), $^{14}\text{N}/^{15}\text{N}$ from HNC (Colzi et al. 2018), and $^{18}\text{O}/^{17}\text{O}$ from $\text{C}^{18}\text{O}/\text{C}^{17}\text{O}$ (Zhang et al. 2020), with the published linear fits and the solar-system value (circled dot; protosolar for N). Each panel spans one decade.

4.4.1. Galactic gradients

In Fig. 8 we show the four best-measured ratios as a function of galactocentric radius, and all four rise outward. The ratio $^{12}\text{C}/^{13}\text{C}$ rises from 27 ± 3 in the Central Molecular Zone to ≈ 60 at the solar circle, and in the disk it follows $(4.77 \pm 0.81)R + 20.76$ with R in kpc (Yan et al. 2023). Local clouds give ≈ 60 – 70 , and earlier CN, CO, and H_2CO measurements give a similar slope (Milam et al. 2005; Yan et al. 2023).

We can understand the sign of these gradients, and roughly their size, with the primary/secondary argument of Section 2.3. If each annulus of the disk evolves as a closed box with the same yields, the isotope ratio in each annulus depends only on its metallicity. Recall that Eq. 3 then gives a primary/secondary ratio proportional to $1/Z$, so the logarithmic gradient of that ratio should be the metallicity gradient with the sign reversed:

$$\frac{d \log(\text{primary/secondary})}{dR} = -\frac{d \log Z}{dR} \approx +0.042 \text{ dex kpc}^{-1}, \quad (9)$$

where we have used the O/H gradient of H II regions as a proxy for Z (Romano 2022). The panels of

Fig. 8 each span one decade, so equal slopes on the page are equal logarithmic gradients. At the solar circle the measured gradients are 0.035–0.040 dex kpc^{−1} for ¹²C/¹³C, 0.022–0.032 for ¹⁴N/¹⁵N, 0.014 for ³²S/³⁴S, and 0.011 for ¹⁸O/¹⁷O. (Wilson & Rood 1994 found 0.048 for ¹⁶O/¹⁸O, but this is a double-isotope ratio whose gradient is almost entirely the ¹²C/¹³C gradient used to derive it.)

The ratio that compares a primary isotope with a mostly secondary one, ¹²C/¹³C, has a gradient of the same order as the closed-box expectation. Nitrogen and sulfur have flatter gradients, because neither isotope of nitrogen is purely primary and ³⁴S is only weakly secondary. The ratio of two secondary isotopes, ¹⁸O/¹⁷O, is the flattest of all.

This ordering is the robust prediction of the argument, and several effects make the magnitude uncertain. Since ¹³C is made from carbon seeds, the C/H gradient is arguably the better proxy, and in the same compilation it is steeper (≈ -0.07 dex kpc^{−1}) and would predict a steeper ¹²C/¹³C gradient. The infall of pristine gas steepens the relation between an isotope ratio and *Z*, while radial gas flows flatten gradients, and the delayed release of ¹³C and ¹⁵N by low-mass stars and novae changes the prediction again. The choice of proxy and the infall of pristine gas can each change the predicted gradient by a factor of up to ~ 2 .

Note that the double-isotope method propagates any error in ¹²C/¹³C into the other ratios. For example, the measured slope of ³²S/³⁴S fell from 3.3 kpc^{−1} (Chin et al. 1996) to 1.56 kpc^{−1} (Yu et al. 2020), and then to 0.73 kpc^{−1} when Yan et al. (2023) measured the sulfur ratio directly (with their new ¹²C/¹³C gradient, the double-isotope method gives 0.75 kpc^{−1}).

We can now read this enrichment gauge in unevolved stars. In 32 nearby M dwarfs with metallicities between -0.4 and $+0.4$ dex, both ¹²C/¹³C and ¹⁶O/¹⁸O decrease with metallicity, as chemical evolution models that include ¹³C from novae predict (González Picos et al. 2025).

4.4.2. Chemical evolution of the solar neighborhood

The solar isotope ratios sample the local interstellar medium of 4.6 Gyr ago, and we can compare them with those of the present-day interstellar gas. The solar-system ¹²C/¹³C of 89 is higher than the local interstellar value of ≈ 60 –70, as expected if ¹³C has continued to accumulate over the past 4.6 Gyr. If the Sun formed farther in, as its metallicity suggests, the implied decrease grows from $\approx 24\%$ to $\approx 30\%$ (for a birth radius of 6.6 kpc; Milam et al. 2005).

The oxygen ratios fit less well, since the solar-system ¹⁸O/¹⁷O of 5.3 lies far above the local interstellar value of ≈ 4 . One proposed explanation is that the Sun’s parent cloud was contaminated by ¹⁸O from nearby massive stars (Wouterloot et al. 2008). However, the presolar oxide grains, which sample the interstellar medium before the Sun formed, suggest that the solar ratio lay within the interstellar range at that time (Nittler 2009). We return to the oxygen isotopes of the Solar System in Section 6.3.

Chemical evolution models reproduce the carbon and nitrogen isotope ratios only with additional sources of ¹³C and ¹⁵N. Without classical novae, the models of Kobayashi et al. (2011) underproduce ¹⁵N by a factor of 6–10, and Romano et al. (2017) calibrate empirical nova yields of $10^{-7} M_{\odot}$ of ¹⁵N per outburst to fit the solar value and the gradient. At low metallicity the carbon isotopes provide an even sharper test. The same models without stellar rotation predict ¹²C/¹³C ≈ 4200 at $[\text{Fe}/\text{H}] = -2.6$, while the unmixed metal-poor subgiant HD 140283 has 33^{+12}_{-6} (Spite et al. 2021) and the most iron-poor carbon-enhanced stars have 39–100 (Molaro et al. 2023). This suggests that

the first stellar generations made ^{13}C as a primary product, for example in rapidly rotating massive stars that mix carbon from the helium-burning core into the hydrogen-burning shell.

4.4.3. *Other galaxies*

We can apply these diagnostics to the molecular gas of other galaxies, where they probe the stellar populations that enriched the gas. ^{13}C is released mostly by long-lived intermediate- and low-mass stars and novae, while ^{18}O comes from massive stars. The ratio $^{13}\text{CO}/\text{C}^{18}\text{O}$ measures the relative contributions of the two. The ratio is $\approx 7\text{--}10$ in the disk of the Milky Way and in nearby spirals. Zhang et al. (2018) found ≈ 1 in four lensed dusty starburst galaxies at $z \approx 2\text{--}3$, and they interpret this low ratio as evidence for a top-heavy stellar initial mass function.

Because ^{13}C is released late, the ratio depends on the age of the burst, although Zhang et al. (2018) argue that no plausible star-formation history reproduces the observed ratio with a normal IMF. The ratio also depends on the optical depth of C^{18}O in dense gas, on stellar rotation, which changes the yields (Romano et al. 2019), and on the mixing of gas components that have been processed to different degrees.

The nuclear starburst of NGC 253 shows how much the answer can depend on the tracer. Resolved optically thin tracers give $^{12}\text{C}/^{13}\text{C} = 21 \pm 6$, which Martín et al. (2019) attribute to stellar nucleosynthesis. In contrast, CCH, whose ^{13}C isotopologues are depleted by chemistry even in Galactic dark clouds (Sakai et al. 2010), gives a lower limit of 81 (Martín et al. 2010). Martín et al. (2019) attribute part of the difference to an extended, less processed gas component that the resolved observations filter out.

4.4.4. *Carbon isotopes across environments*

In Fig. 9 we gather the $^{12}\text{C}/^{13}\text{C}$ ratio across all of these environments, together with the presolar grains of Section 4.5. The ratio spans almost two orders of magnitude, from ≈ 3 in J-type carbon stars, near the CN-cycle equilibrium value of 3.6, to a lower limit of 138 in the nucleus of M82.

At the low end are the stars and grains made of CN-processed material: giants that have undergone extra mixing, J-type carbon stars, and the A+B presolar SiC grains. The Solar System, the local interstellar medium, most AGB carbon stars, most mainstream presolar SiC grains, and the atmospheres of M dwarfs and brown dwarfs all lie within a factor of two of the solar value (Abia et al. 2017; González Picos et al. 2025, 2024; Zhang et al. 2021a). The Galactic Center and the metal-poor stars lie in between. The Galactic Center lies at the inner end of the gradient in Fig. 8, where the gas is most enriched in secondary ^{13}C , and the metal-poor stars formed from gas whose ^{13}C the first stellar generations apparently made as a primary product.

4.5. Presolar grains

Primitive meteorites contain a small population of dust grains that condensed in the outflows of other stars before the Sun formed and survived the formation of the Solar System. We recognize these **presolar grains** by isotope ratios that differ enormously from anything else in the Solar System, and we measure them one grain at a time with ion microprobes. Each grain is a sample of a single star, read with a precision that no telescope can approach (Zinner 1998; Nittler & Ciesla 2016). The grains were discovered by chasing the carriers of isotopically anomalous noble gases, which led to

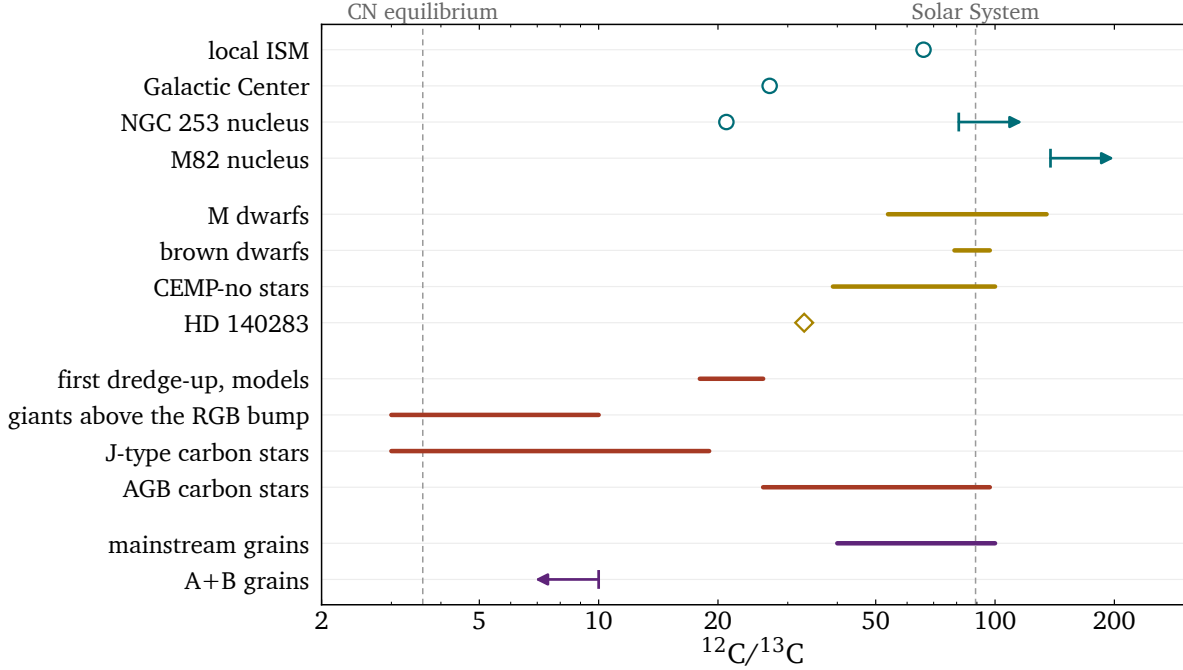


Figure 9: $^{12}\text{C}/^{13}\text{C}$ across environments, with colors grouping interstellar gas, dwarfs and metal-poor stars, giants and carbon stars, and presolar SiC grains. Bars span the measured values of each class (the typical range for mainstream grains, and standard-model predictions for 1–7 $M_{\odot}$ stars after the first dredge-up), open symbols are single values, arrows are limits, and dashed lines mark the CN-cycle equilibrium (3.6) and solar-system (89) ratios.

diamonds carrying anomalous xenon (Lewis et al. 1987) and to SiC in meteorite separates enriched 20 000-fold in anomalous neon and xenon (Bernatowicz et al. 1987).

4.5.1. Silicon carbide grains

Most presolar SiC grains (93%, the “mainstream” grains) have $^{12}\text{C}/^{13}\text{C}$ between 10 and 100 (mostly 40–100; Karakas & Lattanzio 2014) and heavy trace elements enhanced 10–20 times over solar (Lodders & Amari 2005). In Fig. 9 we see that their carbon ratios overlap those of AGB carbon stars. The isotopes of Sr, Zr, Mo, and Ba in single grains match models of 1.5–3 $M_{\odot}$ AGB stars of roughly solar metallicity and exclude more massive AGB stars (Lugaro et al. 2003). The grains have become a laboratory for the s process itself. The ^{135}Ba and ^{138}Ba abundances of mainstream grains (relative to the s-only ^{136}Ba) constrain the structure of the ^{13}C pocket in which the neutrons are released (Liu et al. 2014). Their $^{134}\text{Ba}/^{136}\text{Ba}$ reads the ^{134}Cs branch of Fig. 3b. With the stellar decay rate of ^{134}Cs then in standard use, two grains had $^{134}\text{Ba}/^{136}\text{Ba}$ too low for any AGB model and seemed to require the much higher neutron density of the i process (Liu et al. 2014). New calculations find that at s-process temperatures, where excited nuclear states are thermally populated, ^{134}Cs decays 3–10 times more slowly than had been assumed (Li et al. 2021; Taioli et al. 2022). More of the flow then captures a neutron at ^{134}Cs , and the models account for the Ba isotopes of the grains and the Sun without the i process (Li et al. 2021; Taioli et al. 2022).

The rarer populations of SiC grains point to other sources. The A+B grains (a few percent; Lodders

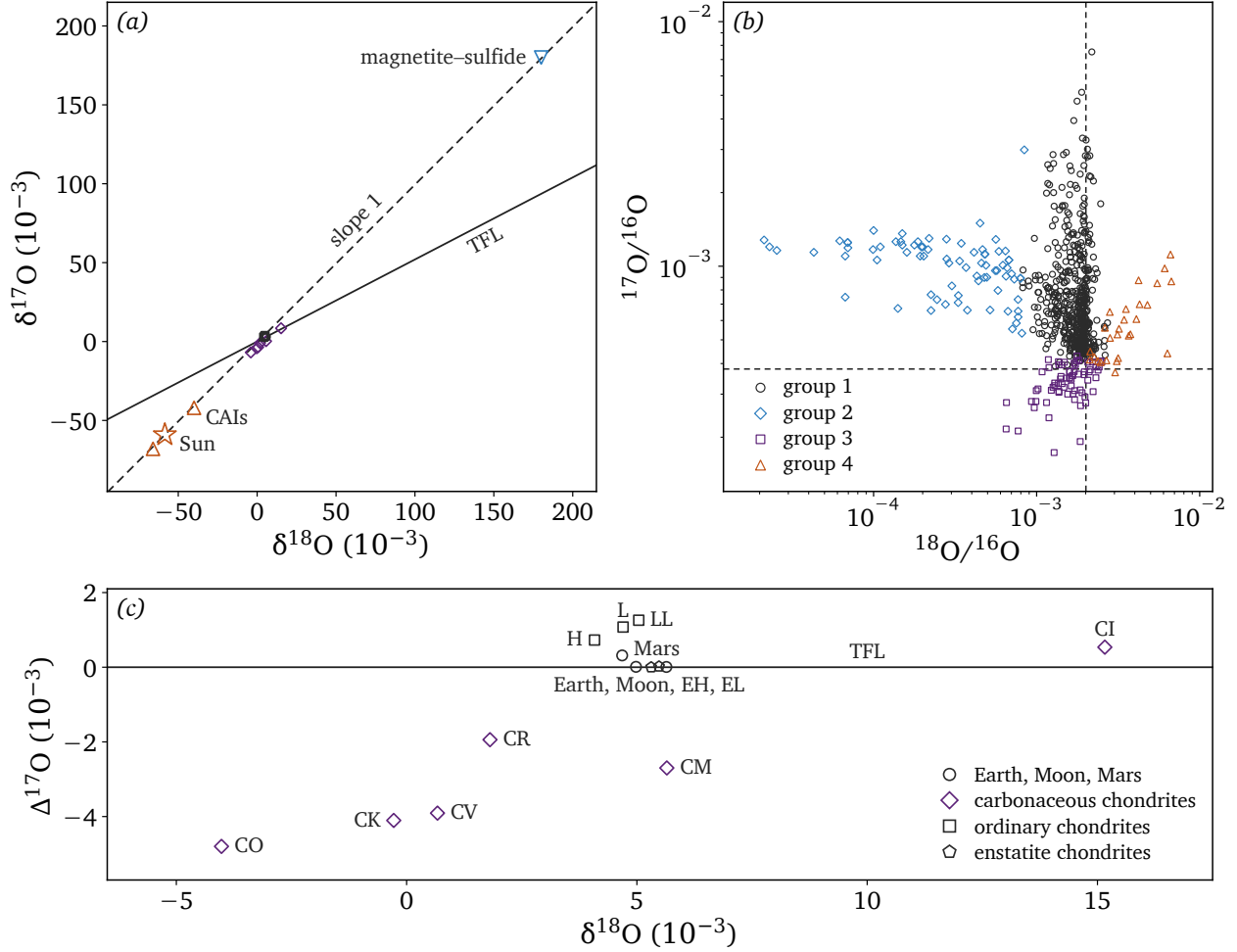


Figure 10: (a) Oxygen three-isotope plot for the Sun (*Genesis*), ^{16}O -rich CAIs from Allende and Isheyevo, magnetite-sulfide from Acfer 094, and the Earth, Moon, Mars, and chondrite-group means, with the terrestrial fractionation line (TFL) and a line of slope 1 through the Sun. (b) Oxygen isotope ratios of 717 presolar oxide grains by group (Hynes & Gyngard 2009), with dashed lines at the terrestrial ratios. (c) $\Delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ for the Earth, Moon, Mars, and chondrite groups.

& Amari 2005) have $^{12}\text{C}/^{13}\text{C} < 10$ and probably come from J-type carbon stars and born-again AGB stars (Amari et al. 2001). The X grains ($\sim 1\%$) carry ^{44}Ca from the decay of ^{44}Ti , a signature of supernovae, and a few grains with $^{12}\text{C}/^{13}\text{C} = 4\text{--}9$ and $^{14}\text{N}/^{15}\text{N} = 5\text{--}20$ are thought to come from novae (Lodders & Amari 2005).

4.5.2. Oxide grains

In Fig. 10b we show the oxygen isotope ratios of 717 presolar oxide grains from the 2010 version of the Presolar Grain Database (Hynes & Gyngard 2009). Among the grains the ratio $^{17}\text{O}/^{16}\text{O}$ spans a factor of ~ 40 and $^{18}\text{O}/^{16}\text{O}$ a factor of ~ 300 . Nittler et al. (1997) divided the grains into four groups. The Group 1 grains, which dominate the sample, have excess ^{17}O and a modest deficit of ^{18}O . This pattern is the signature of red giants after the first dredge-up, which mixes material from the interior, where the CNO cycle has made ^{17}O and destroyed ^{18}O , into the envelope (Section 4.2).

After the dredge-up, $^{17}\text{O}/^{16}\text{O}$ rises with stellar mass to a maximum of $\approx 0.004\text{--}0.006$ near $2\text{--}2.5 M_{\odot}$ (Section 4.2), while $^{18}\text{O}/^{16}\text{O}$ falls to $\approx 80\%$ of its initial value and reflects the composition the star was born with (Nittler et al. 2008). From a Group 1 grain we can therefore learn both the mass and (through a model of Galactic chemical evolution) the metallicity of its parent star. Both inferences have limits, because the mass relation is double valued ($^{17}\text{O}/^{16}\text{O}$ falls again at higher mass) and stars born at the same time scatter in metallicity (Nittler 2009).

The Group 2 grains have much larger ^{18}O deficits, which were attributed to extra mixing below the convective envelope of low-mass AGB stars (Nittler et al. 2008). With the $^{17}\text{O}(\text{p},\alpha)$ rate remeasured at LUNA (Bruno et al. 2016), hot-bottom burning in $4\text{--}8 M_{\odot}$ AGB stars can also reproduce their oxygen isotopes (Lugaro et al. 2017), and the choice between the two sites is debated (Liu et al. 2022). The ^{18}O -rich Group 4 grains probably come from supernovae (Nittler et al. 2008).

Figs. 10a and 10b are plotted on very different scales (Figs. 10a and 10c use the δ notation defined in Section 3.4.3). The presolar grains span factors of 40 and 300, while every bulk reservoir in the Solar System, from the most ^{16}O -rich inclusions to the most ^{16}O -poor water, lies within $\sim 20\%$ of the terrestrial ratios. Nuclear processes change isotope ratios by large factors, while in the bulk oxygen of Solar System bodies chemistry and physics change them by parts per thousand to tens of percent. Clayton (1993) applied this rule to C and N, although we will see in Section 6.3 that fractionation, probably by N_2 self-shielding, produces factor-of-three effects in the nitrogen of comets. We return to the Solar System variations in Section 6.

4.5.3. *Anomalies in bulk meteorites*

Bulk meteorites also carry nucleosynthetic fingerprints, at the much smaller level of parts in 10^4 (“ ϵ units”), left by the incomplete mixing of presolar material in the solar nebula. Warren (2011) showed that in a plot of the ^{54}Cr anomaly vs. the ^{50}Ti anomaly the carbonaceous chondrites (CC) are separated by a wide margin from all other planetary materials (the non-carbonaceous, or NC, group), including the Earth and Mars.

The two reservoirs coexisted and stayed spatially separate for several Myr after the formation of the calcium–aluminum-rich inclusions (CAIs), the oldest solids of the Solar System. The Mo isotopes assign each iron meteorite to one of the two reservoirs, and the Hf–W ages of the carbonaceous iron meteorites (Section 5.5.4) show that the carbonaceous reservoir existed within ≈ 1 Myr of the CAIs. The accretion ages of the carbonaceous chondrites show that this reservoir remained separate until at least 3–4 Myr. The simplest explanation is that Jupiter’s core grew to ~ 20 Earth masses within the first Myr and blocked the inward drift of material (Kruijer et al. 2017). In this application the nucleosynthetic fingerprints become tracers of transport in the protoplanetary disk.

Samples returned from two carbonaceous asteroids have refined this picture. The Hayabusa2 samples of Ryugu have the same ^{54}Fe anomaly as the CI chondrites ($+1 \pm 4$ and $+3 \pm 2$ parts per 10^6), while every other group of carbonaceous chondrites measured has $+15$ to $+40$ (Hopp et al. 2022). Ti, Cr, and O isotopes place Ryugu and the CI chondrites in the carbonaceous reservoir, but Fe sets them apart as a third group, which Hopp et al. (2022) suggest formed farther out in the disk. The 122 g returned by OSIRIS-REx from Bennu (Lauretta et al. 2024) carry a neutron-rich ^{50}Ti anomaly most similar to those of Ryugu and the CI chondrites, together with presolar SiC grains at a concentration of 25 ppm. Barnes et al. (2025) propose that the parent bodies of all three formed from a common but heterogeneous reservoir in the outer disk.

5. Radioactive isotopes as clocks and heat sources

We now turn to the isotopes that carry a timescale set by nuclear physics. A radioactive nucleus decays at a rate that is (with rare exceptions) independent of its environment, so the fraction of nuclei that survives measures the time since they were made. The decays also release energy, and in some settings (e.g., supernova ejecta and the first planetesimals of the Solar System) this energy is the dominant heat source, while in the Earth today radioactivity supplies $\approx 45\%$ of the heat that flows out through the surface.

In Fig. 11 we collect the radionuclides used in astronomy, arranged by half-life along a separate rail for each channel through which they have been measured. The half-lives span 13 decades, from 6.1 days for ^{56}Ni to 107 Gyr for ^{147}Sm . Notice that several channels draw on a narrow band of half-lives. Supernova light curves use nuclides that live for days to decades, the cosmic rays use 0.3–3 Myr, and meteorites use 0.3–100 Myr. The other channels are broad. The γ -ray lines combine the supernova nuclides with the Galactic ^{26}Al and ^{60}Fe , the stellar spectra span everything from ^7Be in novae to the actinides in the oldest stars, the terrestrial archives run from ^{14}C to ^{244}Pu , and geochronology uses the Gyr nuclides together with radiocarbon. In Table 3 we list these nuclides with their decay modes, each followed down its chain to a stable daughter (e.g., $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$).

5.1. Decay as a clock

A radioactive clock compares the surviving amount of a radioactive nuclide with the amount at the start, and the clocks in this subsection differ mainly in how they fix the starting point: radiocarbon dating assumes it and calibrates it with tree rings, an isochron measures it, and the Pb–Pb method cancels it.

5.1.1. Radiocarbon

The most familiar radioactive clock is radiocarbon dating. Cosmic-ray neutrons in the upper atmosphere convert nitrogen into ^{14}C through $^{14}\text{N}(\text{n},\text{p})^{14}\text{C}$, and the ^{14}C is mixed into the carbon of the atmosphere, the oceans, and living organisms. Anderson et al. (1947) first detected this natural radiocarbon. Carbon from living organisms contained 0.95×10^{-12} g of ^{14}C per gram of carbon, while carbon from petroleum, which has been isolated from the atmosphere for far longer than the half-life, showed none. An organism exchanges carbon with its surroundings while alive. After death the ^{14}C in the organism decays ($t_{1/2} = 5.70$ kyr; Kondev et al. 2021) and is not replaced, so the $^{14}\text{C}/^{12}\text{C}$ ratio of the remains measures the time since death (Libby et al. 1949; Arnold & Libby 1949).

Radiocarbon dating also illustrates two complications. The first complication is the initial ratio. The method assumes that the atmospheric $^{14}\text{C}/^{12}\text{C}$ has been constant, but the production rate follows the cosmic-ray flux at the Earth, which the solar wind and the geomagnetic field modulate (Section 5.5). Radiocarbon ages are calibrated against the ^{14}C content of tree rings of known age, a record that now reaches back $\approx 13\,900$ years, while the full calibration curve reaches 55 000 years (Reimer et al. 2020).

The second complication is fractionation. Photosynthesis takes up the lighter $^{12}\text{CO}_2$ slightly faster than $^{13}\text{CO}_2$ and $^{14}\text{CO}_2$, because lighter molecules diffuse and react faster (a kinetic fractionation, Section 6.1). A plant therefore starts life with a lower $^{14}\text{C}/^{12}\text{C}$ than the air it grew from. For wood

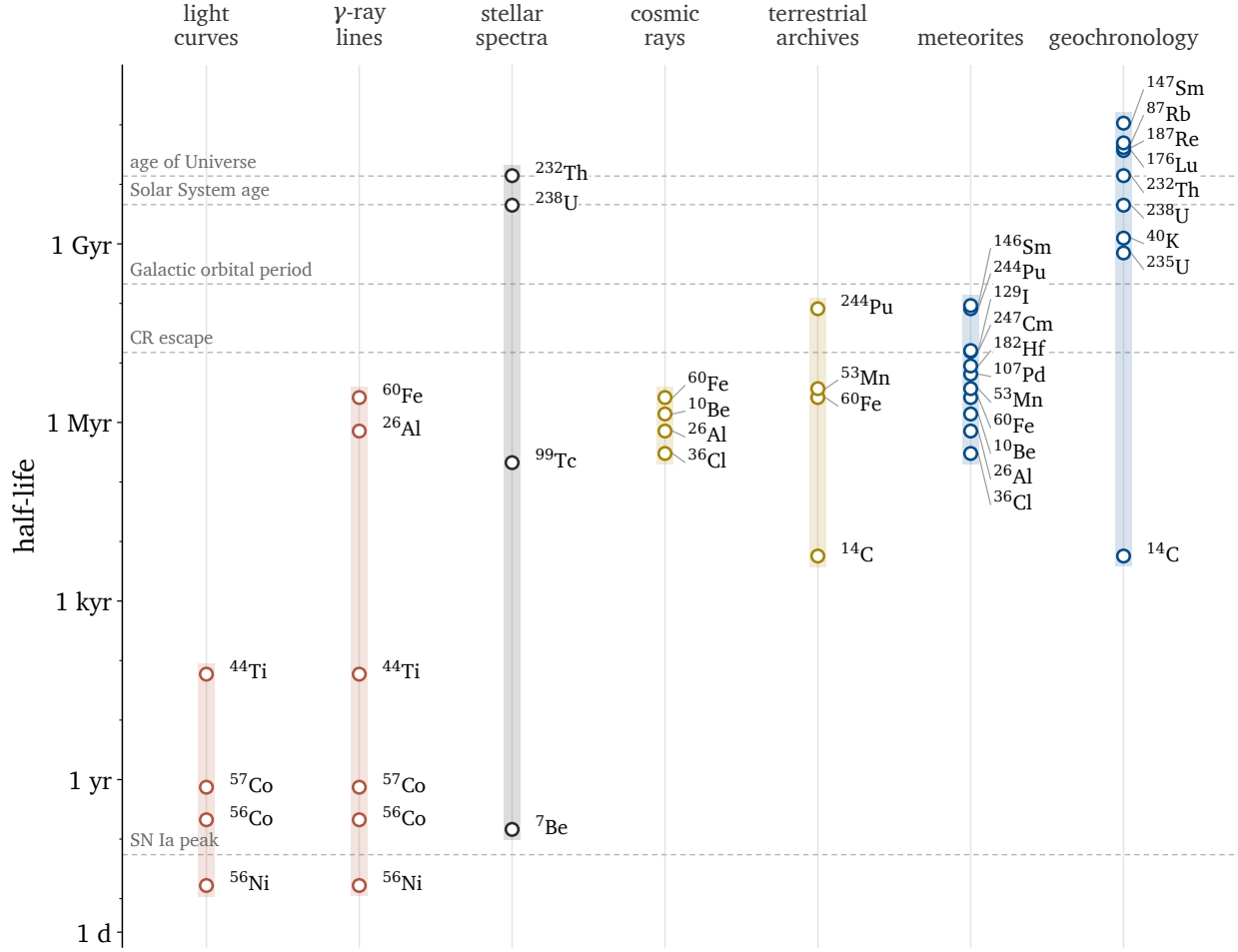


Figure 11: Half-lives of radionuclides used as clocks or heat sources, placed on a vertical rail for each channel through which they have been measured; shaded bars span the half-lives used in each channel. Half-lives are from NUBASE2020 (Kondev et al. 2021) except for ^{146}Sm (Chiera et al. 2024), and dashed lines mark the rise time of SN 2014J, the cosmic-ray confinement time, the Galactic orbital period of the Sun, and the ages of the Solar System and the Universe.

the deficit is equivalent to ≈ 260 years of decay, and other materials fractionate by different amounts, so that marine carbonate would appear 410 years younger than wood of the same age (Stuiver & Polach 1977). The stable isotope ^{13}C provides the correction, because its abundance in a sample reflects the fractionation alone. We measure the $^{13}\text{C}/^{12}\text{C}$ of each sample and scale its ^{14}C activity to the value it would have if its $^{13}\text{C}/^{12}\text{C}$ were that of average wood. The ^{14}C correction is taken to be twice the ^{13}C offset, because mass-dependent fractionation grows with the mass difference and ^{14}C is two mass units heavier than ^{12}C while ^{13}C is one. Each 0.1‰ of ^{13}C offset then shifts the age by 16 years.

Table 3. The radioactive decays in this memo, in order of the parent’s half-life, followed to the stable daughter. Half-lives are from NUBASE2020 (Kondev et al. 2021) except for ^{146}Sm (Chiera et al. 2024); for chains marked $\dots$ the third column gives the net numbers of α and β^- decays, and the last column gives sections.

decay chain	half-lives	decays	where it appears
$^7\text{Be} \rightarrow ^7\text{Li}$	53.2 d	EC	novae, CAIs (5.3, 5.5)
$^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$	6.08 d, 77.2 d	EC/ β^+	supernova light and γ -rays (5.3)
$^{57}\text{Ni} \rightarrow ^{57}\text{Co} \rightarrow ^{57}\text{Fe}$	35.6 h, 272 d	EC/ β^+ , EC	SN Ia and SN 1987A light curves (5.3)
$^{55}\text{Co} \rightarrow ^{55}\text{Fe} \rightarrow ^{55}\text{Mn}$	17.5 h, 2.76 yr	EC/ β^+ , EC	SN Ia light curves (5.3)
$^{44}\text{Ti} \rightarrow ^{44}\text{Sc} \rightarrow ^{44}\text{Ca}$	59.1 yr, 4.04 h	EC, EC/ β^+	SN 1987A, Cas A, X grains (4.5, 5.3)
$^{14}\text{C} \rightarrow ^{14}\text{N}$	5.70 kyr	β^-	radiocarbon (5.1, 5.5)
$^{41}\text{Ca} \rightarrow ^{41}\text{K}$	99.4 kyr	EC	CAIs (5.5)
$^{99}\text{Tc} \rightarrow ^{99}\text{Ru}$	0.211 Myr	β^-	S stars (5.2)
$^{36}\text{Cl} \rightarrow ^{36}\text{Ar}$	0.301 Myr	β^-	cosmic rays, CAIs (5.4, 5.5)
$^{26}\text{Al} \rightarrow ^{26}\text{Mg}$	0.717 Myr	EC/ β^+	γ -rays, CAIs, heat (5.4, 5.5)
$^{10}\text{Be} \rightarrow ^{10}\text{B}$	1.39 Myr	β^-	cosmic rays, CAIs (5.4, 5.5)
$^{60}\text{Fe} \rightarrow ^{60}\text{Co} \rightarrow ^{60}\text{Ni}$	2.62 Myr, 5.27 yr	β^- , β^-	γ -rays, cosmic rays, sea floor (5.4, 5.5)
$^{53}\text{Mn} \rightarrow ^{53}\text{Cr}$	3.7 Myr	EC	sea floor, meteorites (5.5)
$^{107}\text{Pd} \rightarrow ^{107}\text{Ag}$	6.5 Myr	β^-	meteorites (5.5)
$^{182}\text{Hf} \rightarrow ^{182}\text{Ta} \rightarrow ^{182}\text{W}$	8.90 Myr, 115 d	β^- , β^-	meteorites (5.5)
$^{247}\text{Cm} \rightarrow \dots \rightarrow ^{235}\text{U}$	15.6 Myr	3α , $2\beta^-$	meteorites (5.5)
$^{129}\text{I} \rightarrow ^{129}\text{Xe}$	16.1 Myr	β^-	meteorites (5.5)
$^{244}\text{Pu} \rightarrow \dots \rightarrow ^{232}\text{Th}$	81.3 Myr	3α , $2\beta^-$	sea floor, meteorites (5.5)
$^{146}\text{Sm} \rightarrow ^{142}\text{Nd}$	92 Myr	α	meteorites
$^{235}\text{U} \rightarrow \dots \rightarrow ^{207}\text{Pb}$	0.704 Gyr	7α , $4\beta^-$	Pb–Pb ages, heat (5.1, 5.5)
$^{40}\text{K} \rightarrow ^{40}\text{Ca}$ (89%), ^{40}Ar (11%)	1.25 Gyr	β^- ; EC/ β^+	Earth’s heat (5.5)
$^{238}\text{U} \rightarrow \dots \rightarrow ^{206}\text{Pb}$	4.46 Gyr	8α , $6\beta^-$	stellar and Pb–Pb ages, heat (5.1–5.5)
$^{232}\text{Th} \rightarrow \dots \rightarrow ^{208}\text{Pb}$	14.0 Gyr	6α , $4\beta^-$	stellar ages, heat (5.2, 5.5)
$^{176}\text{Lu} \rightarrow ^{176}\text{Hf}$	37.0 Gyr	β^-	geochronology
$^{187}\text{Re} \rightarrow ^{187}\text{Os}$	41.6 Gyr	β^-	geochronology
$^{87}\text{Rb} \rightarrow ^{87}\text{Sr}$	49.7 Gyr	β^-	geochronology
$^{147}\text{Sm} \rightarrow ^{143}\text{Nd}$	107 Gyr	α	geochronology

5.1.2. The decay law

More generally, consider a radioactive isotope R that is produced together with a stable reference isotope S. After a time t we have:

$$\left(\frac{R}{S}\right)_{\text{now}} = \left(\frac{R}{S}\right)_0 e^{-t/\tau}, \quad \tau = \frac{t_{1/2}}{\ln 2}, \quad (10)$$

where τ is the mean life. To read this clock we need the present ratio, which is measured, and the initial ratio $(R/S)_0$, which must come either from a nucleosynthesis model or from an independent anchor. As we will see, most of the controversies in this section concern $(R/S)_0$.

If we differentiate Eq. 10, we find that a fractional measurement error σ_R/R gives an age error $\sigma_t = \tau \sigma_R/R$, independent of t . The fractional precision σ_t/t therefore improves as the clock runs, until the surviving fraction becomes too small to detect. If the error is set by counting the surviving nuclei, $\sigma_R/R \propto e^{t/2\tau}$ and σ_t/t is smallest at $t = 2\tau$, so a clock reads best when the elapsed time is ≈ 2 mean lives. For example, thorium ($t_{1/2} = 14.0$ Gyr) decays by only a factor of two in a Hubble time, so a 0.1 dex error in a stellar Th abundance corresponds to 4.7 Gyr, compared with 1.5 Gyr for uranium ($t_{1/2} = 4.46$ Gyr), as Hill et al. (2002) noted. An error in the initial ratio enters the age in the same way, and in practice this error usually dominates. Note the two coincidences that make actinide dating (Section 5.2) possible at all: the half-life of ^{238}U is close to the age of the Solar System, and that of ^{232}Th is close to the age of the Universe.

5.1.3. Daughter isotopes

If the parent is alive (e.g., Tc in a star, ^{26}Al in the interstellar medium, ^{244}Pu in an ocean crust), we measure its abundance directly. Many of the clocks of the early Solar System have parents that died out long ago. The ^{26}Al in the first solids ($t_{1/2} = 0.72$ Myr) had decayed completely within a few Myr, and the only record it left is an excess of its daughter, ^{26}Mg .

To read such a clock we must separate the radiogenic daughter atoms from those that were present from the start, since ^{26}Mg makes up 11% of ordinary magnesium. The separation relies on chemistry. Minerals that crystallize together from the same melt or gas inherit the same isotopic composition of each element, because chemistry barely distinguishes isotopes, but they take up very different amounts of each element. In a calcium–aluminum-rich inclusion, for example, anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$) contains abundant aluminum and almost no magnesium, while spinel (MgAl_2O_4) contains both. Every mineral therefore starts with the same $^{26}\text{Al}/^{27}\text{Al}$ and the same $^{26}\text{Mg}/^{24}\text{Mg}$, and once the ^{26}Al has decayed each mineral has gained ^{26}Mg in proportion to the aluminum it contained.

If we count atoms, $N(^{26}\text{Mg})_{\text{now}} = N(^{26}\text{Mg})_0 + N(^{26}\text{Al})_0$. We divide by the stable ^{24}Mg , which no decay produces, and write $N(^{26}\text{Al})_0 = (^{26}\text{Al}/^{27}\text{Al})_0 N(^{27}\text{Al})$, where ^{27}Al , the only stable isotope of aluminum, stands in for the ^{26}Al that has since decayed. We then find:

$$\left(\frac{^{26}\text{Mg}}{^{24}\text{Mg}}\right)_{\text{meas}} = \left(\frac{^{26}\text{Mg}}{^{24}\text{Mg}}\right)_0 + \left(\frac{^{26}\text{Al}}{^{27}\text{Al}}\right)_0 \frac{^{27}\text{Al}}{^{24}\text{Mg}}. \quad (11)$$

A plot of the measured $^{26}\text{Mg}/^{24}\text{Mg}$ vs. $^{27}\text{Al}/^{24}\text{Mg}$ for the minerals of one object is therefore a straight line, called an **isochron**, and in Fig. 12a we show an example. The intercept is the initial magnesium isotope ratio, and the slope is the $^{26}\text{Al}/^{27}\text{Al}$ at the moment the minerals formed and stopped exchanging magnesium. If the minerals did not form together, or were later disturbed, the points scatter about the line, so the isochron also tests its own assumptions.

The slope measures time because the $^{26}\text{Al}/^{27}\text{Al}$ of the solar nebula fell as the ^{26}Al decayed. If two objects formed from the same reservoir with initial ratios R_A and R_B , Eq. 10 gives the interval between them, $t_B - t_A = \tau \ln(R_A/R_B)$, with $\tau = 1.03$ Myr. An object whose initial ratio is ten times smaller than that of another formed 2.4 Myr later. This resolution, 4.5 Gyr after the event, is fine enough to show that the CAIs formed within ± 20 kyr of one another (Jacobsen et al. 2008). However, the method gives only relative ages, and it assumes that the ^{26}Al was spread evenly through the nebula. To fix the absolute time we need a clock whose parents are still alive.

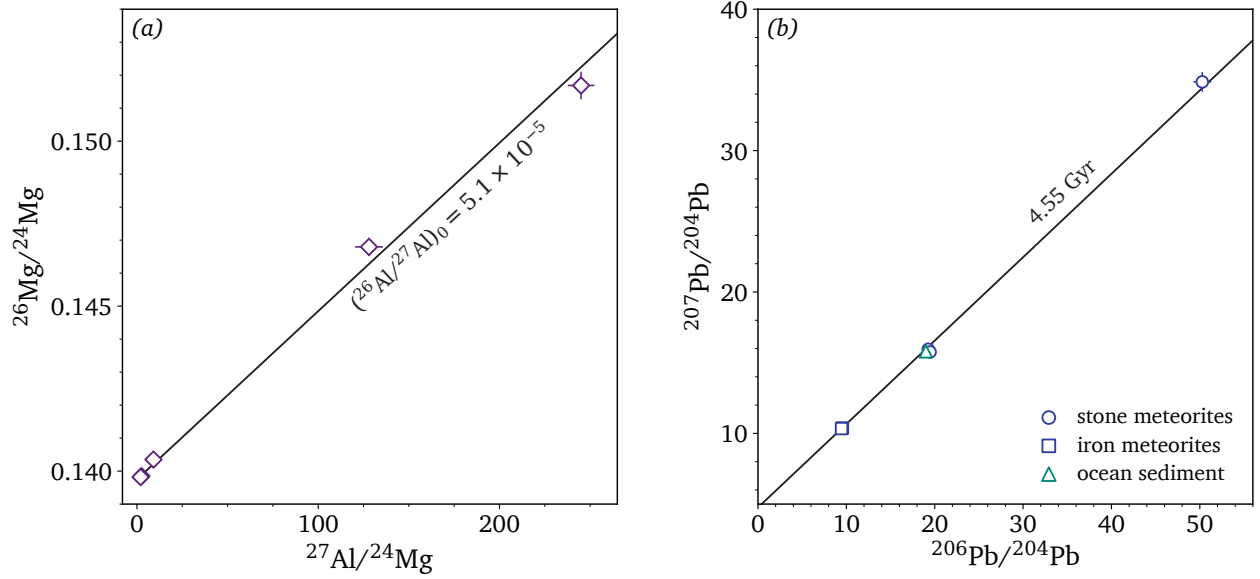


Figure 12: (a) $^{26}\text{Mg}/^{24}\text{Mg}$ vs. $^{27}\text{Al}/^{24}\text{Mg}$ for coexisting minerals (anorthite, melilite, spinel, and fassaite) of the Allende CAI WA, with the published isochron of Lee et al. (1977); two of the five points overlap near the origin. (b) $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ for three stony meteorites, the troilite of two iron meteorites, and recent ocean sediment (Patterson 1956), with the 4.55 Gyr isochron of Eq. 12 through the iron-meteorite lead, computed with Patterson’s decay constants.

The Pb–Pb method supplies that clock. Uranium has two long-lived isotopes, ^{238}U and ^{235}U , which decay through long chains to ^{206}Pb and ^{207}Pb with half-lives of 4.46 and 0.70 Gyr (Table 3), and lead has an isotope, ^{204}Pb , that no decay produces and that plays the role of ^{24}Mg above. The radiogenic lead added since a rock formed, which we denote $^{206}\text{Pb}^*$ and $^{207}\text{Pb}^*$, then satisfies:

$$\frac{^{207}\text{Pb}^*}{^{206}\text{Pb}^*} = \frac{^{235}\text{U}}{^{238}\text{U}} \frac{e^{\lambda_{235}t} - 1}{e^{\lambda_{238}t} - 1}, \quad (12)$$

where $\lambda = 1/\tau$ are the decay constants and the uranium ratio is today’s value. The amount of uranium cancels, so this ratio depends only on the age t , and rocks with different U/Pb ratios lie on a straight line in a plot of $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ whose slope is given by Eq. 12. At the age of the Solar System the radiogenic ratio is 0.62. Because ^{235}U decays faster than ^{238}U , the ratio changes by 0.07% per Myr, so a measurement to 0.1% fixes the age to 1.4 Myr. Patterson (1956) measured the lead in three stony meteorites and in the troilite (FeS) of two iron meteorites, which contains almost no uranium and so preserves the initial lead. All five lay close to one line (Fig. 12b), which gave an age of 4.55 ± 0.07 Gyr, and lead from recent ocean sediments fell on the same line within the errors, so the Earth formed together with the meteorites. Eq. 12 assumes that $^{238}\text{U}/^{235}\text{U}$ is the same everywhere, but the ratio varies among CAIs from 137.41 to 137.89, probably because of live ^{247}Cm , which decays to ^{235}U , in the early Solar System (Brennecka et al. 2010). This 0.35% range shifts the Pb–Pb age by 5 Myr, so modern Pb–Pb ages use the $^{238}\text{U}/^{235}\text{U}$ measured in each sample. The corrected Pb–Pb age of the CAIs (Section 5.5) then anchors the relative ages from ^{26}Al .

5.2. Radioactivity in stellar atmospheres

In a stellar spectrum the isotopes of an element cannot be separated, so an element can serve as a stellar clock only if it consists of a single radioactive nuclide. Technetium in AGB stars and thorium and uranium in old halo stars meet this condition.

5.2.1. Technetium

In 1952 Paul Merrill identified absorption lines of Tc I at 4031, 4238, 4262, and 4297 Å in the spectra of S-type red giants (Merrill 1952). Technetium has no stable isotope, and the isotope made by the s process, ^{99}Tc , has a laboratory half-life of 2.1×10^5 yr. Any technetium present when the star formed would have decayed long before a low-mass star reached the giant branch. The technetium must therefore have been made inside the star and brought to its surface. Merrill's lines were the first direct evidence that nucleosynthesis happens in stars, five years before B²FH.

Technetium also sorts S stars into two classes (Van Eck & Jorissen 1999). Tc-rich S stars are thermally pulsing AGB stars that are dredging up fresh s-process material. Tc-poor S stars are binaries that accreted s-process material from a former AGB companion (now a white dwarf) long enough ago for the Tc to decay. In the Henize sample, 29 of 70 S stars are Tc-rich. (^{99}Tc is also one of the rare cases in which the environment changes a decay rate. At the 3×10^8 K of a thermal pulse its effective half-life falls to ~ 1 yr; Van Eck & Jorissen 1999.)

5.2.2. Uranium and thorium

The oldest stars can be dated by the decay of thorium and uranium, a method known as actinide dating (the actinides are the elements from actinium to lawrencium). The method works because neither element has a stable isotope, and in an old star each consists almost entirely of one long-lived nuclide. Thorium is ^{232}Th ($t_{1/2} = 14.0$ Gyr; its next longest-lived isotope, ^{230}Th , has $t_{1/2} = 75$ kyr), and the uranium in a star 12 Gyr old is almost pure ^{238}U ($t_{1/2} = 4.46$ Gyr), since ^{235}U ($t_{1/2} = 0.70$ Gyr) has decayed by a factor of $\sim 10^5$ (Kondev et al. 2021). A spectral line, which cannot separate the isotopes, therefore measures the abundance of the radioactive nuclide itself. If the radioactive isotope were a minor part of the element, its decay would barely change the elemental abundance. Potassium, for example, is only 0.012% ^{40}K , so the decay of ^{40}K leaves the potassium abundance of a star unchanged. Technetium, which has no stable isotope at all, qualifies for the same reason.

Both ^{232}Th and ^{238}U are made only by the r process. A metal-poor halo star that formed from gas enriched by one or a few r-process events keeps the U and Th that those events made, and since then the faster decay of ^{238}U has lowered U/Th at a known rate. Applying Eq. 10 to each nuclide, we find that $\log(\text{U/Th})$ falls by one dex in 21.8 Gyr, so the measured ratio, compared with the ratio at production, gives the time since the r-process event and hence the age of the star.

Cayrel et al. (2001) detected U II 3859.6 Å in the r-process-enhanced giant CS 31082-001 ($[\text{Fe}/\text{H}] = -2.9$), measured $\log(\text{U/Th}) = -0.74 \pm 0.15$, and derived an age of 12.5 ± 3 Gyr. Because U and Th are chemically and spectroscopically similar, errors in the model atmosphere largely cancel in their ratio. The age also depends on the production ratio, the $(R/S)_0$ of Eq. 10, which must come from models of the r process. Relative to the classical r-process pattern, the zero-age U and Th in CS 31082-001 are enhanced by a factor of ≈ 2.5 , the “actinide boost” (Schatz et al. 2002), and a clock that pairs Th with the stable r-process element Eu gives this star a negative age (Hill et al.

2002). The boost raises U and Th together, so U/Th ages are robust to it while Th/Eu ages are not. The star HE 1523-0901 gives 13.2 Gyr from several chronometers (Frebel et al. 2007), consistent with the age of the Universe.

5.3. Radioactive power in explosions

Radioactive decay powers the light of Type Ia supernovae and kilonovae from the start, and that of core-collapse supernovae once the energy deposited by the shock has been radiated. Their light curves therefore follow the decay rates of the nuclides that the explosions made, and their γ -ray lines identify those nuclides.

5.3.1. Type Ia supernovae

Recall from Section 2.2 that NSE with equal numbers of neutrons and protons makes ^{56}Ni . This nucleus decays to ^{56}Co ($t_{1/2} = 6.08$ d) and then to ^{56}Fe ($t_{1/2} = 77.2$ d), releasing 1.75 and 3.73 MeV per decay, mostly as γ -rays. Colgate & McKee (1969) proposed that this decay chain powers supernova light curves, and Clayton et al. (1969) predicted the γ -ray lines that the decays would produce. The ejecta of a Type Ia supernova are opaque at first, and the light emerges once the diffusion time has fallen to the expansion time. At that moment, the peak of the light curve, the luminosity equals the instantaneous rate of radioactive energy deposition (Arnett’s rule; Arnett 1982). For a rise time of 19 days this gives (Stritzinger & Leibundgut 2005):

$$L_{\text{max}} = (2.0 \pm 0.3) \times 10^{43} \text{ erg s}^{-1} \left(\frac{M_{\text{Ni}}}{1 M_{\odot}} \right), \quad (13)$$

so the peak luminosity of a Type Ia supernova is in effect a measurement of the mass of ^{56}Ni that the explosion made. The first direct test came from the nearby SN 2014J, whose ^{56}Co lines at 847 and 1238 keV implied $0.6 \pm 0.1 M_{\odot}$ of ^{56}Ni (Churazov et al. 2014) ($0.49 \pm 0.09 M_{\odot}$ from a second analysis; Diehl et al. 2015). Arnett’s rule applied to the peak luminosity gives $0.42 \pm 0.05 M_{\odot}$, which Churazov et al. (2014) call marginally consistent with their line value, and the second analysis agrees better.

Late-time light curves add an isotope ratio. The neutron-rich ^{57}Ni (28 protons and 29 neutrons) is made in NSE together with ^{56}Ni , and its yield rises with the neutron excess. Electron captures raise the neutron excess at the high central density of a white dwarf near the Chandrasekhar mass, and a metal-rich progenitor raises it as well (Graur et al. 2016; Tiwari et al. 2022). The decay of its daughter ^{57}Co ($t_{1/2} = 272$ d) is predicted to power most of the light from ≈ 3 yr after the explosion (Graur et al. 2016), as we will see for SN 1987A below. *HST* light curves give $^{57}\text{Ni}/^{56}\text{Ni} = 0.043^{+0.012}_{-0.011}$ for SN 2012cg, approximately twice the solar $^{57}\text{Fe}/^{56}\text{Fe}$ and favoring a near-Chandrasekhar progenitor (Graur et al. 2016). SN 2011fe gives ≈ 0.026 (Shappee et al. 2017), and after the later detection of ^{55}Fe this supernova is consistent with either a low-density near-Chandrasekhar or a sub-Chandrasekhar white dwarf (Tucker et al. 2022). Independent analyses of the same supernova differ by factors of 2–4 (0.035–0.13 for SN 2014J; Yang et al. 2018; Li et al. 2019; Graur et al. 2018), because the analyses treat the bolometric correction and the escape of positrons differently. These systematic errors are as large as the differences between the progenitor models, and four of the five supernovae compiled by Tiwari et al. (2022) are consistent with both near- and sub-Chandrasekhar white dwarfs.

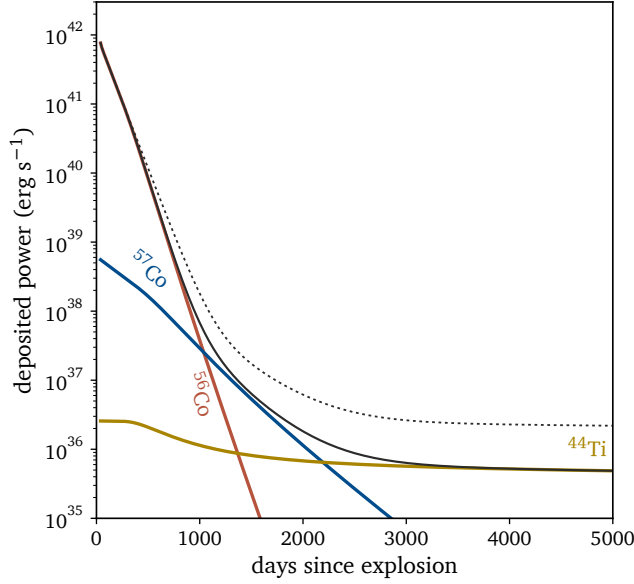


Figure 13: Radioactive power deposited in ejecta resembling those of SN 1987A vs. time since explosion, for the $A = 56, 57$, and 44 decay chains and their sum (black); the dotted curve is the sum if all γ -rays were absorbed. Masses and the γ -ray trapping law are from Seitenzahl et al. (2014).

5.3.2. SN 1987A

SN 1987A in the Large Magellanic Cloud has provided the most complete record of radioactive power. Its light curve required $\approx 0.07 M_{\odot}$ of ^{56}Ni (Arnett et al. 1989), and the *Solar Maximum Mission* detected the 847 keV ^{56}Co line within the first year (Matz et al. 1988), the first direct evidence that supernova light is powered by radioactivity.

In Fig. 13 we show how the power is handed from one decay chain to the next. For the first few hundred days the bolometric light curve declined with the 111-day mean life of ^{56}Co . After day ≈ 1100 the optical decline slowed to an e -folding time of 390 ± 20 days, which matches the 392-day mean life of ^{57}Co (Suntzeff et al. 1992), and after several years the ejecta entered the phase powered by ^{44}Ti ($t_{1/2} = 59.1$ yr; Jerkstrand et al. 2011). Notice that, for similar numbers of nuclei and decay energies, the power at a time t comes mainly from the nuclides whose mean lives are comparable to t . The power released by N nuclei of mean life τ is proportional to $(N/\tau) e^{-t/\tau}$, which at fixed t is largest for $\tau = t$, since shorter-lived nuclides have already decayed and longer-lived ones release their energy too slowly. Because each phase has its own decay rate, the light curve is an isotope-resolved measurement.

The inferred ratio $^{57}\text{Ni}/^{56}\text{Ni}$ is 2.5 ± 1.1 times the solar $^{57}\text{Fe}/^{56}\text{Fe}$ (Seitenzahl et al. 2014), which is itself a fingerprint of how the NSE matter froze out. The ^{44}Ti mass is less secure, and estimates from the light curve, the late-time spectrum, and the γ -ray lines range from 0.55 to $3.1 \times 10^{-4} M_{\odot}$ (Seitenzahl et al. 2014; Jerkstrand et al. 2011; Grebenev et al. 2012). Converting a light curve into masses requires a model of how the ejecta trap and reprocess the decay energy.

5.3.3. ^{44}Ti and the geometry of explosions

In a core-collapse supernova, ^{44}Ti is made in the innermost ejecta, in the layer directly above the newly formed neutron star, so its distribution is a direct probe of the explosion mechanism (Boggs et al. 2015). Its half-life of 59 yr is long enough that its decay lines remain visible for centuries. COMPTEL detected the 1157 keV line, emitted by the daughter ^{44}Sc , from the 340-year-old remnant

Cas A (Iyudin et al. 1994), the first supernova remnant seen in a radioactive line, as Clayton et al. (1969) had predicted. Only 2% of the original ^{44}Ti survives today.

NuSTAR imaged the 68 and 78 keV lines in Cas A and found $(1.25 \pm 0.3) \times 10^{-4} M_{\odot}$ of ^{44}Ti (Grefenstette et al. 2014). The ^{44}Ti map differs from X-ray images of the iron. The Fe-K X-ray lines are emitted only where the reverse shock has heated and ionized the iron, while the ^{44}Ti lines follow each nuclear decay whatever the temperature of the gas. At least 80% of the ^{44}Ti emission lies inside the reverse shock, in ejecta that the shock has not yet heated, and the ^{44}Ti is concentrated in clumps that do not follow the Fe-K emission. The lines are also redshifted, which implies a bulk motion of 1100–3000 km s $^{-1}$ away from us. This distribution rules out both a symmetric explosion and a strongly bipolar one from a rapidly rotating star, and favors large-scale convective instabilities during the explosion (Grefenstette et al. 2014). The ^{44}Ti lines of SN 1987A are likewise redshifted, by ≈ 700 km s $^{-1}$ (Boggs et al. 2015). Because ^{44}Ti decays by electron capture, ions stripped of their inner electrons would decay more slowly (Mochizuki et al. 1999). Since most of the ^{44}Ti lies in unshocked, weakly ionized ejecta, this effect is small (Grefenstette et al. 2014).

5.3.4. Novae and kilonovae

We can identify two further kinds of explosion by their radioactive products. In the nova V339 Del, Tajitsu et al. (2015) detected the Be II resonance doublet near 313 nm, blueshifted by ~ 1000 km s $^{-1}$, 38–48 days after outburst, and inferred a ^7Be mass fraction of $10^{-4.3 \pm 0.3}$. Beryllium-7 decays to ^7Li with $t_{1/2} = 53.2$ d. Novae are therefore a source of the ^7Li that Section 2.2 left unaccounted for.

The kilonova that followed GW170817, a transient predicted by Li & Paczyński (1998), was powered by the decay of freshly made r-process nuclei. These nuclei span a very wide range of half-lives, and a simple argument shows that the sum of their decays is a power law. If the number of nuclei per logarithmic interval of mean life is constant, $dN/d \ln \tau = N_0$, and each releases an energy Q , then we can write the power per unit mass as:

$$\epsilon(t) = \int \frac{QN_0}{\tau} e^{-t/\tau} d \ln \tau = \frac{QN_0}{t} \int_0^{\infty} e^{-x} dx \propto t^{-1}, \quad (14)$$

where $x = t/\tau$. As in SN 1987A, the integrand peaks at $\tau = t$, so at each time the power comes from the N_0 nuclides per unit $\ln \tau$ whose mean lives are $\approx t$, each releasing its energy Q over a time $\sim t$. In reality the decay energy of a β -decaying nucleus is correlated with its lifetime, and the nuclei are not uniformly distributed in $\ln \tau$, so the power law steepens to the network result $\epsilon \approx 10^{10} (t/1 \text{ day})^{-1.3} \text{ erg s}^{-1} \text{ g}^{-1}$ (Metzger et al. 2010; Kasen et al. 2017).

5.4. Radioactivity in the Galaxy

A radioactive nuclide that is made continuously settles at an abundance equal to its production rate times its mean life. A census of the radioactive nuclei in the interstellar medium therefore measures the rates at which stars make them, and the radioactive nuclei among the cosmic rays measure how long the particles have been confined.

5.4.1. ^{26}Al and ^{60}Fe

The 1809 keV line of ^{26}Al ($t_{1/2} = 0.717$ Myr) was discovered by *HEAO 3* toward the inner Galaxy (Mahoney et al. 1984). This line was the first detection of freshly made radioactive nuclei

in the interstellar medium, and showed that nucleosynthesis continues across the Galaxy today. *INTEGRAL*/SPI measured Doppler shifts of the line consistent with Galactic rotation (so the emitting material corotates with the disk) and found a steady-state mass of $2.8 \pm 0.8 M_{\odot}$ of ^{26}Al (Diehl et al. 2006). Because the ^{26}Al mean life (1.03 Myr) is short compared with Galactic timescales, we can write the amount present today as the production rate times the mean life:

$$M_{\text{ss}} = \dot{M} \tau, \quad (15)$$

which gives a production rate of $2.7 M_{\odot}$ per Myr. Eq. 15 is the steady state of Eqs. 1 and 2, with radioactive decay as the destruction process. If we divide this rate by the ^{26}Al yield of a massive star, we obtain a Galactic core-collapse supernova rate of 1.9 ± 1.1 per century (Diehl et al. 2006), revised to 1.3 ± 0.6 per century (as summarized by Diehl et al. 2021). Most of the systematic error in this estimate lies in the stellar yields. The ^{26}Al emission is concentrated toward groups of massive stars (e.g., $1.1 \times 10^{-4} M_{\odot}$ in Sco-Cen; Diehl et al. 2010), and in the most recent all-sky analysis only 16% of the flux comes from the inner Galaxy ($|l| < 30^\circ$; Pleintinger et al. 2023). Converting the flux into a mass requires a model of where the nearby sources lie. The decays of ^{26}Al also supply 10–17% of the positrons that annihilate in the Galaxy (Prantzos et al. 2011).

The lines of ^{60}Fe at 1173 and 1332 keV have also been detected (Harris et al. 2005). Recall that in steady state each isotope decays as fast as it is produced. If the two isotopes come from the same sources and share their distribution on the sky, the ratio of the ^{60}Fe and ^{26}Al line fluxes is therefore the ratio of the production rates by number. This ratio does not depend on the distances to the sources. Wang et al. (2020) measure a flux ratio of 0.184 ± 0.042 for a common sky distribution (0.2–0.4 if each isotope is given its own best-fitting map), which corresponds to a mass production ratio of ≈ 0.43 and tests the yields of massive stars.

5.4.2. Cosmic rays

The Galactic cosmic rays are atomic nuclei accelerated to relativistic energies and confined by the Galactic magnetic field for millions of years before they leak out. Instruments in space, such as the Cosmic Ray Isotope Spectrometer (CRIS) on the *Advanced Composition Explorer* (ACE), measure the charge and mass of each nucleus that arrives. They therefore separate isotopes directly, without any of the difficulties of spectroscopy described in Section 3, and give us isotope ratios for matter from elsewhere in the Galaxy.

Radioactive isotopes in the cosmic rays measure how long the particles are confined. As the cosmic rays cross interstellar gas, collisions break some C, N, and O nuclei into lighter fragments, among them the radioactive ^{10}Be ($t_{1/2} = 1.39$ Myr; Korschinek et al. 2010). Stable fragments, such as boron, accumulate in proportion to the amount of gas traversed, while ^{10}Be also decays, so the fraction of ^{10}Be that survives measures the time the cosmic rays have spent in the Galaxy. The nuclei move at a large fraction of the speed of light, and time dilation lengthens the ^{10}Be lifetime by 8–43% at the energies measured. The ACE/CRIS measurements of ^{10}Be , ^{26}Al , ^{36}Cl , and ^{54}Mn at 70–400 MeV per nucleon give a confinement time of 15.0 ± 1.6 Myr (Yanasak et al. 2001).

The confinement time and the amount of gas traversed together give the mean density along the path, 0.34 cm^{-3} (Yanasak et al. 2001). This density is lower than the mean density of the gas disk, so the cosmic rays must spend much of their lives in the low-density halo above and below the disk, as the first ^{10}Be measurements already suggested (Garcia-Munoz et al. 1975, 1977).

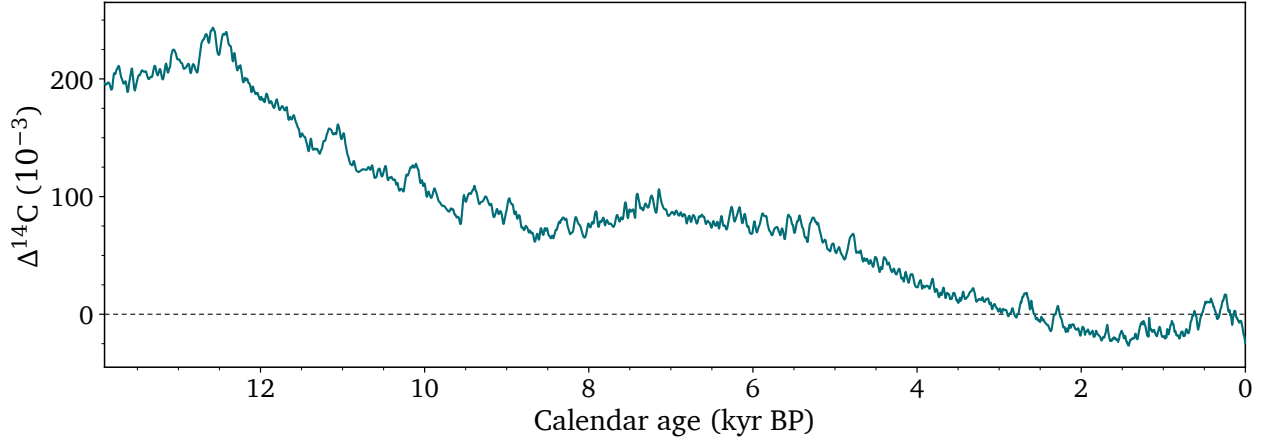


Figure 14: $\Delta^{14}\text{C}$ vs. calendar age before AD 1950 for the tree-ring part of the IntCal20 Northern Hemisphere curve (Reimer et al. 2020).

ACE/CRIS has also detected 15 nuclei of ^{60}Fe among 3.55×10^5 Fe nuclei, which corresponds to $^{60}\text{Fe}/^{56}\text{Fe} = (7.5 \pm 2.9) \times 10^{-5}$ at the source (Binns et al. 2016). This is the isotope whose γ -ray lines we discussed above. Given its 2.6 Myr half-life, these nuclei must have been made and accelerated within the last few Myr, by a source within ~ 1 kpc, plausibly a nearby group of massive stars.

The isotopic composition of the cosmic rays also identifies where they were accelerated. At the cosmic-ray source $^{22}\text{Ne}/^{20}\text{Ne}$ is 5.3 ± 0.3 times the solar-wind value. Together with smaller excesses of $^{12}\text{C}/^{16}\text{O}$ and $^{58}\text{Fe}/^{56}\text{Fe}$, this ratio fits a mixture of $\approx 80\%$ material of solar composition and $\approx 20\%$ material from the winds of Wolf–Rayet stars (Binns et al. 2005). These are massive stars that have lost their hydrogen envelopes, and their winds carry the products of helium burning, including ^{22}Ne made from ^{14}N (Section 2.2). Both the ^{22}Ne and the ^{60}Fe therefore point to groups of massive stars, whose winds and supernovae blow large hot cavities (superbubbles) in which at least part of the Galactic cosmic rays are accelerated.

5.5. The local record

In material that reaches the laboratory we count atoms directly. This record extends from the radiocarbon in tree rings of the last $\approx 14\,000$ years, through supernova debris in deep-sea deposits, to the minerals that formed with the Solar System 4.567 Gyr ago, and it includes the long-lived radionuclides that heat the Earth today.

5.5.1. Cosmic rays and the Sun in tree rings

We return now to radiocarbon. Since the production rate of ^{14}C follows the cosmic-ray flux at the Earth (Section 5.1), the ^{14}C in tree rings traces solar activity, superposed on slower geomagnetic changes (Stuiver & Quay 1980). In Fig. 14 we show $\Delta^{14}\text{C}$ in tree rings over the last 13 900 years, a decline of $\approx 250 \times 10^{-3}$ with century-scale wiggles of $(10\text{--}20) \times 10^{-3}$. Here $\Delta^{14}\text{C}$ is the fractional deviation of the $^{14}\text{C}/^{12}\text{C}$ of a ring from the AD 1950 reference standard. It is corrected for fractionation (to the $^{13}\text{C}/^{12}\text{C}$ of average wood) and for decay since the ring grew, so it records the atmospheric ratio at the time of growth (Stuiver & Polach 1977).

Miyake et al. (2012) measured the ^{14}C in individual annual rings of Japanese cedar and found an increase of $\approx 12 \times 10^{-3}$ between AD 774 and 775, ~ 20 times larger than the changes attributed to ordinary solar modulation. They argued that neither a solar flare nor a local supernova was likely to have been responsible. Later measurements of ^{10}Be and ^{36}Cl in polar ice identified the event as an extreme burst of solar energetic particles, at least five times stronger than any recorded with instruments (Mekhaldi et al. 2015). The production of each radionuclide responds differently to the energy of the incoming particles, so the three together acted as a particle spectrometer.

5.5.2. *Supernova debris on Earth*

The Earth itself has collected debris from nearby supernovae. Knie et al. (1999, 2004) found ^{60}Fe in a deep-sea ferromanganese crust, and Wallner et al. (2016) found it in archives from all the major oceans, deposited in two episodes 1.5–3.2 and 6.5–8.7 Myr ago. Supernova-made ^{53}Mn appears in the same layers (Korschinek et al. 2020). Breitschwerdt et al. (2016) attribute the signal to supernovae 90–100 pc away in the moving group that inflated the Local Bubble. The r-process isotope ^{244}Pu ($t_{1/2} = 81$ Myr) is also present, but at a level ~ 100 times below that expected if the r process operated continuously in ordinary supernovae (Wallner et al. 2015, 2021). We conclude that the r process occurs in rare and prolific events, such as neutron-star mergers or a small subset of supernovae.

The meteorites add a constraint on the last such event before the Sun formed. ^{129}I and ^{247}Cm have almost equal half-lives (16.1 and 15.6 Myr), so their ratio barely changes however long the material took to reach the solar nebula. The early Solar System value, $^{129}\text{I}/^{247}\text{Cm} = 438 \pm 184$, is best matched by the moderately neutron-rich ejecta expected from the disks of neutron-star mergers, although nuclear-physics uncertainties limit the conclusion (Côté et al. 2021).

5.5.3. *The birth of the Solar System*

The first evidence that the Solar System formed with live radioactivities came from Lee et al. (1976), who found a ^{26}Mg excess in a CAI of the Allende meteorite that correlated with its Al/Mg ratio. The isochron of the Allende inclusion WA (Fig. 12a; Lee et al. 1977) gives $(^{26}\text{Al}/^{27}\text{Al})_0 = (5.1 \pm 0.6) \times 10^{-5}$ (2σ), and the canonical initial ratio is $(5.23 \pm 0.13) \times 10^{-5}$ (Jacobsen et al. 2008). The Pb–Pb age of CAIs, corrected with their measured $^{238}\text{U}/^{235}\text{U}$, is 4567.30 ± 0.16 Myr (Connelly et al. 2012). This age is the absolute “time zero” of the Solar System, and the chondrules formed over the following ~ 3 Myr.

We have now identified at least a dozen short-lived radionuclides in this way (e.g., ^{10}Be , ^{53}Mn , ^{107}Pd , ^{129}I , ^{182}Hf , ^{244}Pu , ^{247}Cm ; Lugaro et al. 2018). We can compare their initial abundances with the Galactic steady state of Eq. 15. If we divide the γ -ray mass of ^{26}Al by the mass of ^{27}Al in the interstellar gas, we obtain a present Galactic average $^{26}\text{Al}/^{27}\text{Al}$ between 2×10^{-6} and 1.7×10^{-5} , 3–25 times below the CAI value (Lugaro et al. 2018). Even if the interstellar ^{27}Al abundance had grown in proportion to time, which Diehl et al. (2021) treat as an upper limit on the Galactic ratio 4.6 Gyr ago, the CAI value would remain ≈ 2 times higher. The radionuclides with half-lives longer than ~ 5 Myr, in contrast, are roughly consistent with the decayed Galactic background, while ^{26}Al , ^{36}Cl , and ^{41}Ca lie far above it (Lugaro et al. 2018). The Sun’s ^{26}Al therefore requires a source close to the Sun in space and time. The longer-lived radionuclides date the prehistory of the solar material. ^{107}Pd and ^{182}Hf place the last contribution from an AGB star ~ 10 – 30 Myr before the Sun formed,

and ^{129}I and ^{247}Cm place the last r-process event $\sim 100\text{--}200$ Myr before (Lugaro et al. 2014; Côté et al. 2021).

Stars destroy beryllium (Section 2.2), so the ^{10}Be ($t_{1/2} = 1.39$ Myr) recorded by ^{10}B excesses in CAIs is attributed to spallation, as in the cosmic rays of Section 5.4 (neutrino reactions in a low-mass supernova have also been proposed for the lowest values; Lugaro et al. 2018). An Allende CAI formed with $^{10}\text{Be}/^9\text{Be} \approx 9 \times 10^{-4}$ (McKeegan et al. 2000; Chaussidon et al. 2006). This Allende CAI also shows evidence, not yet definitive, for live ^7Be ($t_{1/2} = 53$ d), which could only have been made by irradiation within the solar disk (Chaussidon et al. 2006; Dauphas & Chaussidon 2011). The energetic particles probably came at least partly from the young Sun, since solar-mass stars 1 Myr old are $\sim 10^3$ times more luminous in X-rays than the Sun today (Dauphas & Chaussidon 2011). Among CAIs ^{10}Be does not correlate with ^{26}Al , as expected if irradiation made the ^{10}Be and stars made the ^{26}Al (Lugaro et al. 2018).

The inventory of short-lived radionuclides also constrains the Sun’s birth environment. If the ^{26}Al and ^{60}Fe came from a single supernova, the Sun formed in a cluster of at least $\sim 10^3$ stars, 0.1–0.3 pc from a massive star that exploded (Adams 2010). The key measurement, the initial $^{60}\text{Fe}/^{56}\text{Fe}$, is disputed. Bulk measurements give 10^{-8} , low enough that no nearby supernova is required, while in-situ measurements give $10^{-7}\text{--}10^{-6}$, which require one (Lugaro et al. 2018).

5.5.4. *The growth of planetesimals and planets*

The decay of ^{26}Al was the main heat source of the first planetesimals. With $^{26}\text{Al}/^{27}\text{Al} = 5 \times 10^{-5}$, a rocky body only 5 km across heats its center to 1500 K within $\sim 4 \times 10^5$ yr in conductive models (Dauphas & Chaussidon 2011), but wholesale melting requires a body large enough to insulate its interior. Bodies that accreted before ≈ 1.3 Myr and were large enough to insulate themselves (radius above ≈ 20 km at the time of the CAIs, rising to ≈ 80 km at 1.3 Myr) melted and formed metal cores (Hevey & Sanders 2006). The differentiated meteorites therefore come from bodies that accreted within $\approx 1.5\text{--}2$ Myr of the CAIs, and the chondrites from bodies that accreted at $\approx 2\text{--}4$ Myr, when too little ^{26}Al remained to melt them (Hevey & Sanders 2006; Kruijer et al. 2017). This heating also drove water out of the planetesimals. In models in which rocky planets grow from planetesimals, systems with the Sun’s initial ^{26}Al make planets with water mass fractions of order 1% or less, while systems without ^{26}Al mostly make ocean worlds (Lichtenberg et al. 2019).

The $^{182}\text{Hf}\text{--}^{182}\text{W}$ clock ($t_{1/2} = 8.9$ Myr) dates the formation of the metal cores. Hafnium stays in silicate rock while tungsten follows the metal, so a core that forms while ^{182}Hf is alive keeps the tungsten isotope composition of that moment, and the mantle it leaves behind continues to accumulate radiogenic ^{182}W . The method is an isochron on the scale of a planet, with core formation in place of crystallization. Relative to the Earth’s mantle, the Solar System began with $\epsilon^{182}\text{W} = -3.49$ (in the ϵ units of Section 4.5, measured in CAIs), the chondrites today have -1.91 , and the iron meteorites have -3.45 to -3.18 , close to the initial value (Kruijer et al. 2017). Their cores formed 0.3–2.8 Myr after the CAIs, in parent bodies that had accreted within $\approx 0.1\text{--}1$ Myr (Kruijer et al. 2014, 2017). The Earth’s mantle has $\epsilon^{182}\text{W} \approx +2$ relative to the chondrites, which in the simplest model dates the Earth’s core formation to ~ 30 Myr (Kleine et al. 2002; Yin et al. 2002). Models with prolonged accretion, in which the cores of impactors did not fully equilibrate with the mantle, give ~ 30 to more than 100 Myr (Kleine et al. 2009). The Earth therefore grew over tens of Myr, while Mars reached approximately half its present size within ~ 2 Myr (Dauphas & Pourmand 2011).

Table 4. Radioactive heating of the bulk silicate Earth for $U = 20 \text{ ng g}^{-1}$, $Th = 80 \text{ ng g}^{-1}$, and $K = 280 \text{ } \mu\text{g g}^{-1}$ (McDonough & Sun 1995; Arevalo et al. 2009), the specific heat production of Ruedas (2017), and the NUBASE2020 half-lives. The third column gives the fraction of the decay energy carried off by antineutrinos, which is excluded from the heat.

nuclide	half-life (Gyr)	energy to antineutrinos	heat today (TW)	heat 4.5 Gyr ago (TW)
^{238}U	4.46	7.8%	7.6	15
^{235}U	0.704	4.3%	0.33	27
^{232}Th	14.0	5.2%	8.5	11
^{40}K	1.25	49%	3.9	47
total			20	100

5.5.5. The Earth's internal heat

In Table 4 we list the heat that the long-lived radionuclides ^{238}U , ^{235}U , ^{232}Th , and ^{40}K release today in the bulk silicate Earth (the mantle and crust). We computed these values from the specific heat production of each nuclide (Ruedas 2017) and the abundances $U = 20 \text{ ng g}^{-1}$, $Th = 80 \text{ ng g}^{-1}$, and $K = 280 \text{ } \mu\text{g g}^{-1}$ (McDonough & Sun 1995; Arevalo et al. 2009). The total today is 20 TW, in agreement with the $19.9 \pm 3.0 \text{ TW}$ of McDonough et al. (2020), and ^{238}U and ^{232}Th supply 79% of it. The Earth loses 44–47 TW through its surface (Pollack et al. 1993; Davies & Davies 2010), so radioactivity supplies $\approx 45\%$ of the present heat loss. The remainder is the secular cooling of an interior that still holds heat from the formation of the Earth (KamLAND Collaboration et al. 2011).

When the Earth formed, 4.5 Gyr ago, these nuclides released $\approx 100 \text{ TW}$, five times more than today, and ^{40}K and ^{235}U , whose half-lives are short compared with the age of the Earth, supplied 47% and 27% of it.

The escaping antineutrinos also measure the heat production directly. Detectors record them through inverse beta decay on protons, $\bar{\nu}_e + p \rightarrow n + e^+$, which has a threshold of 1.806 MeV. Only four β decays in the U and Th chains (of ^{234m}Pa , ^{214}Bi , ^{228}Ac , and ^{212}Bi) emit antineutrinos above this threshold. Each chain is in secular equilibrium, so these decays occur at the rates of the long-lived parents, and the detectors measure ^{238}U and ^{232}Th but are blind to ^{40}K and ^{235}U (Agostini et al. 2020). The combined KamLAND and Borexino data gave $20.0^{+8.8}_{-8.6} \text{ TW}$ from ^{238}U and ^{232}Th (KamLAND Collaboration et al. 2011), consistent with the 16 TW of Table 4. The latest analyses disagree, with $10.6^{+5.2}_{-4.2} \text{ TW}$ from KamLAND (Abe et al. 2022) and $31.7^{+14.4}_{-9.2} \text{ TW}$ from Borexino (Agostini et al. 2020). However, Sammon & McDonough (2022) find three times more uranium in the upper crust near Borexino than had been assumed, and with this correction the combined data favor $\approx 20 \text{ TW}$ in total. KamLAND excludes an Earth heated entirely by radioactivity at 4–5 σ (Abe et al. 2022).

6. Isotopes as tracers of fractionation

In Section 4 we assumed that the isotopes of an element travel together, so that a ratio set in a star survives unchanged until we measure it. The assumption is usually excellent, because the isotopes of an element have identical electron shells and therefore almost identical chemistry. The differences in behavior come from the mass of the nucleus (and occasionally from its size or spin), and they separate isotopes by amounts that are small but measurable. We call any process that separates isotopes in this way **isotope fractionation**: without any nuclear reaction, it leaves two reservoirs of the same element (e.g., two molecules, two phases, or the gas that escapes and the gas that remains) with different isotope ratios. As we saw in Fig. 10, for the bulk oxygen of Solar System bodies these effects are orders of magnitude smaller than the nuclear ones. Hydrogen is the exception, because the relative mass difference between D and H is so large that fractionation determines the D/H ratio of all the water in the Solar System, as we saw in Fig. 1. In other elements, too, fractionation can change an isotope ratio by a factor of several or more. This happens at temperatures low enough that the zero-point energy difference exceeds kT (Section 6.2), and in reservoirs that hold a small fraction of an element, such as a rare isotopologue (Section 6.3), the residue of an escaping atmosphere (Section 6.4), or the atmosphere of a star (Section 6.8).

Fractionation is also a source of systematic error for measurements such as the fingerprints of Section 4, since fractionation biases double-isotope ratios, disk masses measured from C^{18}O , and ^{15}N measured from HCN. In this section we describe the physics of mass-dependent fractionation, and then the chemical, photochemical, and escape processes that separate isotopes in astrophysical settings. We then discuss the origin of the Earth’s water and the isotopes now being measured in brown dwarfs, giant planets, and an interstellar comet, and we close with the separation of isotopes in stars.

6.1. Mass-dependent fractionation

We begin with the physics of equilibrium isotope fractionation, which is a quantum effect (Bigeleisen & Mayer 1947). In classical statistical mechanics, chemical equilibrium cannot separate the isotopes of an element. The positions of the atoms are distributed according to the Boltzmann factor of the potential energy, which in the Born–Oppenheimer approximation is the same for every isotope. Their momenta follow a Maxwell distribution whose normalization depends on the mass of the atom but not on the molecule or phase that holds it. The mass therefore cancels from any exchange equilibrium. Classical physics can, however, separate isotopes in a gravitational field or in one-way processes such as evaporation, as we will see below.

Quantum mechanics changes this through the vibrational levels. The force constant k_f of a chemical bond does not depend on which isotope is present, so the vibrational frequency scales as $\omega = (k_f/\mu)^{1/2}$ and the zero-point energy $\hbar\omega/2$ drops when a heavier isotope is substituted. The drop is larger for a stiffer bond, so at equilibrium the heavy isotope concentrates where it is most tightly bound. For molecular hydrogen, for example, the reduced mass rises from 1/2 to 2/3 of the atomic mass unit when D replaces H (the same scaling as Eq. 7), and the zero-point energies of H_2 and HD differ by 287 cm^{-1} , or 412 K.

At high temperature most of the zero-point energy difference is offset by the more closely spaced levels of the heavier isotope. What survives is the leading quantum correction to the free energy of

each oscillator, $(\hbar\omega)^2/24kT = \hbar^2 k_f/24\mu kT$, and the isotopes behave almost classically, much as the Planck function approaches the Rayleigh–Jeans law at low frequency. If we divide this correction by kT and sum over the bonds that hold an atom, we obtain the fractionation factor between two substances A and B (Bigeleisen & Mayer 1947; Dauphas et al. 2012):

$$\ln \alpha_{AB} \approx \left(\frac{1}{m} - \frac{1}{m^*} \right) \frac{\hbar^2 (\langle F \rangle_A - \langle F \rangle_B)}{8k^2 T^2}, \quad (16)$$

where α_{AB} is the ratio of the (heavy/light) isotope ratios of A and B, m and m^* are the masses of the light and heavy isotopes, and $\langle F \rangle$ is the mean force constant of the atom in each substance, i.e., its restoring force per unit displacement averaged over the three directions in which it can move (for simple bonds, one third of the sum of their force constants). In the opposite limit, where kT is small compared with the vibrational quantum, α approaches the Boltzmann factor $e^{\Delta E/kT}$, where ΔE is the zero-point energy released by the exchange (232 K in Eq. 18). This factor governs the cold-cloud chemistry of Section 6.2.

Eq. 16 has several important consequences. First, equilibrium fractionation falls rapidly with temperature. Between quartz and water, for example, the $^{18}\text{O}/^{16}\text{O}$ fractionation is 36×10^{-3} at 300 K but 0.9×10^{-3} at 1000 K (Sharp et al. 2016). Equilibrium processes in the hot inner solar nebula therefore produce small mass-dependent effects. (Real calibrations include lower-order terms, so this fractionation falls by a factor of 40 between the two temperatures, whereas a pure $1/T^2$ law gives 11.) Second, the mass factor is 0.5 for H and D but 0.0069 for ^{16}O and ^{18}O , so hydrogen fractionates ≈ 70 times more than oxygen in the same bond. Third, for an element with three isotopes the ratio of the two fractionations depends in the high-temperature limit only on the masses, independent of temperature and chemistry. For oxygen we find:

$$\frac{\ln \alpha_{17}}{\ln \alpha_{18}} = \frac{1/m_{16} - 1/m_{17}}{1/m_{16} - 1/m_{18}} = 0.5305, \quad (17)$$

while kinetic processes such as evaporation and diffusion, whose rates scale as $m^{-1/2}$, give an exponent set by the logarithms of the masses, 0.516 (Young et al. 2002; Schauble & Young 2021). Terrestrial rocks and waters fall between the two, with exponents of 0.525 and 0.528, and the conventional slope of the terrestrial fractionation line (Section 3.4.3) is 0.52.

The temperature dependence of Eq. 16 also makes fractionation a thermometer, an application that goes back to Urey (1947). For calcite and water, $\ln \alpha$ is 33.7×10^{-3} at 0 °C and 28.1×10^{-3} at 25 °C (Tarutani et al. 1969), so a measurement to 10^{-4} fixes the temperature at which the calcite grew to ≈ 0.5 K. This classical method needs the isotopic composition of the water, which is unknown for ancient or extraterrestrial fluids. The clumping of ^{13}C and ^{18}O into the same carbonate ion removes this requirement, since the clumping is an equilibrium within a single mineral and follows the same $1/T^2$ law (Ghosh et al. 2006). Clumped isotopes show that water altered the parent bodies of the CM chondrites at 20–71 °C (Guo & Eiler 2007).

Many fractionating processes are one-way, or **kinetic**, among them evaporation into a vacuum, diffusion, photosynthesis (Section 5.1), and atmospheric escape (Section 6.4). In a kinetic process the isotopes differ in how fast they move or react. At a given temperature the thermal speed scales as $m^{-1/2}$, so the lighter isotope diffuses and evaporates faster, and a bond to a lighter isotope breaks more easily because its zero-point energy sits higher in the potential well. The ratio of thermal

speeds does not depend on temperature, so a fractionation set by thermal speeds does not fade in hot gas. When Davis et al. (1990) evaporated molten forsterite (Mg_2SiO_4) into a vacuum until 12% of its mass remained, the residue was enriched in the heavy isotopes of Mg by $\approx 3 \times 10^{-2}$ per mass unit. Some CAIs, which formed in the hot inner nebula, carry mass-dependent fractionations of Mg, Si, and Cr of up to a few percent per mass unit, and Davis et al. (1990) conclude that these inclusions must have been at least partially molten if evaporation produced the fractionation.

Any mass-dependent process moves a sample parallel to the TFL, so a $\Delta^{17}\text{O}$ (Section 3.4.3) well above the spread produced by the different mass-dependent laws (of order 10^{-4} ; Young et al. 2002) marks either a nuclear effect or a mass-independent one. The mass is not the only property of a nucleus that enters chemistry. In the heaviest elements the field shift of Section 3.1 has a chemical counterpart. A larger nucleus binds the s electrons that penetrate it slightly less tightly. At equilibrium the larger (and heavier) isotopes therefore concentrate in species with little electron density at the nucleus, such as Hg^{2+} , which has no $6s$ electrons, relative to neutral Hg. For mercury and thallium this **nuclear-volume** fractionation reaches 3×10^{-3} at room temperature, larger than the mass-dependent fractionation of $(0.5\text{--}1) \times 10^{-3}$ (Schauble 2007). The nuclear spin of odd- A isotopes also changes the rates of some radical reactions (the magnetic isotope effect, seen in mercury; Dauphas & Schauble 2016). Neither effect follows the mass law of Eq. 17.

6.2. Chemical fractionation in cold interstellar gas

In cold molecular clouds, at ≈ 10 K, the Boltzmann factor $e^{\Delta E/kT}$ of Section 6.1 turns a modest zero-point energy difference into an enormous fractionation.

6.2.1. Deuterium

The key reaction is (Gerlich et al. 2002; Parise et al. 2009):



At equilibrium the ratio $\text{H}_2\text{D}^+/\text{H}_3^+$ would exceed HD/H_2 by a few times 10^8 at 10 K (the Boltzmann factor $e^{232/10} \approx 10^{10}$, reduced by spin and partition-function factors; Gerlich et al. 2002). In Fig. 15a we compare the Boltzmann factors of three exchange reactions. At 10 K the factor is 10^{10} for H_2D^+ but 32 and 17 for the carbon and nitrogen exchanges, and for deuterium it falls by nine orders of magnitude between 10 and 100 K. In practice kinetics limits the enhancement. The back reaction with ortho- H_2 , which carries ≈ 170 K of rotational energy, is fast unless the ortho-to-para ratio is very low, and CO and O destroy H_2D^+ until they freeze out onto grains (Ceccarelli et al. 2014).

In the cold, CO-depleted centers of prestellar cores the deuterium is then passed from H_2D^+ to other molecules. In a survey of 31 starless cores, Crapsi et al. (2005) found $\text{N}_2\text{D}^+/\text{N}_2\text{H}^+$ from 0.03 to 0.44, correlated with the depletion of CO, and in L1544 the ratio is 0.23, $\sim 10^4$ times the elemental D/H. In Fig. 15b we plot $\text{N}_2\text{D}^+/\text{N}_2\text{H}^+$ vs. the CO depletion factor for the 14 cores in which both were measured. The grain-surface chemistry that follows produces multiply deuterated molecules, such as CD_3OH with 1.4% of the abundance of CH_3OH in the protostar IRAS 16293-2422 (Parise et al. 2004).

Primitive meteorites preserve a diluted record of low-temperature chemistry of this kind, made in the parent cloud or in the cold outer disk (Busemann et al. 2006; Aléon 2010). The insoluble

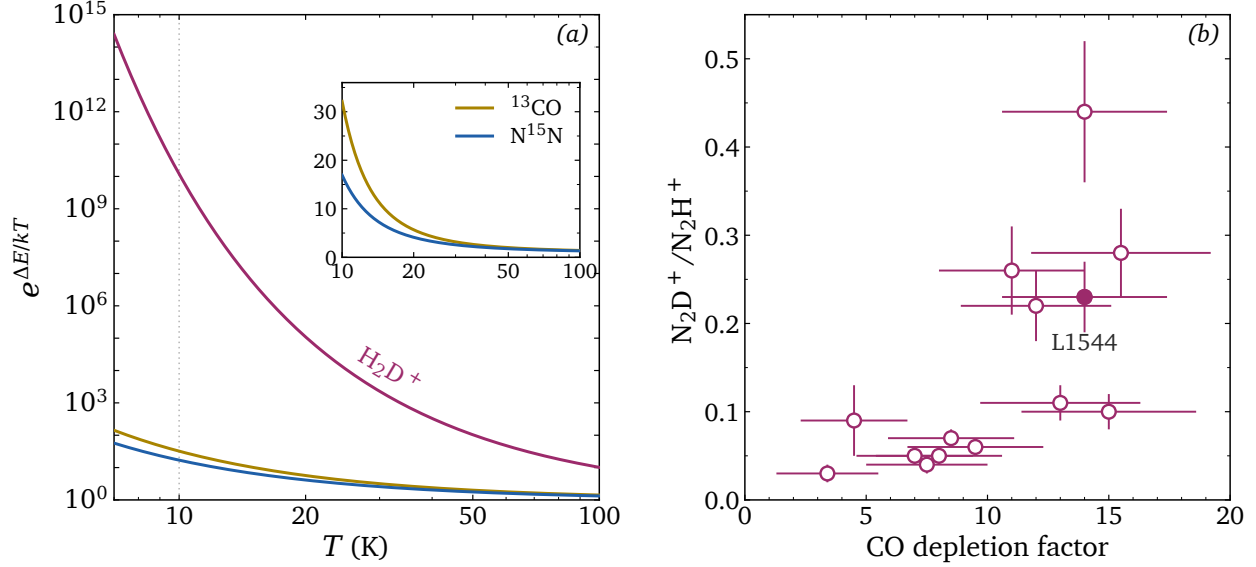


Figure 15: (a) Boltzmann factor $e^{\Delta E/kT}$ vs. temperature for $\text{H}_3^+ + \text{HD} \rightarrow \text{H}_2\text{D}^+ + \text{H}_2$ ($\Delta E/k = 232$ K), $^{13}\text{C}^+ + \text{CO} \rightarrow ^{12}\text{C}^+ + ^{13}\text{CO}$ (34.7 K), and $^{15}\text{N}^+ + \text{N}_2 \rightarrow ^{14}\text{N}^+ + \text{N}^{15}\text{N}$ (28.3 K), each labeled by the enriched molecule; the inset shows the carbon and nitrogen exchanges on a linear scale, and the dotted line marks 10 K. (b) $\text{N}_2\text{D}^+/\text{N}_2\text{H}^+$ vs. the CO depletion factor (canonical over observed CO abundance along the line of sight) toward the N_2D^+ peaks of 14 starless cores (Crapsi et al. 2005), with 1σ errors; the filled circle is L1544.

organic matter of the CR chondrites has bulk $\text{D}/\text{H} \approx (5.5\text{--}6.4) \times 10^{-4}$ and $^{14}\text{N}/^{15}\text{N} \approx 221\text{--}237$ (Nomura et al. 2023). It also contains micron-sized hotspots, a few percent of the material, with D/H up to 3×10^{-3} (Busemann et al. 2006; Aléon 2010), 20 times the ocean value, and up to 6×10^{-3} in later measurements (Nomura et al. 2023). Other hotspots reach $^{14}\text{N}/^{15}\text{N} \approx 65$, more than twice as ^{15}N -rich as cometary CN (Aléon 2010). The most D-rich and the most ^{15}N -rich hotspots are different grains (Aléon 2010), which suggests two separate processes, deuteration by the chemistry of Eq. 18 and a separate nitrogen fractionation. The self-shielding of N_2 (Section 6.3) is one candidate for the second, although Aléon (2010) prefers a chemical origin in the disk.

6.2.2. Carbon and nitrogen

The heavier elements fractionate much less, because their exchange reactions release less energy. The reaction $^{13}\text{C}^+ + ^{12}\text{CO} \rightleftharpoons ^{12}\text{C}^+ + ^{13}\text{CO} + 35$ K concentrates ^{13}C into CO (Watson et al. 1976), and the other carbon-bearing molecules become correspondingly ^{13}C -poor ($\text{HCN}/\text{H}^{13}\text{CN} = 114$ in a model dense core, compared with an elemental ratio of 68; Roueff et al. 2015). This bias propagates into the double-isotope method of Section 4.4, and a $^{14}\text{N}/^{15}\text{N}$ ratio that we measure from HCN can be off, typically by up to a factor of 2 (Colzi et al. 2020). Comets have $^{14}\text{N}/^{15}\text{N} \approx 148$ (CN in 18 comets; Manfroid et al. 2009), a factor of three more ^{15}N -rich than the protosolar value, and this enrichment is difficult to explain with chemistry. Gas-phase chemistry with realistic reaction barriers leaves most nitrogen-bearing molecules close to the elemental ratio (Roueff et al. 2015). The leading explanation is the isotope-selective photodissociation of N_2 (Heays et al. 2014), to which we now turn.

6.3. Isotope-selective photodissociation

Isotope-selective photodissociation separates the isotopologues of a molecule according to their abundances, through the saturation of their absorption lines. It therefore does not follow the mass law of Eq. 17, and it moves oxygen along a line of slope ≈ 1 in the three-isotope plot.

6.3.1. Self-shielding

Consider carbon monoxide, which is photodissociated only by the absorption of ultraviolet photons in discrete lines between 912 and 1118 Å (Visser et al. 2009). Once the column density of ^{12}CO exceeds $\sim 10^{14}\text{--}10^{15} \text{ cm}^{-2}$ these lines saturate and ^{12}CO shields itself. This is the same saturation that prevents us from measuring isotope ratios with ^{12}CO lines (Section 3.3), acting here on the molecules themselves. The lines of the rare isotopologues are shifted by $\gtrsim 0.3 \text{ Å}$ ($\approx 100 \text{ km s}^{-1}$), well outside the saturated ^{12}CO lines, so the rare isotopologues continue to be photodissociated deeper into a cloud (Bally & Langer 1982). Toward X Per, for example, $N(^{12}\text{CO})/N(\text{C}^{18}\text{O}) = 3000 \pm 600$, compared with an elemental $^{16}\text{O}/^{18}\text{O}$ of 557. C^{17}O and C^{18}O are both depleted by a factor of ≈ 5 , while ^{13}CO is not, because carbon isotope exchange replenishes it (Visser et al. 2009). Isotope-selective photodissociation also biases the gas masses of protoplanetary disks that we measure from C^{18}O , which can be underestimated by factors of 1.8 to 67 in T Tauri disk models (Miotello et al. 2014). Note that the depth at which each isotopologue shields itself is set by its abundance. Where ^{12}CO is self-shielded while C^{17}O and C^{18}O are both far too rare to shield themselves, the two rare isotopologues are dissociated at almost the unattenuated rate. Self-shielding therefore removes ^{17}O and ^{18}O from CO in similar proportions, i.e., along a line of slope ≈ 1 in the three-isotope plot (Visser et al. 2009). An admixture of pure ^{16}O would move material along the same line, so a slope-1 line by itself cannot distinguish a photochemical effect from a nuclear one.

6.3.2. The oxygen isotopes of the Solar System

Self-shielding connects to one of the classic problems of meteoritics. Clayton et al. (1973) discovered that the CAIs of carbonaceous chondrites are enriched in ^{16}O , with ^{17}O and ^{18}O depleted by similar amounts, and interpreted the anomaly as nuclear, an admixture of almost pure ^{16}O carried by presolar dust. In Fig. 10a we see that the most ^{16}O -rich CAIs lie far from the TFL, close to a line of slope 1 through the Sun, while the planets and bulk chondrites cluster near the TFL. CAI minerals commonly reach $\delta^{18}\text{O} \approx -40 \times 10^{-3}$, and a few CAIs reach -66×10^{-3} (Gounelle et al. 2009). Thieme & Heidenreich (1983) then showed in the laboratory that the formation of ozone produces a mass-independent fractionation, with a slope close to 1 like that of the CAIs, so chemistry by itself can move oxygen along a slope-1 line. The leading explanation invokes the symmetry of the molecule. Symmetry restricts how $^{16}\text{O}_3$ can share the energy of its formation among its rotational and vibrational states, so the symmetric molecule is stabilized less often than the asymmetric molecules that contain one ^{17}O or one ^{18}O . The two heavy isotopes are therefore enriched by similar amounts (Gao & Marcus 2001). For decades it was unclear which composition we should regard as normal. As late as 1993 Clayton wrote that the solar composition “is not known with enough precision to be useful” (Clayton 1993). The *Genesis* mission settled which composition is normal by capturing the solar wind and returning it to Earth. The Sun has $\delta^{17}\text{O} \approx \delta^{18}\text{O} \approx -59 \times 10^{-3}$ and $\Delta^{17}\text{O} \approx -28 \times 10^{-3}$, like the CAIs. The anomalous bodies are therefore the Earth, Moon, Mars, and the bulk meteorites, all enriched in ^{17}O and ^{18}O by $\approx 7\%$ (McKeegan et al. 2011). The photospheric

CO lines give $\delta^{18}\text{O} = (-50 \pm 11) \times 10^{-3}$, in agreement (Lyons et al. 2018).

Self-shielding in O_2 was suggested for the CAI anomaly a decade after its discovery (Kitamura & Shimizu 1983; Navon & Wasserburg 1985). The favored explanation today is CO self-shielding, which Clayton (2002) proposed for the inner solar nebula and later work placed in the parent molecular cloud or at the surface of the nebula (Yurimoto & Kuramoto 2004; Lyons & Young 2005). Ultraviolet light preferentially dissociates C^{17}O and C^{18}O , and the liberated ^{17}O and ^{18}O end up in water ice. The water later reacts with rock and moves it up the slope-1 line, away from the solar composition. The ^{16}O -poor end member has been found in magnetite–sulfide intergrowths formed by the oxidation of metal with water, at $\delta^{17}\text{O} \approx \delta^{18}\text{O} \approx +180 \times 10^{-3}$ (Sakamoto et al. 2007). Several lines of evidence from outside the three-isotope plot support this photochemical reading, including the laboratory demonstration of mass-independent chemistry, the composition of the Sun (McKeegan et al. 2011), the ^{16}O -poor water (Sakamoto et al. 2007), and the absence of correlated nuclear anomalies in other elements (Lyons & Young 2005).

6.3.3. Planets, comets, and disks

Once formed, each planetary body carries its own $\Delta^{17}\text{O}$, as Fig. 10c shows (e.g., $+0.32 \times 10^{-3}$ for Mars; Greenwood et al. 2017), and the value survives melting and differentiation because those processes are mass dependent. The Earth and Moon differ by only -1 ± 5 parts per million (Young et al. 2016), a tight constraint on the giant impact that formed the Moon. We find the same pattern in nitrogen. The protosolar $^{14}\text{N}/^{15}\text{N}$ is 441 ± 5 (Marty et al. 2011), the inner Solar System (planets and bulk meteorites) has 272 ± 15 , and cometary ices have 147 ± 11 . The ^{15}N enrichment increases with distance from the Sun (Füri & Marty 2015), as expected if N_2 self-shielding in the outer disk produced ^{15}N -rich ices. We can now see these processes at work in protoplanetary disks. In the disk of TW Hya, $\text{HCN}/\text{HC}^{15}\text{N}$ is 121 ± 11 at 20 au, close to the cometary value, and rises to 339 ± 28 at 45 au, while CN in the same disk gives 323 ± 30 . Two nitrogen reservoirs coexist where comets form, as N_2 self-shielding would produce, although the models do not yet reproduce the values (Hily-Blant et al. 2019). In the outbursting young star V883 Ori, the disk’s water has sublimated out to ≈ 80 au, and $\text{HDO}/\text{H}_2\text{O} = (2.26 \pm 0.63) \times 10^{-3}$, i.e., $\text{D}/\text{H} = (1.13 \pm 0.32) \times 10^{-3}$. This value is comparable to those of protostellar envelopes and comets and is 7 times that of the Earth’s oceans (a 3.1σ excess). The disk has therefore inherited its water from the cloud (Tobin et al. 2023).

6.4. Atmospheric escape

Hydrogen escapes from planetary atmospheres, and deuterium, being heavier, escapes less efficiently. Suppose that an atmosphere loses hydrogen and that deuterium is lost with a relative efficiency f , i.e., the escape flux of D is f (D/H) times that of H. Then $d \ln N_{\text{D}} = f d \ln N_{\text{H}}$, and if we integrate we obtain the **Rayleigh distillation** law (Cangi et al. 2020):

$$\frac{(\text{D}/\text{H})_{\text{now}}}{(\text{D}/\text{H})_0} = \left(\frac{N_{\text{H},0}}{N_{\text{H},\text{now}}} \right)^{1-f}, \quad (19)$$

so the D/H ratio of the remaining water records how much has been lost. The *Pioneer Venus* mass spectrometer measured $\text{D}/\text{H} = (1.6 \pm 0.2) \times 10^{-2}$ in the atmosphere of Venus, a hundredfold enrichment relative to the Earth’s oceans (Donahue et al. 1982). Mars has $\text{D}/\text{H} \approx 6$ times terrestrial

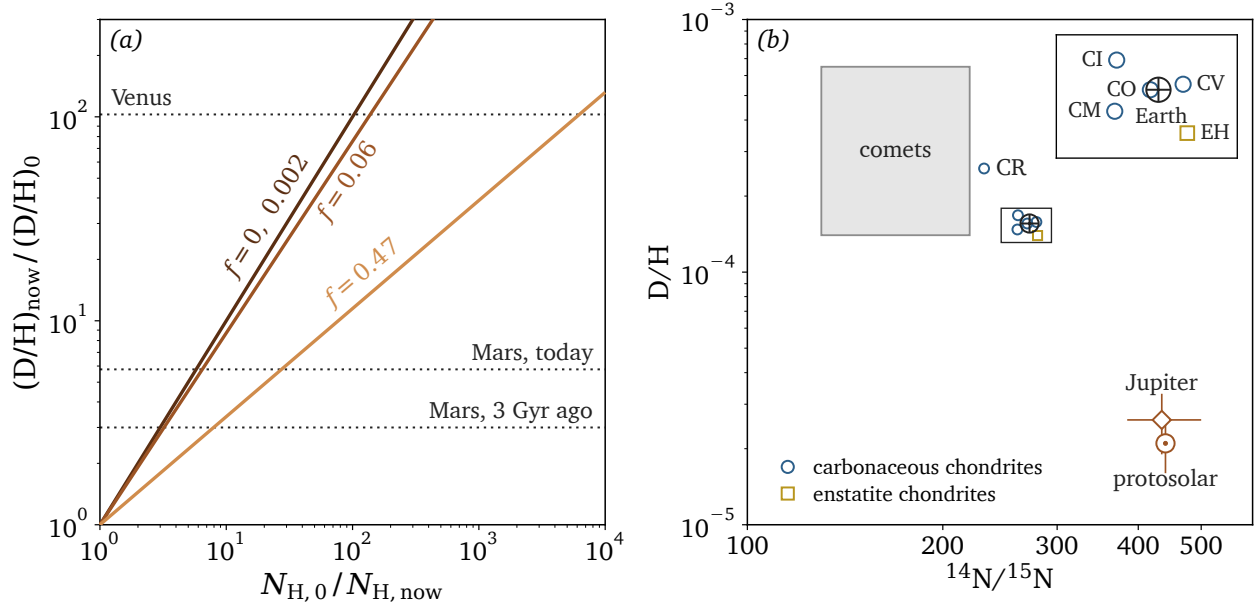


Figure 16: (a) $(D/H)_{\text{now}} / (D/H)_0$ vs. $N_{H,0} / N_{H,\text{now}}$ from Eq. 19 for $f = 0, 0.002, 0.06$, and 0.47 ; dotted lines mark the measured D/H of Venus and Mars relative to the Earth’s oceans, taken as $(D/H)_0$. (b) D/H vs. $^{14}\text{N}/^{15}\text{N}$ for protosolar gas, Jupiter (1σ errors), the Earth (oceans and air), carbonaceous-chondrite group means (Alexander et al. 2018), and enstatite chondrites (Piani et al. 2020); the box spans the D/H of cometary water and the $^{14}\text{N}/^{15}\text{N}$ of cometary CN, measured in different comets, and the inset enlarges the outlined region 3.6 times.

today (Owen et al. 1988; Villanueva et al. 2015), and 3 times terrestrial in 3 Gyr old clays at Gale crater (Mahaffy et al. 2015).

The efficiency f depends on how the atoms escape. In thermal (Jeans) escape the escaping atoms come from the tail of the Maxwell distribution above the escape speed. The fraction of atoms in that tail falls as $e^{-\lambda}$, where the escape parameter $\lambda = GMm/kTr$ is the ratio of the gravitational binding energy of an atom at the exobase (radius r) to kT . A deuterium atom has twice the mass, and therefore twice the λ , of a hydrogen atom, so its escape is suppressed by a further factor $\sim e^{-\lambda_H}$. Above the homopause, where molecular diffusion dominates over turbulent mixing, each species also follows its own scale height, so the gas at the exobase is already depleted in the heavier isotopes (as *MAVEN* found for argon on Mars; Jakosky et al. 2017). For Mars, Cangi et al. (2020) find $f = 0.002$ for thermal escape and $f = 0.06$ once non-thermal processes are included, which give atoms energies well above the escape energy and discriminate less between the isotopes. In a vigorous hydrodynamic outflow the escaping hydrogen drags heavier species with it (Zahnle & Kasting 1986), and f approaches unity. The D/H ratio therefore records mainly the slower, later phases of escape.

For Mars, with these values of f , Eq. 19 implies that the planet has lost at least 66–122 m of water averaged over its surface (Cangi et al. 2020). The argon isotopes independently show that 66% of the Martian atmospheric argon has been lost to space (Jakosky et al. 2017). For Venus the result depends strongly on f . A hundredfold enrichment requires the water inventory to have shrunk by a factor of 100 if $f = 0$, but by a factor of ~ 6000 if $f = 0.47$, the value for escape solely by collisional ejection (Gurwell & Yung 1993). In Fig. 16a the Venus line crosses the $f = 0$ curve at

a hundredfold loss and the $f = 0.47$ curve near 6000, while for small f the Martian values need a loss of a factor of 3–6. D/H by itself cannot fix the initial water inventory of Venus. Rayleigh distillation also assumes that no water is added, and the Venus D/H can instead be read as a steady state in which water is continuously supplied and lost (Grinspoon 1993).

6.5. The origin of the Earth’s water

Isotope ratios provide most of what we know about where the Earth’s water came from. We now return to Fig. 1, which collects D/H across the environments discussed in this memo. The protosolar D/H of $(2.1 \pm 0.5) \times 10^{-5}$, somewhat below the primordial value, comes from the solar wind by an indirect route. The Sun burned its deuterium into ^3He before reaching the main sequence, so the $^3\text{He}/^4\text{He}$ of the solar wind measures the protosolar $(\text{D} + ^3\text{He})/\text{H}$. Subtracting the protosolar $^3\text{He}/\text{H}$ measured in Jupiter and in the “planetary” noble gases of meteorites gives D/H (Geiss & Reeves 1972; Geiss & Gloeckler 1998). The *Galileo* probe measured a consistent value directly in the atmosphere of Jupiter (Mahaffy et al. 1998). The Earth’s oceans have $\text{D}/\text{H} = 155.76 \times 10^{-6}$ (Hagemann et al. 1970), 7 times more, so they cannot be mostly nebular gas captured by the young Earth unless their D/H was later raised by escape (Genda & Ikoma 2008). Deep-mantle lavas with low D/H do suggest that some nebular hydrogen survives in the Earth’s interior (Hallis et al. 2015). Most of the water probably arrived in solids enriched in deuterium, as water is when it forms or equilibrates at low temperature (Section 6.2). Uranus and Neptune show the same effect on a larger scale, with twice the protosolar D/H, presumably because they accreted large amounts of deuterium-rich ice (Feuchtgruber et al. 2013). The two candidate sources are the asteroids, which we sample through the carbonaceous chondrites, and the comets. The carbonaceous chondrites have bulk D/H between 120 and 230 ppm (excluding the CR group), with a mean of 156 ppm (Jacquet & Robert 2013), the same as the oceans. Comets from the Oort cloud average twice the ocean value (Hartogh et al. 2011). The Jupiter-family comets range from ocean-like (103P/Hartley 2, Hartogh et al. 2011; 46P/Wirtanen, Lis et al. 2019) to three times the ocean value (67P; Altwegg et al. 2015; Müller et al. 2022). Lis et al. (2019) found that the comets with ocean-like D/H are hyperactive (an active fraction above 0.5), probably because much of their water sublimates from icy grains in the coma.

Nitrogen gives an independent test of the cometary contribution, since cometary ices are far richer in ^{15}N than the Earth (Section 6.3). The Earth’s nitrogen matches the chondrites and lies far from the comets. If comets had delivered most of the Earth’s volatiles, the Earth’s nitrogen would be much richer in ^{15}N than it is. In Fig. 16b the Earth sits among the chondrites in both ratios, while the comets overlap it in D/H only.

Taken together, the hydrogen and nitrogen isotopes exclude comets as the main source of the Earth’s water. The carbonaceous chondrites, from the outer part of the protoplanetary disk, match both ratios, but so do the enstatite chondrites, which are the closest match to the Earth’s nucleosynthetic makeup and may carry enough hydrogen to supply several oceans (Piani et al. 2020). The nucleosynthetic anomalies of Section 4.5 show that the Earth is made mostly of non-carbonaceous material, with a carbonaceous fraction that Warren (2011) estimated very roughly at 24%. This fraction is consistent with carbonaceous water but does not require it. Comets may have supplied a significant part of the Earth’s noble gases, since $22 \pm 5\%$ of the atmospheric xenon is cometary (Marty et al. 2017). The ocean-like comets show that D/H by itself cannot exclude a cometary source. The D/H argument

also assumes that the D/H of the oceans has not changed since the water arrived, and the atmospheres of Venus and Mars (Section 6.4) show how large such changes can be. Exchange between H₂ and water at low temperature also raises the D/H of the water, so D/H is a degenerate tracer of the source (Young et al. 2023).

6.6. Isotopes in brown dwarfs and giant planets

We can now measure isotope ratios in the atmospheres of brown dwarfs and giant planets, where they record how these objects formed and where the fingerprint and fractionation roles appear together. D/H tests whether an object has burned its deuterium, while ¹⁴N/¹⁵N and ¹²C/¹³C test whether the object formed from gas or accreted ices, which carry the fractionation of Section 6.2. *JWST* spectra of the Y dwarf WISE J1828 give ¹⁴N/¹⁵N = 670⁺³⁹⁰₋₂₁₁ from ¹⁵NH₃, consistent with the protosolar value and inconsistent with cometary ice, as expected for an object that formed like a star (Barrado et al. 2023). CH₃D in the cold brown dwarf WISE 0855–0714 gives D/H ≈ (1.2–1.6) × 10^{−5} (Rowland et al. 2024), close to protosolar. This brown dwarf has not burned its deuterium, which places its mass below ≈ 12 Jupiter masses (Rowland et al. 2024). The brown dwarfs and wide companions measured so far have ¹²C/¹³C close to the interstellar value, e.g., 81⁺²⁸₋₁₉ and 79⁺²⁰₋₁₄ in TWA 28 and 2MASS J0856 (González Picos et al. 2024). The M7.5 companion HIP 55507 B has ¹²C/¹³C = 98⁺²⁸₋₂₂, and H₂¹⁶O/H₂¹⁸O = 240⁺¹⁴⁵₋₈₀ (Xuan et al. 2024). In contrast, the young super-Jupiter TYC 8998-760-1 b has ¹²C/¹³C = 31⁺¹⁷₋₁₀ (90% confidence), 2.5σ below the local interstellar value of ≈ 68, which Zhang et al. (2021b) interpret as the accretion of ¹³C-rich ices beyond the CO snowline.

6.7. An interstellar comet

The comet 3I/ATLAS, discovered in 2025, is the third interstellar object found in the Solar System, after 1I/ʻOumuamua and 2I/Borisov (Seligman et al. 2025), and the first in which isotope ratios have been measured (2I/Borisov was too faint; Opitom et al. 2021). Its water gives the first D/H of a comet that formed around another star. *JWST* spectra of the coma give D/H = (0.98 ± 0.06)% in water and ¹²C/¹³C of 141–191 in CO₂ and 123–172 in CO (Cordiner et al. 2026). ALMA gives a consistent lower limit, D/H > 6.6 × 10^{−3} (Salazar Manzano et al. 2026), and the CN band gives ¹²C/¹³C = 151⁺¹¹⁰₋₄₄ and ¹⁴N/¹⁵N = 363⁺⁶³³₋₁₅₃ (3σ; Opitom et al. 2026). In Fig. 1 the D/H of 3I/ATLAS lies far above that of every Solar System comet (34 times their mean of 0.029%; Cordiner et al. 2026) and 63 times above that of the oceans. The comet carries both a fractionation signature and a nucleosynthetic fingerprint. Through the H₂D⁺ chemistry of Eq. 18, a water D/H near 1% requires that most of the ice formed below ≈ 30 K (Cordiner et al. 2026; Salazar Manzano et al. 2026). The high ¹²C/¹³C, read as the enrichment gauge of Section 2.3, places the birth of the comet in gas less enriched in ¹³C than the local interstellar medium. In chemical evolution models of the nearby disk, the interstellar gas last had such a high ratio 11–12 Gyr ago (Cordiner et al. 2026). The age depends on these models and on where the comet formed. Even the outermost disk in Fig. 8 has ¹²C/¹³C ≈ 80–120 today, but enrichment there is slower, and an origin in the outer disk would shorten the age to ~ 10–11 Gyr (Cordiner et al. 2026). The ¹⁴N/¹⁵N, close to the solar value and far above the ≈ 150 of Solar System comets, suggests that the self-shielding of N₂ (Section 6.3) was weak where the ices of 3I/ATLAS formed (Opitom et al. 2026).

6.8. Isotope separation in stars

In the quiet atmospheres of some stars, and in the degenerate interiors of white dwarfs, gravity and radiative forces separate isotopes without any chemistry.

6.8.1. Chemically peculiar stars

The clearest cases of isotope separation in stars are found among the chemically peculiar B and A stars. The HgMn stars are slowly rotating late B stars whose atmospheres show mercury overabundant by factors of 6×10^3 to 4×10^5 (White et al. 1976). Recall from Section 3.1 that the isotopic components of Hg II 3984 Å are separated by a large field shift, spanning 0.302 Å from ^{196}Hg to ^{204}Hg . White et al. (1976) found line centroids distributed uniformly from the terrestrial mixture to nearly pure ^{204}Hg among 30 stars. In χ Lupi the mercury is 99% ^{204}Hg (Hubrig et al. 1999), and platinum shows the same behavior, with only ^{198}Pt present in HR 1800. The Ca II infrared triplet of the HgMn star HD 175640 is consistent with pure ^{48}Ca , which makes up 0.187% of terrestrial calcium (Castelli & Hubrig 2004). In magnetic Ap stars the heavy calcium isotopes are concentrated in the outer atmospheric layers when the field is below 4–5 kG, and the separation disappears above 6–7 kG (Ryabchikova et al. 2008). The helium isotopes can also be separated, and in the star 3 Cen A the ratio $^3\text{He}/(^3\text{He}+^4\text{He})$ reaches 0.84 (Sargent & Jugaku 1961). Diffusion in these quiet, radiative atmospheres can separate the elements in $\sim 10^4$ yr, much less than the 10^8 yr main-sequence lifetime (Michaud 1970). Mercury accumulates at the depth where gravity and the radiative force balance to within a few percent, and at this depth the small mass difference between the mercury isotopes becomes decisive (Michaud et al. 1974).

The radiative forces on the isotopes also differ, because their lines are shifted relative to one another and absorb different parts of the stellar flux. In **light-induced drift** (Atutov & Shalagin 1988), an isotope whose line sits on the red side of the blended line, where the flux rises, absorbs mainly photons that are Doppler-shifted into resonance by motion toward the source. These excited atoms collide more often than ground-state atoms, which slows their inward motion, while the unexcited atoms moving outward are impeded less, so the isotope drifts along the flux and rises. In each of the cases above the enriched isotope is the one whose lines lie to the red (heavy Hg, heavy Ca, and light He), as light-induced drift predicts (Ryabchikova et al. 2008). However, the importance of the drift for mercury is disputed (LeBlanc & Michaud 1993). These stars show that an extreme isotope ratio in a stellar spectrum is not necessarily a nucleosynthetic signature.

6.8.2. White dwarfs

In white dwarfs isotope separation becomes a heat source. The nucleus ^{22}Ne , the helium-burning product of ^{14}N (Section 2.2), has two more neutrons than a nucleus of the same charge with $A = 2Z$. In the degenerate interior, where an electric field supports the ions against gravity, each ^{22}Ne nucleus feels a net downward force of $\approx 2m_p g$ and sinks (Bildsten & Hall 2001). *Gaia* showed that $\sim 6\%$ of white dwarfs of $1.08\text{--}1.23 M_\odot$ need an extra cooling delay of ~ 8 Gyr (Cheng et al. 2019). The separation of ^{22}Ne during crystallization, in which ^{22}Ne -poor crystals float up and ^{22}Ne -rich liquid sinks, can supply the energy (Blouin et al. 2021; Bédard et al. 2024). A single nuclide here is a fingerprint of helium burning, subject to mass separation, and a heat source that distorts a clock (the white-dwarf cooling age).

7. Summary

We have organized this memo around three roles that an isotope ratio can play, and we close with the lessons that cut across them. The first lesson concerns measurement. Which isotope ratios we know is determined by spectroscopic physics as much as by astrophysical interest (Section 3), and outside the chemically peculiar stars the isotopes of the elements heavier than the iron peak have been measured in only a handful of stars. The isotope ratios that do appear in stellar spectra are often encoded in subtle line asymmetries, and as we saw with ${}^6\text{Li}$ in halo stars and the barium isotopes in HD 140283, such ratios are only as reliable as the model of the line profile. In the interstellar medium the corresponding problems are optical depth and chemistry, which bias any ratio measured from a single molecule. Both the stellar and the interstellar ratios would benefit from better laboratory atomic and molecular data.

The second lesson concerns scale. Nuclear processes change the isotope ratios of the bulk of an element by large factors, from the factor of ≈ 25 between the terrestrial and CN-equilibrium ${}^{12}\text{C}/{}^{13}\text{C}$ to the factors of hundreds in presolar grains. Chemical and physical processes redistribute isotopes among reservoirs, and in warm material they change ratios by parts per thousand to tens of percent. Changes by large factors occur only for hydrogen, whose isotopes differ in mass by a factor of two; at temperatures low enough that the zero-point energy difference exceeds kT ; and in reservoirs that hold a small fraction of an element (e.g., a rare isotopologue or the atmosphere of an HgMn star). The Solar System oxygen isotopes, once read as a nucleosynthetic anomaly, are now thought to record photochemistry in the solar nebula or its parent cloud. The D/H of the Earth's oceans records, in diluted form, ice chemistry at low temperature. Distinguishing a fingerprint from fractionation is the first step in interpreting any ratio. An element with at least three isotopes (e.g., oxygen, magnesium, sulfur) provides a first test, because mass-dependent processes move its ratios in a fixed proportion. However, a slope-1 anomaly can be either nuclear or photochemical, as the history of oxygen shows, and separating the two requires other evidence.

The third lesson is that the most informative ratios are built so that the unknown quantities cancel. A ratio of two isotopes of one element cancels the chemistry, and the U/Th ratio of a star also cancels most errors in the model atmosphere and the actinide boost. The quantity $\Delta^{17}\text{O}$ cancels mass-dependent fractionation, and the ${}^{13}\text{C}$ correction of radiocarbon cancels the fractionation by photosynthesis. The ratio of the ${}^{60}\text{Fe}$ and ${}^{26}\text{Al}$ line fluxes cancels the distances to the sources, and ${}^{129}\text{I}/{}^{247}\text{Cm}$ cancels the time between production and incorporation into the Solar System. The most robust clocks also cancel the initial abundance of the daughter (an isochron) or the amount of the parent (the Pb–Pb method). In stellar spectra, where the isotopes of an element cannot be separated, only elements that consist of a single radioactive nuclide (Tc, Th, and U) can serve as clocks. What cannot cancel is the reference with which a measured ratio is compared, and many of the disputes we have described concern that reference. Examples are the r-process production ratio behind actinide ages, the initial ${}^{60}\text{Fe}/{}^{56}\text{Fe}$ that bears on the birth environment of the Sun, and, until *Genesis* returned the solar wind, the oxygen isotope ratios of the Sun itself.

The fourth lesson is that many isotope abundances are set by a steady state, in which a nuclide accumulates until it is destroyed as fast as it is made. The CN-equilibrium ${}^{12}\text{C}/{}^{13}\text{C}$, the σN rule of the s process, the $2.8 M_{\odot}$ of ${}^{26}\text{Al}$ in the Galaxy, and the daughters in secular equilibrium are all examples. In each case a measured abundance gives a production rate, or a ratio of abundances gives a ratio of rates, independent of the earlier history. For comparable numbers of nuclei and

decay energies, radioactive heating at a time t comes mainly from nuclides whose mean lives are comparable to t , and once these have decayed even much rarer, longer-lived nuclides take over. This rule describes the order in which ^{56}Co , ^{57}Co , and ^{44}Ti power SN 1987A, the power-law heating of a kilonova, the melting of the first planetesimals by ^{26}Al , and the shift of the Earth's heat sources from ^{40}K and ^{235}U to ^{238}U and ^{232}Th .

The fifth lesson is that the same ratio can play different roles in different places. $^{12}\text{C}/^{13}\text{C}$ is a probe of mixing in red giants, a gauge of chemical enrichment across the Galaxy, and a tracer of the stellar populations of starburst galaxies, as Figs. 7, 8, and 9 show, and it is altered by ion–molecule chemistry in cold clouds (Section 6.2). D/H is a probe of the baryon density, a measure of stellar processing, and a record of ice chemistry and atmospheric escape, as Fig. 1 shows. ^3He constrains the baryon density and is thought to drive the mixing that lowers $^{12}\text{C}/^{13}\text{C}$ in giants. ^{22}Ne is a fingerprint of helium burning in the cosmic rays and a heat source in white dwarfs.

Many important problems remain open, and we end with a partial list:

- The predicted primordial D/H depends at the few-percent level on laboratory cross sections for $d(d,n)^3\text{He}$ and $d(d,p)t$, which now limit the comparison with the CMB.
- The extra mixing that lowers $^{12}\text{C}/^{13}\text{C}$ in red giants is well established observationally, but in hydrodynamic simulations the salt fingers of the proposed thermohaline mechanism are more than ten times too short, so the mixing is roughly two orders of magnitude too weak.
- The polluters behind the Mg isotope variations of globular clusters are unidentified.
- The $^{13}\text{CO}/\text{C}^{18}\text{O}$ ratios of dusty starburst galaxies suggest a top-heavy initial mass function, but the interpretation depends on uncertain yields, optical depths, burst ages, and gas mixing.
- The low ^{244}Pu in deep-sea deposits and the meteoritic $^{129}\text{I}/^{247}\text{Cm}$ point to rare and prolific r-process events, but the site or sites of the r process, and the variation of the actinide production from one event to the next (the actinide boost), are uncertain.
- The $^{12}\text{C}/^{13}\text{C}$ of the most metal-poor stars is far below the predictions of chemical evolution models without rotation (33 in HD 140283, compared with ≈ 4200), which suggests that the first stars made ^{13}C as a primary product, but the source has not been identified.
- The initial $^{60}\text{Fe}/^{56}\text{Fe}$ of the Solar System, which decides whether a nearby supernova enriched the solar nebula, is disputed by one to two orders of magnitude.
- The geoneutrino measurements of the radiogenic heat from ^{238}U and ^{232}Th in the Earth differ by a factor of three between KamLAND and Borexino, and neither experiment can detect ^{40}K .
- The ^{15}N enrichment of comets and of the organic hotspots of meteorites is not reproduced by gas-phase chemical models, and the roles of chemistry and N_2 self-shielding are unsettled.
- The sources of the Earth's water (carbonaceous, enstatite-like, cometary, or nebular) are unsettled.
- Atomic line lists lack isotopic structure for many important species, including Mg, Cr, and Fe.

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